Who Drives the U.S. Treasury Premium?

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Abstract

I quantify the contribution of different investor countries and sectors to the U.S. Treasury premium using a model of sovereign portfolio choice with sector-specific Treasury preferences, estimated on a new panel of Treasury holdings. The estimates point to Treasury inconvenience: preferences beyond Treasuries' risk-return profile raise the return every investor requires, and the model-implied premium averages −78 basis points over 2013–2025. An investor's contribution depends both on the size of its holdings and on how far Treasury preferences alter its demand elasticity. Domestic and foreign non-bank financial institutions and the foreign official sector, chiefly China, account for over two-thirds of it. U.S. debt sustainability is thus exposed to geopolitical realignment and to return-sensitive investors.

JEL-Classification: E43, F30, G11, G21, G22.
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1 Introduction

U.S. Treasuries are among the most important assets in the world. With a market capitalisation of $39 trillion, they sit at the centre of the international financial system as its principal safe asset, providing liquidity, a store of value and collateral for dollar-denominated transactions. These features have long earned Treasuries a premium, the convenience yield, allowing the U.S. government to borrow at lower rates than other advanced economies.[1]

The convenience yield plays a critical role for U.S. fiscal sustainability (Jiang et al., 2024d, Choi et al., 2024), but this funding advantage has eroded since the Great Financial Crisis, even turning negative (Du et al., 2018a, 2026, Jiang et al., 2026). This development is particularly concerning at a time of rapidly increasing public debt exceeding 120% of GDP and persistent budget deficits regularly above 5% of GDP, levels not seen outside of wartime or major crises. In this environment, even yield movements of a few basis points can suffice to tip the fiscal trajectory onto an unsustainable path.

It is therefore crucial to understand which investor sectors drive the Treasury premium and its reversal. The strategic implications for the U.S. government differ starkly depending on the answer. If the financial sector drives the premium, its evolution follows economic considerations such as liquidity demand and collateral value, which are broadly predictable. If instead it rests in the hands of geopolitically motivated foreign governments, the United States faces a strategic vulnerability. The question has gained urgency as official reserves diversify away from the dollar (Ito and McCauley, 2020) and as the geopolitical uncertainty surrounding the second Trump administration leads even longstanding allies to contemplate shedding Treasury holdings in response to perceived hostility (Subacchi and van den Noord, 2026).

This paper asks which investors drive the Treasury premium and its reversal. Between 2013 and 2025 the premium averaged about -78 basis points; domestic and foreign non-bank financial institutions, together with the foreign official sector (chiefly China), account for more than two-thirds of it.

The contribution of each investor to the Treasury premium depends both on its holding size and on the extent to which Treasury preferences alter its demand elasticity. These results qualify the rankings of investors' contributions to Treasury yields based on their size and elasticity (Eren et al., 2026, Fang et al., 2025): both matter, but neither is a sufficient statistic for an investor's pricing impact.

How far a sector's demand moves the convenience yield depends on its own preferences and on the preferences and responses of every other sector in equilibrium. Isolating each investor's contribution to the Treasury premium therefore requires a structural model.

To this end, I develop a model of portfolio choice in which investors across countries and sectors allocate their foreign sovereign portfolio between U.S. Treasuries and other foreign government bonds. Investors are mean-variance optimisers over wealth, augmented with a country- and sector-specific concave preference term for Treasury holdings, governed by a level and a curvature parameter. This term is a parsimonious, reduced-form representation of the benefits that different classes of investors derive from Treasuries beyond their risk-return profile: liquidity and collateral value for banks, liability-matching properties for pension funds, and reserve management objectives for official institutions. It allows the comparison of Treasury preferences across sectors without a separate micro-foundation for each. Concavity in Treasury preferences implies

[1] The literature refers to this phenomenon as either the Treasury premium (Du et al., 2018a) or the convenience yield (Jiang et al., 2025, 2024c). I use the two terms interchangeably throughout.

that the marginal benefit of holding Treasuries declines with the portfolio share, which, other things equal, dampens the sensitivity of portfolios to yield changes. In equilibrium, investors who value Treasuries beyond their risk-return profile accept a lower yield relative to comparable assets, generating a Treasury premium; the model also allows for Treasury "inconvenience", which has the opposite effect.

To bring the model to the data, I assemble a quarterly dataset of Treasury holdings from Q4 2013 to Q4 2025 for domestic investors and the five largest foreign holders: the euro area, Japan, the United Kingdom, China and Canada, jointly 56% of reported foreign holdings at the end of the sample. Country holdings from the U.S. Treasury International Capital (TIC) data are restated to attribute positions booked in financial centres to their ultimate investors, and split into banks, non-bank financial institutions (NBFIs), households and non-financial corporations (HH/NFC), and the official sector using the ECB's Securities Holdings Statistics, the IMF's Portfolio Investment Positions and the Federal Reserve's Financial Accounts. This yields 17 investor cells: four sectors each for the euro area, Japan and the United Kingdom, country aggregates for China and Canada, and three private U.S. sectors.

I estimate the Treasury preference parameters and risk aversion cell by cell via Generalised Method of Moments (GMM), using the first-order conditions of the investor's problem as moment conditions. Identification relies on granular instrumental variables (GIV) from Gabaix and Koijen (2024): the idiosyncratic demand shocks of investors in other countries and sectors ("donor" cells), purged of common drivers of Treasury demand, move equilibrium returns independently of the investor's own demand. Rather than averaging donor residuals into a single instrument as in Gabaix and Koijen (2024), I use each one separately, giving enough moment conditions to identify three structural parameters per investor.

The estimates point to Treasury *inconvenience* rather than convenience. The level parameter is negative in every cell, and the curvature parameter sits at its concavity bound in 9 of the 17 cells, including every euro-area and Japanese cell. Risk aversion is estimated at zero in 6 cells (all British cells, China and U.S. banks), which behave as if risk neutral. At their average holdings in partial equilibrium, Treasury preferences raise every investor's required Treasury excess return, by 76 basis points per annum on average.

I then compare portfolio shares and equilibrium returns at the estimated parameters with a counterfactual without Treasury preferences, in which investors value Treasuries only for their risk-return profile. Without Treasury preferences, every investor would hold a larger Treasury share, by between 12 and 57 percentage points: Treasury holdings are depressed, not inflated, relative to their risk-return benchmark. Portfolio shares are also more sensitive to excess returns at the estimated parameters in every cell except Canada, U.S. banks and U.S. households and non-financial corporations. This result stems mostly from counterfactual portfolios sitting more often at a corner, where they do not respond to returns.

The model-implied Treasury premium is negative, averaging -78 basis points per annum. In the model, the equilibrium Treasury excess return has two components that work in opposite directions. Treasuries' hedging properties earn them a negative risk premium: on risk-return grounds alone, investors would hold them at an expected return below that of other sovereign bonds. Treasury preferences instead raise the return investors require, and this second effect dominates, leaving the overall excess return positive. The sign matches the post-crisis reversal of the Treasury premium documented by Du et al. (2018a) and Du et al. (2026). The magnitude exceeds these currency-hedged, spread-based measures, consistent with a structural model capturing dimensions of Treasury convenience that a single spread does not.

The sectoral decomposition of the Treasury premium is the central finding of the paper. A cell's contribution depends on the size of its Treasury position and, crucially, on how much Treasury preferences affect its demand elasticity. China holds 11% of modelled Treasury holdings but accounts for 38% of the premium, because Treasury preferences sharply raise its elasticity. U.S. NBFIs show the opposite pattern: they are the largest modelled holder, with 43% of holdings, but contribute only 16% of the premium, as Treasury preferences barely move their elasticity.

Across sectors, NBFIs contribute 30% of the premium. Attributing China's contribution to its official sector in proportion to its official share of holdings, official investors account for 39%, the largest share of any sector. This qualifies the reading in Jiang et al. (2024b) and Tabova and Warnock (2026) of foreign official investors as price-inelastic buyers of safe dollar assets who accept low returns: in the model, China's weight in the premium comes from the elasticity of its demand, even though the other official cells, Japan's excepted, are among the least elastic.[2]

The sustainability of the burgeoning U.S. public debt is thus threatened not only by the erosion of Treasuries' global safe-asset status, but also by return-sensitive investors.

Related literature. This paper is situated in the literature on the convenience yield of U.S. Treasuries. A strand of this literature is concerned with the measurement of convenience yields, typically through a spread between Treasuries and comparably safe or liquid assets such as AAA corporate bonds (Krishnamurthy and Vissing-Jorgensen, 2012), repurchase agreements (Nagel, 2016), overnight indexed swaps (He et al., 2022) and foreign government bonds, adjusted for credit risk differentials and deviations in covered interest parity (Du et al., 2018a). I contribute by developing a model-based methodology to estimate the convenience yield of U.S. Treasuries relative to other foreign sovereign bonds.

Several papers underscore the significant contribution of Treasury convenience to a broad range of macroeconomic phenomena, including the dynamics of nominal and real exchange rates (Engel and Wu, 2023, Valchev, 2020, Jiang et al., 2021, 2024c, Graziano et al., 2026), fiscal sustainability (Jiang et al., 2024d, Choi et al., 2024, Jiang et al., 2024e), and global business and financial cycles (Bodenstein et al., 2023, Georgiadis et al., 2023, Jiang et al., 2024a). I expand the literature on convenience yields and fiscal sustainability by identifying which investors' Treasury preferences affect U.S. debt financing costs the most.

A more recent generation of studies investigates the origins and drivers of convenience yields. Acharya and Laarits (2023) emphasises the hedging properties of Treasuries as safe assets, showing that the convenience yield is inversely proportional to the correlation between Treasury and stock returns. Corell et al. (2025) decomposes the sources of convenience yields for European government bonds, identifying regulatory capital advantages as the largest component. I add another dimension to this line of inquiry by asking not *when* convenience yields arise or *what* they are, but rather *who* drives them.

My modelling approach – estimating an asset pricing model that combines price and quantity data – borrows from the demand system asset pricing framework of Koijen and Yogo (2019), adopted by a rapidly growing literature (Gabaix and Koijen, 2021, Bretscher et al., 2026, Haddad et al., 2025, Gabaix et al., 2024, Nenova, 2025). This paper goes beyond taking estimated elasticities as primitives, and instead recovers the underlying risk aversion and Treasury preference parameters

[2]The appendix reports two further equilibrium experiments. When one investor's Treasury preferences are switched off, U.S. banks and China, the investors with the largest absorption shares, take up most of the change in its position on average, and Treasury preferences raise the sensitivity of the equilibrium return to Treasury supply on average, mainly through China, although those of the Japanese and of the U.S. cells lower it.

directly, taking a step towards understanding the nature of demand heterogeneity at the core of the Koijen and Yogo (2019) model.

A strand of this literature applies the demand system framework to the market for U.S. Treasuries. Chaudhary et al. (2026) decomposes Treasury yield changes by sector, finding that since 2008 the Federal Reserve's role has grown while that of the foreign sector as a whole has diminished. Jansen et al. (2026) estimates maturity-specific demand functions in an equilibrium model with arbitrageurs, finding that Treasury market elasticity declines steeply with maturity and that investment funds and banks are more price elastic than insurers, pension funds and foreign official investors. Eren et al. (2026) finds that domestic banks and mutual funds display elastic demand for Treasuries, while foreign official demand is inelastic. My approach complements these studies by explaining cross-sectoral differences in elasticities through the lens of convenience yields: it decomposes them into risk aversion and Treasury preference components, and quantifies both the Treasury premium and each sector's contribution to it.

Closest to this paper is Cavaleri (2025), which models Treasury convenience explicitly as a feature of investor preferences within a demand system, estimates it on sectoral holdings data, and recovers a ranking of Treasury preferences across sectors: among domestic investors, banks and broker-dealers enjoy the largest convenience yields, while households, insurers and pension funds value convenience little. My paper differs in subject and method. I measure the convenience yield of U.S. Treasuries against foreign government bonds rather than domestic AAA corporate bonds. The two are related but distinct, and the former speaks more directly to the U.S. government's funding advantage and hence to fiscal policy. Methodologically, my approach recovers point estimates of Treasury preference parameters rather than rankings, opening the door to quantitative counterfactual experiments on the role of Treasury convenience for demand and yields.

Roadmap. The remainder of the paper is structured as follows. Section 2 lays out a model of sovereign portfolio choice with Treasury preferences and derives its comparative statics. Section 3 describes the construction of the holdings dataset. Section 4 details the estimation procedure that maps the model to the data, and Section 5 reports the estimated Treasury preference parameters. Section 6 carries out counterfactual experiments quantifying the effect of Treasury preferences on portfolios and equilibrium returns, and decomposes the Treasury premium across countries and sectors. Section 7 concludes. The appendix contains the model derivations, the details of the dataset construction and of the estimation, additional experiments and robustness tests.

2 A model of portfolio choice with Treasury convenience

2.1 Model setup

Investors, portfolio shares and returns. Consider a world populated by a finite cross-section of mean-variance investors, indexed by country and sector. Foreign investors are indexed by (c, i), where $c \in \mathcal{C}^F \coloneqq \{\text{EA}, \text{JPN}, \text{GBR}, \text{CHN}, \text{CAN}\}$ denotes the country of residence and $i \in \mathcal{I} \coloneqq \{\text{B}, \text{N}, \text{H}, \text{O}\}$ denotes the sector within that country: banks (B), non-bank financial institutions (N), households and non-financial corporations (H), and the official sector (O; central banks plus general government). Each pair (c, i) is represented by a single mean-variance investor, who chooses between U.S. Treasuries and a portfolio of foreign government bonds, defined as the market-capitalisation-weighted portfolio of sovereign bonds issued by countries other than the

United States and country c itself.[3] Alongside foreign investors, the model features three sector-specific domestic U.S. investors, indexed (US, i) for $i \in \{\mathrm{B}, \mathrm{N}, \mathrm{H}\}$—banks, non-bank financial institutions, and households plus non-financial corporations—each of whom chooses between U.S. Treasuries and the foreign sovereign portfolio. The full investor universe is therefore $\mathcal{J} := \{(c, i) : c \in \mathcal{C}^F, i \in \mathcal{I}\} \cup \{(US, i) : i \in \{\mathrm{B}, \mathrm{N}, \mathrm{H}\}\}$. In the empirical implementation, China and Canada enter the estimation as country-level aggregates (a single cell per country) rather than being disaggregated into the four sectors of $\mathcal{I}$, because sector-level data for these two countries are either too short or too coarse to support sector-by-sector identification; the euro area, Japan, and the United Kingdom are disaggregated by sector as in the model definition.

Each investor $j \in \mathcal{J}$ chooses a *portfolio share* of U.S. Treasuries,

$$s_{j,t} \;:=\; \frac{H^{UST}_{j,t}}{H^{UST}_{j,t} + H^{F\text{-}gov}_{j,t}} \;\in [0, 1], \tag{1}$$

where $H^{UST}_{j,t} \geq 0$ is the U.S.-dollar value of Treasuries held by cell j and $H^{F\text{-}gov}_{j,t} \geq 0$ the U.S.-dollar value of its holdings of foreign sovereign bonds. The choice of a foreign rather than domestic benchmark aims at avoiding the issue of home bias. Global bond portfolios display a pervasive home bias (Coeurdacier and Rey, 2013, Maggiori et al., 2020, Florez-Orrego et al., 2024), rooted in currency matching of liabilities, regulation, information and institutional frictions that have nothing to do with the special status of U.S. Treasuries. Measuring Treasury holdings against the domestic government bond would load this home bias onto the convenience parameters, conflating the premium investors attach to Treasuries with the premium they attach to any foreign asset relative to their own sovereign. By excluding the domestic bond from the benchmark, the portfolio share isolates the allocation *within* the foreign sovereign portfolio, so that the convenience parameters measure the value of U.S. Treasuries relative to other foreign sovereigns of comparable safety and liquidity.

The investor's *portfolio wealth* is defined as the value of its foreign sovereign portfolio,

$$W_{j,t} \;:=\; H^{UST}_{j,t} + H^{F\text{-}gov}_{j,t} \;>\; 0. \tag{2}$$

For each cell j, the foreign sovereign portfolio has gross return

$$R^F_{j,t+1} \;:=\; \sum_{k \in \mathcal{C}^F} w^F_{j,k}\, R_{k,t+1}, \tag{3}$$

where $R_{k,t+1}$ is the gross return on country-k sovereign bonds and the weights $w^F_{j,k} \geq 0$, summing to one, are fixed to market capitalisation with $w^F_{(c,i),c} = 0$ for foreign cells. Therefore, I assume a fixed allocation within the benchmark foreign portfolio excluding the U.S.

For each foreign country c, the expected excess return, return variance, and covariance of U.S. Treasuries with the benchmark return are

$$E_{c,t} := \mathbb{E}_t[R_{US,t+1} - R^F_{c,t+1}], \quad V_{c,t} := \mathbb{V}_t[R_{US,t+1} - R^F_{c,t+1}], \quad C_{c,t} := \mathbb{C}_t[R_{US,t+1} - R^F_{c,t+1}, R^F_{c,t+1}],$$

where $R_{US,t+1}$ is the gross return on U.S. Treasuries and $R^F_{c,t+1}$ the gross return on the foreign sovereign portfolio of country-c investors. All returns are unhedged and measured in the investor's currency, so that both legs of the portfolio carry exchange rate risk. The pair $(V_{c,t}, C_{c,t})$ is proxied

[3] In the empirical implementation, this portfolio is the ex-U.S., ex-domestic World Government Bond Index (WGBI).

with sample equivalents from the data; $E_{c,t}$ is determined endogenously through global market clearing.

The three domestic U.S. investors face common moments $(E_{US,t}, V_{US,t}, C_{US,t})$, defined relative to the common benchmark $R^F_{US,t+1}$ in U.S. dollars.

Because both assets are foreign to investor c, foreign exchange movements affect returns only through the dollar's exchange rate *vis à vis* the reference basket currencies. Let $X_{c,t+1}$ denote the gross change in the price of a U.S. dollar in units of currency c. Translating both legs into currency c gives

$$R_{US,t+1} - R^F_{c,t+1} \;=\; X_{c,t+1}\big(R^{\$}_{US,t+1} - R^{F,\$}_{c,t+1}\big),$$

where the superscript \$ denotes dollar returns. To a first-order approximation, and exactly in log returns, the home currency cancels: the excess return of country-c investors is the dollar excess return of Treasuries over the foreign sovereign portfolio, and $(E_{c,t}, V_{c,t})$ differ across countries only through the exclusion of country-c bonds from the benchmark weights $w^F_{c,k}$.

Three implications follow. First, the Treasury excess return is close to a common global object across cells, which is the natural counterpart of a single market-clearing U.S. Treasury return and places the foreign and domestic U.S. blocks on the same footing. Second, the covariance term $C_{c,t}$ does not cancel, because the benchmark return in currency c carries the exchange rate risk of the basket against currency c; cross-country heterogeneity in the risk-return trade-off therefore enters mainly through $C_{c,t}$. Third, with excess returns nearly common, identification of sector- and country-specific convenience parameters rests on cross-sectional variation in portfolio shares at a given excess return, a point I return to below.

Preferences. Each investor j has an exponential-utility objective in terminal wealth augmented by an additive convenience term $h_j(s_{j,t})$. As in the canonical mean-variance derivation under conditionally normal returns, this reduces to a mean-variance objective in the terminal portfolio return:

$$\max_{s_{j,t}\in[0,1]} \; \mathbb{E}_t[r^{port}_{j,t+1}] - \frac{\gamma_j}{2}\mathbb{V}_t[r^{port}_{j,t+1}] + h_j(s_{j,t}),$$

where γ_j is the coefficient of absolute risk aversion, and terminal wealth per unit of portfolio value $W_{j,t}$ is the share-weighted average of the two assets' gross returns: $r^{port}_{j,t+1} := s_{j,t}(R_{US,t+1} - R^F_{j,t+1}) + R^F_{j,t+1}$, with $R^F_{j,t+1} \equiv R^F_{c,t+1}$ for foreign cells $j = (c,i)$ and $R^F_{j,t+1} \equiv R^F_{US,t+1}$ for the domestic U.S. cells. Appendix A.1 provides the full derivation.

The convenience function h_j is a reduced-form representation of the non-monetary payoffs that investors receive from Treasuries. The underlying mechanisms giving rise to a convenience yield are highly heterogeneous across sectors. Banks and investment funds value the liquidity of Treasury bills to meet redemptions of on-demand dollar claims (Doerr et al., 2023), and value notes or bonds as repo collateral (Krishnamurthy, 2002, Laarits et al., 2026). Foreign pension funds, especially in low-interest-rate countries, value long-maturity Treasuries over other long-maturity safe sovereigns as they provide safe returns that match the maturity of their liabilities and fit within their regulatory perimeter (Bernanke, 2005, Greenwood and Vissing-Jorgensen, 2018, Corell et al., 2025). Central bank reserve managers seek Treasuries as a safe and liquid container for their dollar holdings, ultimately motivated by exchange-rate policy or geopolitical aims (Bernanke et al., 2011, Krishnamurthy and Lustig, 2019).

Providing a separate micro-foundation for each sector would significantly increase the complexity of the model without adding much insight. As pointed out in Krishnamurthy and Ma (2025),

nothing fundamental is lost by adopting a reduced-form approach: the equilibrium convenience yield simultaneously reflects the combined incentives of all marginal investors, rather than the simple sum of distinct components attributable to liquidity, collateral value, or other features.

To obtain an analytical solution of the investor's problem, I adopt the specification

$$h_j(s_{j,t}) \;=\; \psi_{0,j}\, s_{j,t} \;+\; \psi_{1,j} \log s_{j,t},$$

with parameters $\psi_{0,j}$ (level) and $\psi_{1,j}$ (curvature). I assume $\psi_{1,j} \geq 0$ for concavity, implying a declining marginal benefit of Treasury convenience. This functional form follows Krishnamurthy and Vissing-Jorgensen (2012). Since $s_{j,t}$ is the share of Treasuries within the foreign sovereign portfolio, h_j captures the convenience of U.S. Treasuries *relative to other foreign sovereigns*, such as German, Japanese or British government bonds. The parameters $(\psi_{0,j}, \psi_{1,j})$ therefore measure the specific premium attached to the dollar safe asset, net of any safe-asset advantage other government bonds in the reference basket may enjoy.

Risk aversion γ_j is also heterogeneous and estimated for each investor. I assume $\gamma_j \geq 0$ to rule out risk-loving investors, but the model remains well defined in the risk-neutral limit $\gamma_j \to 0$, as shown in Appendix A.4. At the risk-neutral limit, two cases arise depending on the curvature parameter. If $\psi_{1,j} > 0$, diminishing marginal convenience replaces risk aversion as the source of an interior portfolio, and Treasury demand is driven only by excess returns and convenience motives. If $\psi_{1,j} = 0$, so that both risk aversion and curvature are absent, the portfolio choice collapses to a corner solution: the investor holds only Treasuries if the equilibrium excess return exceeds the required return implied by $\psi_{0,j}$, none if it falls short, and is indifferent between all Treasury shares if the two coincide.

2.2 Optimal portfolio shares

Solution. The first-order condition of cell j's problem with respect to $s_{j,t}$ is

$$f(E_{j,t}; \psi_{0,j}, \psi_{1,j}) \;:=\; E_{j,t} - \gamma_j V_{j,t}\, s_{j,t} - \gamma_j C_{j,t} + \psi_{0,j} + \frac{\psi_{1,j}}{s_{j,t}} \;=\; 0, \tag{4}$$

where $j = (c, i)$, and the parameters $(\psi_{0,j}, \psi_{1,j}) \equiv (\psi_{0,c,i}, \psi_{1,c,i})$ are sector-and-country specific.

Equation (4) can be rearranged into a quadratic in $s_{j,t}$ that admits an explicit closed-form positive root, reported in Proposition 1. I select the positive root because $s_{j,t} > 0$ is required both by the log term and by the implicit no-short-selling constraint.

Proposition 1. *The optimal non-negative portfolio share $s^*_{j,t}$ of investor j's problem is*

$$s^*_{j,t} \;=\; \frac{E_{j,t} - \gamma_j C_{j,t} + \psi_{0,j} \;+\; \sqrt{(E_{j,t} - \gamma_j C_{j,t} + \psi_{0,j})^2 + 4\gamma_j V_{j,t} \psi_{1,j}}}{2\gamma_j V_{j,t}}. \tag{5}$$

Proof. In Appendix A.2. □

Absent a convenience yield, with $\psi_{0,j} = \psi_{1,j} = 0$, the expression collapses to the standard optimal portfolio prescription of a mean-variance problem. If $\psi_{0,j} \neq 0$ and $\psi_{1,j} = 0$, so that the utility benefit of holding Treasuries is linear in the portfolio share, the presence of a convenience yield (or "inconvenience" for $\psi_{0,j} < 0$) causes a level shift in the optimal share but leaves its sensitivity to excess returns unaffected in an interior solution. In the general case $\psi_{0,j} \neq 0$ and $\psi_{1,j} \neq 0$, the convenience yield affects both the level of the optimal share and its return sensitivity through $\psi_{1,j}$.

If $\psi_{0,j} > 0$ and $\psi_{1,j} > 0$, $s^*_{j,t}$ is larger than its counterpart in the standard mean-variance problem. Investors who value Treasuries beyond their monetary returns are willing to hold a larger portfolio share for any combination of risk and returns than those who do not. The presence of non-monetary payoffs also implies that the problem admits a non-zero solution for $E_{j,t} - \gamma_j C_{j,t} < 0$, that is when the risk-return trade-off of Treasuries is particularly poor.

This case of $\psi_{0,j} < 0$ should not be discarded *a priori*: Treasury "inconvenience" has been documented both in specific acute episodes such as the COVID pandemic (He et al., 2022), and more chronically as a trend starting after the Great Financial Crisis (Klingler and Sundaresan, 2023, Du et al., 2026, Jiang et al., 2026). I allow for Treasury "inconvenience" by leaving $\psi_{0,j}$ unrestricted, while I impose $\psi_{1,j} \geq 0$ to preserve concavity in the convenience function, which guarantees a closed-form solution.

The presence of Treasury preferences allows for an interior solution of the portfolio choice problem even for risk-neutral investors. As long as $\psi_{1,j} > 0$, the diminishing marginal benefit of Treasury convenience ensures concavity in the objective function even with $\gamma_j = 0$. Therefore, all results below extend to the risk-neutral limit (Appendix A.4).

The different implications of the level and curvature parameters on optimal holdings reveal the limitation of modelling convenience yield as a residual latent demand term in a log-linear regression, as for example in Koijen and Yogo (2026) within a demand-system framework.

Portfolio-share sensitivity. To qualify these statements more precisely, I compute the derivatives of the optimal portfolio share with respect to expected excess returns and the convenience parameters, which will also be useful as ingredients for comparative statics on the equilibrium return.

Proposition 2. *The total derivative of the optimal portfolio share $s^*_{j,t}$ with respect to the equilibrium excess return $E_{j,t}$ is*

$$\frac{\mathrm{d}s^*_{j,t}}{\mathrm{d}E_{j,t}} = \frac{1}{\gamma_j V_{j,t} + \psi_{1,j}/(s^*_{j,t})^2}. \tag{6}$$

$\frac{\mathrm{d}s^*_{j,t}}{\mathrm{d}E_{j,t}} > 0$ *if* $\gamma_j V_{j,t} + \psi_{1,j}/(s^*_{j,t})^2 > 0$.

Proof. In Appendix A.3.1. □

A diminishing marginal utility of holding Treasuries, captured by $\psi_{1,j} > 0$, reduces the sensitivity of portfolio shares to excess returns.

Higher exposure to Treasuries implies a steeply declining marginal benefit, reducing investors' incentive to rebalance in the face of higher returns. In this sense, a concave Treasury convenience yield plays the same role as risk aversion. A precise, model-consistent quantification of the U.S. Treasury premium therefore cannot rely solely on the cross-sectoral comparison of elasticities because it cannot disentangle the roles of risk aversion and convenience yields, as pointed out in Cavaleri (2025), and requires direct estimation of structural preference parameters.

Proposition 3. *The partial derivative of the optimal portfolio share $s^*_{j,t}$ with respect to the level parameter $\psi_{0,j}$ at a fixed return $R_{US,t+1}$ (and hence at fixed $E_{j,t}$) is*

$$\frac{\partial s^*_{j,t}}{\partial \psi_{0,j}}\bigg|_R = \frac{1}{\gamma_j V_{j,t} + \psi_{1,j}/(s^*_{j,t})^2}. \tag{7}$$

$\frac{\partial s^*_{j,t}}{\partial \psi_{0,j}}\Big|_R > 0$ *if* $\gamma_j V_{j,t} + \psi_{1,j}/(s^*_{j,t})^2 > 0$.

Proof. In Appendix A.3.2. □

Note that $\frac{\partial s_{j,t}}{\partial \psi_{0,j}}\Big|_R = \frac{\partial s_{j,t}}{\partial E_{j,t}}$. Both excess returns and the linear component of Treasury preferences produce a level shift in the optimal portfolio share, so their marginal effect coincides.

Proposition 4. *The partial derivative of the optimal portfolio share $s^*_{j,t}$ with respect to the curvature parameter $\psi_{1,j}$ at a fixed return $R_{US,t+1}$ (and hence at fixed $E_{j,t}$) is*

$$\frac{\partial s^*_{j,t}}{\partial \psi_{1,j}}\Big|_R = \frac{1/s^*_{j,t}}{\gamma_j V_{j,t} + \psi_{1,j}/(s^*_{j,t})^2}. \tag{8}$$

$\frac{\partial s^*_{j,t}}{\partial \psi_{1,j}}\Big|_R > 0$ *if* $\gamma_j V_{j,t} + \psi_{1,j}/(s^*_{j,t})^2 > 0$.

Proof. In Appendix A.3.3. □

The optimal share is increasing in both the level and curvature parameters, provided the regularity condition $\gamma_j V_{j,t} + \psi_{1,j}/s^2_{j,t} > 0$ is met.[4]

The convenience yield acts as an additional incentive to hold U.S. Treasuries beyond their risk-return trade-off. Excess returns and the level parameter enter the first-order condition identically, as additive shifts in the marginal benefit of holding Treasuries, so their effects on the optimal share coincide. The curvature parameter instead has a stronger effect when Treasuries are scarce in the portfolio because of the diminishing marginal benefits induced by a concave convenience function.

2.3 Equilibrium

I now characterise the equilibrium and analyse the properties of the equilibrium U.S. Treasury return, focusing on the consequences of Treasury convenience preferences.

Definition. *An equilibrium is defined by the set of portfolio shares $\{s_{j,t}\}_{j\in\mathcal{J}}$ and the equilibrium U.S. Treasury return $R_{US,t+1}$ such that*

1. *$s_{j,t}$ satisfies the optimality condition (4) for every $j \in \mathcal{J}$, given $E_{j,t}$, determined by $R_{US,t+1}$ and the (exogenous) benchmark return for cell j;*
2. *the global market for U.S. Treasuries clears.*

Market clearing. The 17 modelled investor cells together with an exogenous, inelastic residual investor $H^{\text{Res}}_t \geq 0$ absorb the U.S. Treasury supply S_t. The residual bucket collects three groups of non-modelled holders: (a) the Federal Reserve's System Open Market Account (SOMA), which held approximately USD 4–6 trillion in U.S. Treasuries at its post-QE peak and whose holdings are set exogenously by monetary policy (QE/QT programs) rather than by portfolio optimisation; (b) the U.S. official sector excluding SOMA (general government and public pension holdings, averaging on the order of tens of billions of dollars over the sample); and (c) non-dataset foreign investors—small reporters in offshore and financial-centre jurisdictions and all other foreign holders outside the five modelled countries. The residual is taken from the data as the difference between total supply and the modelled holdings, and treating H^{Res}_t as exogenous is the minimum necessary data-driven closure to make global market clearing hold. Because the

[4] Under the assumptions $\gamma_j \geq 0$, $\psi_{1,j} \geq 0$, and $s_{j,t} \in [0,1]$ and with a non-negative variance $V_{j,t} \geq 0$, $\gamma_j V_{j,t} + \psi_{1,j}/s^2_{j,t} \geq 0$, with the edge case equal to 0 for cells with $\gamma_j = 0$ and $\psi_{1,j} = 0$.

residual bucket is inelastic, it does not enter the existence-and-uniqueness argument, which still relies on monotonicity of modelled demand.

The global market-clearing condition, stated in dollar holdings, is

$$G_t(R_{US,t+1}) \;\coloneqq\; \sum_{c\in\mathcal{C}^F,\,i} H^*_{c,i,t}(R_{US,t+1}) \;+\; \sum_{i\in\{\mathrm{B,N,H}\}} H^*_{US,i,t}(R_{US,t+1}) \;+\; H^{\mathrm{Res}}_t \;-\; S_t \;=\; 0, \quad (9)$$

with $H^*_{j,t}(R_{US,t+1}) = W_{j,t}\, s^*_{j,t}(R_{US,t+1})$ from the closed-form solution (5), and where each cell's excess return is an affine function of $R_{US,t+1}$ with slope one $E_{c,t} = R_{US,t+1} - \mathbb{E}_t[R^F_{c,t+1}]$ for foreign cells, $E_{US,t} = R_{US,t+1} - \mathbb{E}_t[R^F_{US,t+1}]$ for the domestic U.S. cells.

Equilibrium U.S. Treasury return. The unique scalar unknown of the global system is the equilibrium U.S. Treasury return $R^*_{US,t+1}$; all cell-specific excess returns follow once it is known. Because $s_{j,t} \in [0,1]$, each cell's dollar demand $W_{j,t}s^*_{j,t}$ lies in $[0, W_{j,t}]$ and is non-decreasing in $R_{US,t+1}$, strictly so wherever the share is interior (Proposition 2). As $R_{US,t+1} \to -\infty$ every share tends to zero and $G_t \to H^{\mathrm{Res}}_t - S_t < 0$; as $R_{US,t+1} \to +\infty$ every share reaches one and $G_t \to \sum_j W_{j,t} + H^{\mathrm{Res}}_t - S_t$, which is positive whenever observed shares are interior, since then $\sum_j W_{j,t} > \sum_j W_{j,t}s_{j,t} = S_t - H^{\mathrm{Res}}_t$. A root therefore exists, and it is unique whenever at least one cell is interior at the root, so that G_t is strictly increasing there. The argument covers the risk-neutral limit $\gamma_j = 0$ with $\psi_{1,j} > 0$. When a cell with $\gamma_j = \psi_{1,j} = 0$ has step demand at its indifference return, a unique equilibrium still exists, as shown in Appendix A.4.

2.4 Comparative statics

The most directly observable effect of the special role of U.S. Treasuries is the comparatively low return historically required by investors, especially foreign ones, to hold U.S. government debt (Jiang et al., 2024d,b). The Treasury premium is often proxied by the spread between U.S. and other government bonds adjusted for credit risk and CIP deviations (Du et al., 2018a), paralleled in the model by the gap between the equilibrium U.S. Treasury return and the benchmark foreign sovereign return.

The convenience yield also affects the absorption capacity of the market for additional debt issuance, with a robustly negative relationship between Treasury supply and convenience yields (Krishnamurthy and Vissing-Jorgensen, 2012, Du et al., 2018a, Jiang et al., 2026) and evidence that the United States can issue more debt than other countries for a given increase in return because of the global safe-asset status of Treasuries (Jiang et al., 2024e).

I confirm that the model captures the first effect by analysing comparative statics for the effect of Treasury preferences on the equilibrium return. The second, the effect of Treasury preferences on the sensitivity of the return to debt supply, is characterised in Appendix A.6.

Before moving to the comparative statics, it is useful to define some quantities used throughout the section. Define the cell-level demand-slope denominator

$$D_{j,t} \;\equiv\; \gamma_j V_{j,t} + \frac{\psi_{1,j}}{s^2_{j,t}},$$

and the aggregate wealth-scaled demand slope denominator

$$\mathcal{A}_t \;\equiv\; \sum_{j\in\mathcal{J}} \frac{W_{j,t}}{D_{j,t}},$$

which is strictly positive under the regularity condition $D_{j,t} > 0$ for all $j \in \mathcal{J}$ — the maintained assumption throughout the comparative statics that follow.

Finally, define cell j's *absorption share*

$$a_{j,t} \equiv \frac{W_{j,t}/D_{j,t}}{\mathcal{A}_t}, \qquad \sum_{j \in \mathcal{J}} a_{j,t} = 1,$$

the fraction of aggregate dollar demand sensitivity accounted for by cell j. Under the regularity condition, $a_{j,t}$ governs how any convenience shock is redistributed across holders in equilibrium, as shown in Proposition 7 in Appendix A.6.

All derivatives of the equilibrium return in this section are *total* derivatives, denoted $\mathrm{d}R^*_{US,t+1}/\mathrm{d}(\cdot)$: they are obtained by differentiating the market-clearing condition (9) in equilibrium, so that the re-optimisation of every cell in $\mathcal{J}$ in response to the induced return change is accounted for. Because the residual bucket is inelastic, the entire adjustment runs through the modelled cells. Partial derivatives at a fixed return, denoted $\partial(\cdot)/\partial(\cdot)|_R$, are used only as intermediate objects and are labelled explicitly.

Treasury preferences and the equilibrium return.

Proposition 5. *The total derivative of the equilibrium U.S. Treasury return $R^*_{US,t+1}$ with respect to the level parameter $\psi_{0,j}$ is*

$$\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}\psi_{0,j}} = -\frac{W_{j,t}/D_{j,t}}{\mathcal{A}_t} = -a_{j,t}. \tag{10}$$

If $D_{j,t} > 0$ for all j, then $\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}\psi_{0,j}} \leq 0$.

Proof. In Appendix A.5.1. □

As investors value U.S. Treasuries beyond their risk-return profile, they require a lower monetary compensation in equilibrium the stronger their preferences. If, instead, investors attach a strong enough "inconvenience yield" to Treasuries to offset risk aversion ($D_{j,t} < 0$), they will require a higher return. The magnitude of the effect is proportional to cell j's absorption share $a_{j,t}$: the larger and more elastic a cell's dollar demand at the margin, the larger its contribution to the equilibrium return. The level parameter $\psi_{0,j}$ shifts demand without entering the demand slope $D_{j,t}$ directly; it affects individual sensitivities to excess returns only indirectly, through the induced change in $s_{j,t}$.

Equation (10) has an impact-multiplier structure. The numerator, $W_{j,t}/D_{j,t} = W_{j,t}\,\partial s_{j,t}/\partial\psi_{0,j}|_R$, is the excess dollar demand that cell j would generate at the initial return; the denominator is the aggregate slope of demand, which measures how much every cell—including j itself—cuts its holdings when the return falls. Their ratio, the absorption share $a_{j,t}$, is therefore the return change needed for the market to re-absorb cell j's additional demand once all investors have re-optimised.

Proposition 6. *The total derivative of the equilibrium U.S. Treasury return $R^*_{US,t+1}$ with respect to the curvature parameter $\psi_{1,j}$ is*

$$\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}\psi_{1,j}} = -\frac{W_{j,t}/(s_{j,t}\,D_{j,t})}{\mathcal{A}_t} = -\frac{a_{j,t}}{s_{j,t}}. \tag{11}$$

If $D_{j,t} > 0$ for all j, then $\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}\psi_{1,j}} \leq 0$.

Proof. In Appendix A.5.2. □

The same properties of the derivative with respect to $\psi_{0,j}$ carry through to $\psi_{1,j}$. Both the level shifter and the curvature of Treasury preferences reduce the equilibrium return in proportion to the cell's demand elasticity. The factor $1/s_{j,t}$ arises because a unit increase in $\psi_{1,j}$ raises marginal convenience at the current allocation by $\partial h'_j/\partial \psi_{1,j} = 1/s_{j,t}$. With log convenience, marginal convenience is highest where Treasuries are scarce, so curvature shocks move the return most through cells with low Treasury shares.

The model-implied Treasury premium. I define the model-implied Treasury premium TP_t as the counterfactual change in the equilibrium U.S. Treasury return obtained by switching off the convenience channel:

$$TP_t \;\coloneqq\; R^*_{US,t+1}\Big|_{\psi=0} \;-\; R^*_{US,t+1}\Big|_{\psi^*}, \tag{12}$$

with both terms evaluated at the same primitives $(V_{j,t}, C_{j,t}, S_t, H_t^{\text{Res}})$ and the same residual bucket. The first term is the equilibrium return that would prevail without convenience preferences ($\psi_{0,j} = \psi_{1,j} = 0$ for every j); the second is the equilibrium return at the realised ψ^*. Under $\psi_{0,j} \geq 0$, $\psi_{1,j} \geq 0$ with at least one parameter strictly positive, $TP_t > 0$: a strictly positive return gap arises because convenience preferences raise the demand schedule at any return, so the market-clearing return with convenience is strictly lower than without.

Equation (12) reflects a general-equilibrium counterfactual: when convenience is removed, each investor re-optimises and portfolio shares move from $s^*_{j,t}$ to a counterfactual $s^0_{j,t}$ at which the global market re-clears with the same supply S_t and the same residual H_t^{Res}. This Treasury premium is thus a theoretical object that depends on the structural parameters $\{\psi_{0,j}, \psi_{1,j}\}_j$.

It can therefore differ from empirical proxies such as Treasury-AAA corporate spreads (Krishnamurthy and Vissing-Jorgensen, 2012) or Treasury-foreign government bond spreads (Du et al., 2018a), offering an alternative perspective on the measurement of Treasury convenience. Because the benchmark is a portfolio of foreign sovereigns, TP_t is conceptually closest to the latter class of proxies, with one difference: the spreads in Du et al. (2018a) are hedged for currency risk, while the model's excess return is unhedged and TP_t is defined net of the risk premium on the dollar against the basket of benchmark currencies, which the mean-variance term prices explicitly.

I explore two further comparative statics in Appendix A.6: the redistribution of a convenience shock across holders, governed by the absorption shares $a_{j,t}$, and the effect of Treasury preferences on the sensitivity of the equilibrium return to Treasury supply.

3 Data

Taking the portfolio problem of Section 2 to the data requires holdings of U.S. Treasuries by investor country and by investor sector, together with the sovereign portfolio against which each investor allocates. No single source provides this. I therefore assemble a quarterly dataset covering the five largest foreign holders of U.S. Treasury securities—the euro area, Japan, the United Kingdom, China and Canada, jointly 56% of all reported foreign holdings at the end of the sample—together with the United States itself, from Q4 2013 to Q4 2025. Where the sources permit, holdings are split into four investor sectors: banks, NBFIs, HH/NFC, and the official sector. This section describes the construction strategy and the patterns in the resulting data;

the full methodology, the per-jurisdiction treatments and the maintained assumptions are set out in Appendix B.

3.1 Construction strategy

The dataset is built in three layers: country-level Treasury holdings, a sector decomposition of holdings, and a denominator measuring the sovereign portfolio of Treasuries and other foreign government bond holdings. Each layer draws on the most direct source available for each jurisdiction.

Country holdings and financial-centre restatements. The baseline source for Treasury holdings is the U.S. Treasury's TIC series of foreign holdings, which records positions by the country of the immediate holder. This is not the country of the ultimate investor. A handful of jurisdictions host round-trip fund vehicles or act as custodians for non-resident clients, so that holdings appear in the financial centre of record rather than with the investor who bears the risk, a well-known limitation of this data source (Bertaut and Judson, 2023, Judson and Kim, 2026). In order to alleviate this issue, I apply a sequence of restatements aimed at attributing Treasury holdings to their ultimate owner wherever possible.

Offshore conduits whose reported positions predominantly reflect fund vehicles are removed outright, which excludes the Cayman Islands and Bermuda among major Treasury holders. Luxembourg and Belgium both host large securities custodians (Clearstream and Euroclear). Therefore, their total is scaled to its non-custodial fraction, estimated from the ratio of its ultimate-ownership bilateral position in U.S. debt securities from the IMF's Portfolio Investment Position (PIP) database to its reported holdings of U.S. debt from TIC. The Luxembourg and Belgian custodial gaps are pooled and reallocated to the non-euro-area dataset countries in proportion to their shares of world official reserve assets, on the grounds that the clientele of these two clearing houses is predominantly official; the part of the pool accruing to economies outside the dataset is dropped. In addition, Luxembourg and Ireland host a large investment fund industry, whose security holdings partly reflect ultimate owners outside of the euro area. The non-euro-area component of Luxembourg and Irish investment-fund holdings is then netted out of the euro-area total using the ultimate-ownership look-through of Beck et al. (2026). Holdings booked in Hong Kong and Switzerland are reallocated to ultimate investor countries in proportion to bilateral cross-border banking positions, following Alstadsæter et al. (2018). The same logic identifies a custodial component in the United Kingdom, which is removed without reallocation, because UK-resident positions blend custody, fund management and international banking in proportions that public data cannot separate. Canada's total holdings do not rely on TIC: its holdings of U.S. general-government debt are observed directly in Canadian national accounts and combined with the dollar component of official reserves.

Sector decomposition. Sector shares come from the best investor-country source in each jurisdiction, or from PIP whenever such a source is not available. For the euro area I use the European Central Bank's Securities Holdings Statistics (SHS) database, which reports resident holdings by holder sector and, until 2022, separately by issuer country; from 2022 the issuer-country dimension is discontinued, and the sector composition is recovered from PIP and calibrated on the overlap window in which both are observed. For Japan and the United Kingdom, sector shares come from PIP, which reports bilateral holdings by holder sector semi-annually, interpolated to quarterly. For the United States, sector holdings are taken directly from the Federal Reserve's Financial Accounts. Appendix B.2 documents the sectoral mapping procedure

country by country.

China and Canada enter the estimation as country aggregates rather than as sector panels. The split available for both countries in PIP is significantly shorter than the sample. Furthermore, the sectoral decomposition is available only for total foreign holdings with no country breakdown for Canada, and against the United States but without a breakdown by security type for China. Descriptive sector decompositions are built for both countries and shown below, but do not enter the estimation.

The official sector. Official holdings are the hardest cell to measure, because balance-of-payments convention excludes reserve assets from portfolio surveys. Where reserve data permit I measure the official cell directly, applying an estimated dollar share of reserves to reported holdings of reserve securities and adding resident general-government holdings. The dollar share comes from the currency-composition evidence of Ito and McCauley (2020), extended with euro-area national central banks' own published accounts, for the euro area; from the Bank of England's published currency breakdown for the United Kingdom; and from the Exchange Fund Account for Canada. Japan is the one case in which the official cell remains a residual, the difference between the restated Treasury holdings and the private sectors measured on a matched instrument scope, clamped at zero. The U.S. official sector, which includes the Federal Reserve's System Open Market Account, is not modelled as a cell at all; it is absorbed into the inelastic residual of the market-clearing condition of Section 2.3, because its positions are set by monetary policy rather than by portfolio choice.

The portfolio share. The model's choice variable is the weight of U.S. Treasuries in the investor's sovereign portfolio. The numerator is the cell's Treasury holdings described above; the denominator is the cell's total holdings of sovereign debt issued outside its own jurisdiction, so that the share measures Treasuries against the alternative foreign sovereign assets the investor could hold rather than against its portfolio as a whole. Two considerations govern its construction. First, an investor's own government's debt is excluded from both legs: including domestic sovereign holdings would confound Treasury preferences with the home bias documented separately in the literature. Second, the denominator is measured wherever possible from the same resident-side source as the numerator, so that the two legs are consistent by construction. This holds for the euro area, where extra-euro-area sovereign holdings are observed directly by holder sector; for Canada, where both legs come from national sources; and for the United States, where domestic Treasury holdings are combined with reported foreign general-government holdings by sector. For Japan, the United Kingdom and China no resident-side sovereign portfolio is observed, and the denominator is imputed from the mirror side: bilateral portfolio positions against each destination are scaled by that destination's sovereign share of externally held debt, summed across destinations, and augmented by the investor's own reserve securities. The imputation follows the logic of Arslanalp and Tsuda (2014) and Fang et al. (2025), who allocate destination-side sovereign liabilities across holder types under a proportionality assumption. Appendix B.3 sets out the construction and its caveats.

3.2 Holdings of U.S. Treasuries

Figure 1 plots total holdings by country over the sample for foreign investors. Japan and China are the two largest holders at the start of the sample, together accounting for 43% of all foreign holdings in Q4 2013, and by its end for 20%. The reordering is driven from both ends: Chinese holdings are broadly flat between 2016 and 2021 and fall sharply thereafter as the People's

Bank of China draws down and diversifies reserves, while euro-area holdings grow by a factor of more than two and UK holdings grow markedly after 2018. The fourth and fifth largest holders, the United Kingdom and Canada, constitute only a small share of modelled foreign holdings throughout the sample. The inequality of the foreign holdings distribution justifies the exclusion of other countries from the explicitly modelled investors, as it would require extensive data collection for a long tail of small holders that are unlikely to materially affect equilibrium Treasury returns.

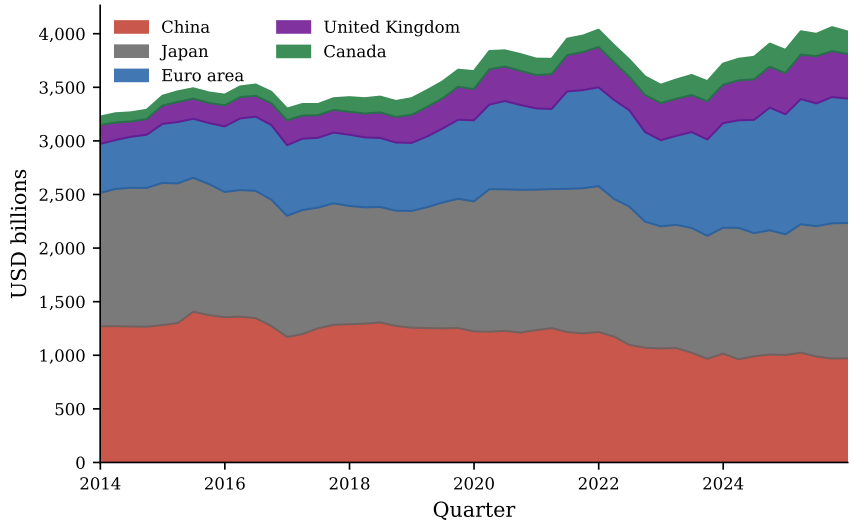


Figure 1. Holdings of U.S. Treasuries by country/region

Notes: Quarterly holdings of U.S. Treasury securities by the five modelled foreign investors (euro area, Japan, United Kingdom, China and Canada), stacked, in USD billions, Q4 2013–Q4 2025. Holdings are restated for the financial-centre distortions described in Section 3.1. *Sources:* Author's calculations based on U.S. Treasury (TIC); IMF (Coordinated Portfolio Investment Survey (CPIS), PIP, IRFCL); Board of Governors of the Federal Reserve System (H.10 foreign exchange rates release, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026).

The sector decomposition, plotted in Figure 2, shows substantial variation in sectoral holdings both across countries and over time within countries. Three patterns stand out.

First, official sector holdings, largely reflecting foreign reserves by central banks, have shifted substantially over time. Japan's official share fluctuates between 46% and 68% over the sample without trending. Chinese holdings are predominantly official throughout the window in which a sector decomposition can be constructed, although the official share declines over that window as private intermediaries take up a marginally larger part of a shrinking aggregate (Figure 2, panel (d)). Therefore, China's country aggregates used in the estimation reflect mostly the official sector. The decline in China's holdings persists even after accounting for custodial distortion, which is a major source of bias in TIC Treasury holdings attributed to China (Setser, 2026), suggesting an actual drop in demand contrary to other estimates (Setser, 2023). The euro area moves in the opposite direction: the official share falls from 37% in Q4 2013 to 21% in Q4 2025, so that the growth of euro-area Treasury holdings is almost entirely a private-sector phenomenon.

Second, non-bank financial institutions are the largest private holder in the euro area, Japan, the United Kingdom and Canada. Euro-area NBFI holdings exceed bank holdings in 47 of 49 quarters and rise in absolute value over time, consistent with the shift of European non-banks

away from euro-area government debt targeted by the ECB's Public Sector Purchase Programme and into foreign government debt reported in Bergant et al. (2020). Japan's NBFI share rises from 18% in Q4 2013 to 23% by Q4 2019, consistent with the role of life insurers, pension funds and investment funds in intermediating domestic savings into long-duration dollar fixed income (Bernanke, 2005). Canada is NBFI-dominated throughout, consistent with the depth of its pension and investment-fund sectors (Figure 2, panel (e)). The United Kingdom is likewise non-bank dominated, with an NBFI share of 56% in Q4 2013 and 54% at the end of the sample; non-banks lead in all 49 of 49 quarters.

Third, bank holdings grow in both the euro area and the United Kingdom: the euro-area bank share rises from 16% to 39%, and the UK share rises from 27% to 30%. Households and non-financial corporations, by contrast, are negligible everywhere abroad: their share never exceeds 8% in the euro area and stays below 2% in the United Kingdom.

The picture of a stagnating or declining demand by the official sector and burgeoning NBFI and bank holdings is consistent with the findings of foreign private investors absorbing a large fraction of Treasury issuance in Cordes and Ferris (2024), Fang et al. (2025), and Eren et al. (2026).

The domestic U.S. picture is different (Figure 2, panel (f)): non-bank financial institutions are by a wide margin the largest modelled domestic holder, but households and non-financial corporations hold a non-negligible and sharply rising amount of Treasuries, likely driven by the classification of hedge funds within this sector, including some offshore funds (Barth et al., 2025). All three sectors grow by a factor of 2.0–2.2 relative to their Q4 2019 level after 2020, showing that the post-pandemic expansion in Treasury supply was absorbed not only by the Federal Reserve but also to a substantial degree by private investors.

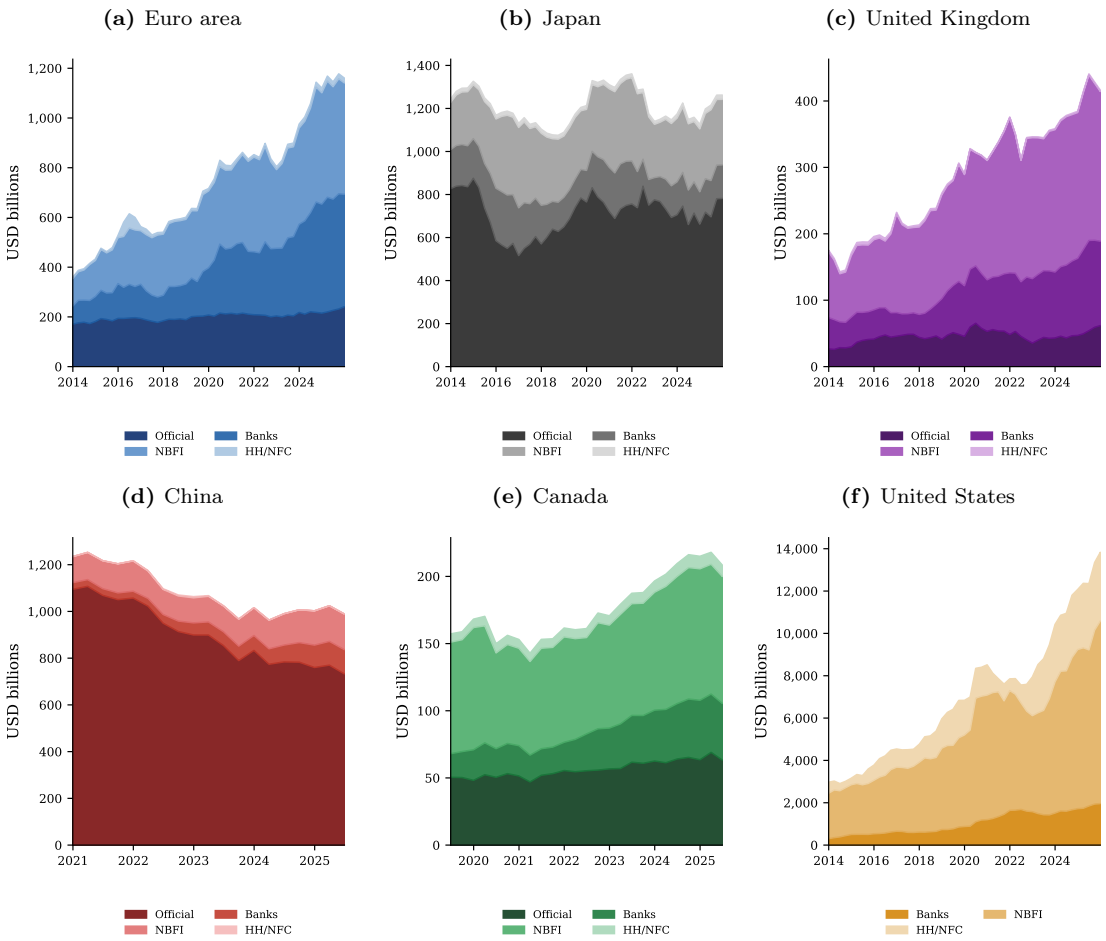


Figure 2. Holdings of U.S. Treasuries by investor sector

Notes: Quarterly holdings of U.S. Treasury securities by investor sector, stacked, in USD billions, Q4 2013–Q4 2025. Euro-area sector holdings are taken from SHS through Q1 2022 and split using PIP from Q2 2022 (Appendix B.2.1). Panels (d) and (e) are descriptive decompositions available over shorter windows. Panel (f) shows the three modelled U.S. private sectors; U.S. official holdings, including the Federal Reserve's System Open Market Account, are excluded. *Sources:* Author's calculations based on U.S. Treasury (TIC); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026).

The two dimensions of heterogeneity the model is designed to exploit are therefore visible in the raw data. Official investors dominate where Treasuries are held for reserve-management and geopolitical reasons; non-bank intermediaries dominate where long-duration liabilities and diversification motives prevail; and both the level and the dynamics of the sector mix differ sharply across countries holding similar aggregate amounts.

3.3 The estimation panel

Combining the foreign cells with the three modelled U.S. sectors gives 17 investor cells: four sectors each for the euro area, Japan and the United Kingdom, country aggregates for China and Canada, and three private sectors for the United States. Portfolio shares are available for every cell in every quarter from Q4 2013 to Q4 2025, or 833 cell-quarter observations. Summary statistics for the variables entering the estimation are reported alongside the estimation design in Section 4.3.

4 Empirical strategy

The model of Section 2 delivers a single first-order condition per cell that links the equilibrium portfolio share, the country-level excess return, and the structural Treasury preference parameters. I take this first-order condition directly to the data as a set of moment restrictions and estimate its parameters by GMM, cell by cell, across the 17 modelled investors (14 foreign and 3 domestic U.S.). Estimation exploits time-series variation in portfolio shares and the first and second moments of excess returns within each cell. Identification relies on a GIV strategy built from the demand residuals of investors resident in other countries and belonging to other sectors ("donor" cells), which corrects the simultaneity between portfolio shares and excess returns. The instrumental variable departs from standard GIV to account for the estimation of three structural parameters per investor, rather than a single elasticity parameter. I call this modified version of the instrument "Diffuse GIV", as it does not collapse residuals from donor cells into a single instrument. The estimation sample spans 49 quarters ending in Q4 2025.

4.1 Estimating equations

FOC moment conditions. Let $\varepsilon_{c,i,t}(\theta_{c,i})$ denote the residual of the first-order condition (4) evaluated at the data-observed portfolio share $s_{c,i,t}$, with $\theta_{c,i} = (\psi_{0,c,i}, \psi_{1,c,i}, \gamma_{c,i})$:

$$\varepsilon_{c,i,t}(\theta_{c,i}) \;\coloneqq\; E_{c,t} - \gamma_{c,i} V_{c,t}\, s_{c,i,t} - \gamma_{c,i} C_{c,t} + \psi_{0,c,i} + \frac{\psi_{1,c,i}}{s_{c,i,t}}. \tag{13}$$

The residual is nonlinear in the observables, through the reciprocal share and the product of variance and share, but linear in $\theta_{c,i}$. Rearranging (13) gives

$$E_{c,t} \;=\; -\psi_{0,c,i} + \gamma_{c,i}\, x_{c,i,t} - \psi_{1,c,i}\, \frac{1}{s_{c,i,t}} + \varepsilon_{c,i,t}(\theta_{c,i}), \qquad x_{c,i,t} \;\coloneqq\; V_{c,t}\, s_{c,i,t} + C_{c,t}, \tag{14}$$

a linear equation in the regressors $x_{c,i,t}$ and $1/s_{c,i,t}$ with the FOC residual as its error term. The moment conditions below are therefore linear in $\theta_{c,i}$, and each step of the GMM estimator reduces to a least-squares problem under the parameter bounds introduced below, which has an exact solution.

The country-specific excess return $E_{c,t} = \mathbb{E}_t[R_{US,t+1} - R^F_{c,t+1}]$ is sector-invariant within a country, since all sectors resident in c face the same non-U.S. foreign sovereign benchmark, and so

are the variances and covariances $V_{c,t}$ and $C_{c,t}$. Identification of the cell-specific parameters therefore rests on two distinct sources of variation: within-cell, time-series variation in $s_{c,i,t}$, which separates the level parameter from the curvature parameter because the two enter the residual with different powers of the share; and within-country, time-series variation in the first and second moments of excess return: $E_{c,t}$, $V_{c,t}$, and $C_{c,t}$. Exploiting time-series variation in the second moments, especially the covariance, is crucial for the success of the estimation strategy as it permits the separate identification of parameters $\gamma_{c,i}$ and $\psi_{0,c,i}$, which would otherwise be conflated in an intercept if $C_{c,t}$ were kept as a fixed parameter.

Inference. I apply the standard two-step efficient GMM estimator to the sample analogue of moment condition $\mathbb{E}[z_{c,i,t}\,\varepsilon_{c,i,t}(\theta_{c,i})] = 0$, where $z_{c,i,t}$ is the Diffuse GIV instrument defined in Section 4.2, cell by cell under the non-negativity restrictions $\psi_{1,c,i} \geq 0$ and $\gamma_{c,i} \geq 0$ imposed by the model to rule out convexity of the convenience function and risk-loving investors, respectively. The first step weights the moments by the identity matrix, the second by the inverse of the Newey–West (HAC) estimate of the long-run variance of the moments with four lags (Newey and West, 1987), computed from the first-step residuals. Because the moment conditions are linear in $\theta_{c,i}$, the problem simplifies to a standard bounded linear two-stage least-squares procedure.

Inference uses the standard GMM sandwich variance with Newey–West (HAC) standard errors with four lags (Newey and West, 1987), which account for both heteroskedasticity and autocorrelation. The choice of four lags reflects a trade-off between accommodating the serial correlation induced by the persistence of Treasury holdings and of expected excess returns, and remaining short relative to the length of the estimation sample. Reported confidence intervals are ninety-percent asymptotic Wald intervals. Because each cell is estimated on its own time series, no cross-cell clustering enters the reported standard errors.

4.2 Identification

Endogeneity. The primary identification challenge consists of the endogeneity of portfolio shares and excess returns in (4), which are jointly determined in equilibrium, an application of the classic problem of estimating demand equations. Suppose the demand of investor j depended not only on excess returns and Treasury preferences as posited by the model, but also on a demand shock $\varepsilon_{c,i,t}$, expressed in excess return units so we can rewrite (4) as

$$E_{c,t} - \gamma_{c,i} V_{c,t}\, s_{c,i,t} - \gamma_{c,i} C_{c,t} + \psi_{0,c,i} + \frac{\psi_{1,c,i}}{s_{c,i,t}} \;=\; \varepsilon_{c,i,t}. \tag{15}$$

A positive shock raises the excess return the investor requires to hold a given share, and hence lowers its Treasury demand. Comparing (15) with (13), the FOC residual evaluated at the true parameters $\theta^0_{c,i}$ is the demand shock itself, $\varepsilon_{c,i,t}(\theta^0_{c,i}) = \varepsilon_{c,i,t}$.

These shocks include pure idiosyncratic demand shocks and investor-specific reactions to any common driver of demand. One prominent example would be a change in Treasury supply, which is assumed fixed in the model. Realisations of $\varepsilon_{c,i,t}$ move $s_{c,i,t}$ and $E_{c,t}$ jointly in equilibrium, so the share is correlated with the shock, $\mathbb{E}[s_{c,i,t}\,\varepsilon_{c,i,t}] \neq 0$, and so, generically, are the share-dependent regressors of (14): $\mathbb{E}[x_{c,i,t}\,\varepsilon_{c,i,t}] \neq 0$ and $\mathbb{E}[\varepsilon_{c,i,t}/s_{c,i,t}] \neq 0$. Therefore, identification requires an instrument $z_{c,i,t}$ that moves these regressors independently of investor j's demand shock.

Diffuse Granular Instrumental Variables. To overcome this challenge, I use granular instrumental variables in the sense of Gabaix and Koijen (2024). The intuition of this approach rests on exploiting idiosyncratic demand shocks from other investors ("donor" cells) as instruments for an investor's demand elasticity. These shocks are analogous to supply shifters, as they affect equilibrium returns by changing the quantity available to other investors, and are uncorrelated with demand shocks from the estimand investor. The canonical GIV instrument is the difference between a size-weighted and an equally weighted average of the other investors' demand residuals. The equally weighted leg contributes to *exogeneity*, along with residualisation: it purges any common factor with equal loading, crucially including the endogenous equilibrium price itself, to which every investor's observed quantity responds. The aggregation into a size-weighted average targets *relevance*: it tilts the combination towards the cells whose idiosyncratic shocks are large enough relative to the market to move that price.

My setting requires estimating the three structural parameters governing demand elasticity in the theoretical model, rather than the elasticity alone as common in GIV applications. Therefore, collapsing donor demand residuals into a single instrument does not provide enough moment conditions to identify all parameters. Portfolio shares from donor cells therefore enter the moment condition *jointly*, without size weighting or aggregation, after undergoing a series of steps to isolate their idiosyncratic component. I dub this instrument "Diffuse GIV", as it keeps information embedded in each donor's residual separate and distinct. Its main advantage is the ability to estimate multiple parameters while satisfying the same exogeneity conditions as canonical GIV, at the cost of giving up the relevance gains from efficient aggregation. An additional benefit is the possibility to test exclusion restrictions, as using multiple moment conditions from donor shares leaves the model over-identified.

Exclusion restriction. Valid instruments must satisfy the exclusion restriction $\mathbb{E}[z_{c,i,t}\,\varepsilon_{c,i,t}] = 0$, which is the moment condition of Section 4.1 evaluated at the true parameters, implying that they are uncorrelated with the estimand investor's idiosyncratic demand shock. Other investors' raw portfolio shares by themselves do not satisfy this condition, as they all respond endogenously to equilibrium excess returns, which in turn respond to the estimand investor's demand shock. Furthermore, they may be exposed to common shocks such as changes in Treasury supply and global risk-off events.

Therefore, I apply three residualisation steps adapted from the canonical Gabaix and Koijen (2024) GIV formulation to extract the idiosyncratic demand component $\hat{\eta}_{l,t}$, in portfolio-share units, for all investors l in the donor set for investor j, as defined below.

The first step consists of removing investors from the same country and the same sector, to remove common exposure to country- and sector-specific demand shocks. These include global regulatory changes such as the introduction of Basel III macroprudential requirements for the banking sector, or a recession that lowers asset demand across all sectors in a given country. As a result, the instrument set is cell-specific, because the number of cells surviving the own-country and own-sector exclusions varies across cells. The donor set for cell $j = \{c, i\}$ is defined as $\mathcal{L} = \{(c', i') : c' \neq c,\ i' \neq i\}$ with $|\mathcal{L}|$ elements, ranging from 10 to 16 cells across investors.[5]

The second step consists of extracting a residual from a regression of donor cell portfolio shares on a set of likely common factors. Cell-by-cell within each donor set, I run regression

$$s_{l,t} = \alpha_l + X_{l,t}\boldsymbol{\beta}_l + Y_t\boldsymbol{\delta}_l + u_{l,t} \tag{16}$$

[5]Canada and China are not affected by this step, as they enter the estimation as country aggregates.

where $X_{l,t}$ is a set of cell-specific variables and Y_t a set of common variables, and extract estimated residuals $\hat{u}_{l,t}$.

This step is particularly important because the heterogeneous demand elasticities implied by my model introduce correlation between demand residuals through differential exposure to excess returns. Therefore, it is necessary to residualise portfolio shares with respect to drivers of equilibrium returns, but not to the returns themselves, through a method that allows for investor-specific coefficients, as regression in (16) does.

The choice of residualising variables strikes a balance between removing endogeneity and avoiding saturation in time-series regressions with 49 observations. I include four variables: (i) the quarterly relative return of U.S. Treasuries with respect to the reference leg of the portfolio; (ii) the Federal Funds Rate; (iii) the quarterly growth of U.S. Treasury marketable debt at face value; and (iv) the CBOE VIX index.

Variable (i) accounts for valuation effects that mechanically induce correlation in demand residuals across investors, including through exchange rate fluctuations between the dollar and currencies in the reference basket. Importantly, this variable also captures Treasury demand shocks stemming directly from movements of each investor's domestic currency *vis à vis* the dollar, as the domestic sovereign debt of investor i features in the reference basket of all its donor cells.

Variable (ii) captures the main source of variation in short-term Treasury returns exogenous to the portfolio choice of individual investor sectors, hence controlling for heterogeneous reactions of demand to expected excess returns at the short end of the yield curve.

Variable (iii) likewise summarises one of the leading drivers of both short- and long-term Treasury returns, as well as controlling for differential reactions to changes in Treasury supply, a major threat to identification as the theoretical model assumes a fixed supply.

Variable (iv) allows for heterogeneous responses to global shifts in risk aversion, one of the main drivers of Treasury demand during flight-to-safety episodes.

The third and final step consists of cross-sectionally de-meaning residuals from the second step to account for any remaining common exposure of investor demand to underlying unobserved factors. Importantly, this step purges the instruments from common exposure to equilibrium excess returns, the main source of endogeneity. The de-meaned residuals are computed as

$$\hat{\eta}_{l,t} = \hat{u}_{l,t} - \frac{1}{|\mathcal{L}|} \sum_{l' \in \mathcal{L}} \hat{u}_{l',t}. \tag{17}$$

Relevance. Conceptually, relevance of a granular instrumental variable rests on the ability of donor demand residuals to affect the estimand investor's demand through equilibrium prices. In my setting, this notion can be operationalised through the linear form (14): at the true parameters its error term is the demand shock $\varepsilon_{c,i,t}$, so both regressors $x_{c,i,t}$ and $1/s_{c,i,t}$ are endogenous. Therefore, I test instrument strength by running first-stage OLS regressions of $x_{c,i,t}$ and $1/s_{c,i,t}$ on the Diffuse GIV instruments.

Instrument construction. The instrument vector for cell j is $z_{j,t} = (1, \hat{\boldsymbol{\eta}}_{l,t}^{\top})^{\top}$, which stacks a constant with a block $\hat{\boldsymbol{\eta}}_{l,t}$ of residuals from cell j's donors built according to the procedure above.[6] Intuitively, the unconditional moment associated with the constant identifies the level

[6] The de-meaned residuals in (17) sum to zero across donors at every date, so the block of $|\mathcal{L}|$ columns has rank at most $|\mathcal{L}| - 1$ and would make the covariance matrix of the moment conditions singular. I therefore replace it

parameter $\psi_{0,j}$ through the mean of the FOC residual, while orthogonality of the FOC residual to the GIV block uses variation in portfolio shares and the first and second moments of excess returns to identify the curvature parameter $\psi_{1,j}$ and the risk-aversion parameter γ_j.

4.3 Estimation data

The estimation requires four inputs: portfolio shares of U.S. Treasuries, country-level expected excess returns, conditional return variances and covariances, and the observable controls used in the second step of the instrument construction. The construction of each is summarised below; full sources are listed in Appendix B.5.

Portfolio shares. The portfolio share is the weight of U.S. Treasuries in the cell's portfolio of sovereign debt issued outside its own jurisdiction,

$$s_{c,i,t} \;=\; \frac{H^{UST}_{c,i,t}}{W_{c,i,t}} \;\in [0,1].$$

The numerator $H^{UST}_{c,i,t}$ is the U.S.-dollar value of cell (c,i)'s Treasury holdings, constructed as described in Section 3. The denominator $W_{c,i,t}$ is the cell's total holdings of sovereign debt issued outside its own jurisdiction, built from the same resident-side source as the numerator wherever available and by mirror imputation otherwise, as described in Section 3.1 and Appendix B.3. The investor's own sovereign is excluded from both legs, so that $s_{c,i,t}$ measures the allocation between U.S. Treasuries and other foreign sovereign debt, so it is free from the home bias that affects the split between domestic and foreign assets. The denominator $W_{c,i,t}$ is also the wealth scale of cell (c,i) in the global market-clearing condition (9). Figure A5 in Appendix C plots the share cell by cell over the full holdings window to visualise the time-series variation in individual cells' portfolio shares used in the estimation.

Expected excess returns. Country excess return $E_{c,t}$ is the expected return on U.S. Treasuries in excess of the expected return on a reference portfolio of non-U.S. foreign sovereign bonds, measured over a one-year holding period and expressed at an annual rate. The reference portfolio for investor c is the FTSE World Government Bond Index universe excluding the United States and excluding c's own sovereign, with the basket re-determined at the start of each quarter as index membership changes. U.S. Treasury returns are computed from the U.S. component of the WGBI for consistency. Index members are weighted by their index market value in U.S. dollars at the start of the quarter,

$$w^{F}_{c,k,t} \;=\; \frac{MV^{USD}_{k,t}}{\sum_{k' \in \mathcal{B}^c_t} MV^{USD}_{k',t}},$$

where $\mathcal{B}^c_t$ is investor c's eligible basket. The choice of a generic bond index reflects the inability to observe foreign debt holdings at a destination-country definition for most investor countries in the sample. The implicit assumption is that the relative country allocation within the reference portfolio follows the weights of the WGBI basket, which is consistent with the approach of modelling only the choice between U.S. Treasuries and other foreign sovereign debt in the aggregate. Each basket member's return is its yield to maturity at the start of the window, so that expected

by its principal components with non-zero singular values, which span the same space with $|\mathcal{L}| - 1$ columns; the discarded direction is identically zero and carries no identifying variation. The instrument block therefore runs from 9 to 15 columns, so each cell uses between 10 and 16 moment conditions.

capital gains are zero under the martingale approximation standard in the bond portfolio-choice literature (Koijen et al., 2021). Sovereign credit risk is netted out of both legs using the five-year sovereign credit-default-swap spread of the relevant issuer, deducted in U.S. dollars before any currency translation. The excess return is then

$$E_{c,t} \;=\; R^{US}_{c,t} \;-\; R^{F}_{c,t},$$

with both legs scaled to an annual basis. Both legs are measured in U.S. dollars, so no expected currency appreciation enters either leg, consistent with a random-walk process for the exchange rate (Fama, 1984). The martingale assumption for both own-currency capital gains and valuation effects is motivated by the lack of data on investors' subjective expectations at the country-sector level.

The United States is treated symmetrically to the five foreign investor countries: its reference basket is the same index universe excluding the United States, and its Treasury leg simply carries no currency translation. Figure A2 in Appendix C plots the baseline series by investor country over the estimation sample, showing that expected returns display substantial time-series variation but correlate highly across countries, as they vary only in the composition of the reference basket.

Variance and covariance. The conditional variance and covariance enter the first-order condition through the risk terms $\gamma_{c,i}V_{c,t}s_{c,i,t}$ and $\gamma_{c,i}C_{c,t}$. Both are built from rolling twenty-quarter windows of *realised* annual total returns, using the realised total-return counterpart of the excess-return construction above:

$$V_{c,t} \;=\; \frac{1}{M-1}\sum_{\tau=P-1}^{P+M-2}\left(r^{\text{ex}}_{c,t-\tau} - \bar{r}^{\text{ex}}_{c,t}\right)^2, \qquad C_{c,t} \;=\; \frac{1}{M-1}\sum_{\tau=P-1}^{P+M-2}\left(r^{\text{ex}}_{c,t-\tau} - \bar{r}^{\text{ex}}_{c,t}\right)\left(r^{F}_{c,t-\tau} - \bar{r}^{F}_{c,t}\right),$$

with $M = 20$ quarters the window length, P the length in quarters of the one-year holding period, $r^{\text{ex}}_{c,q}$ the realised annual excess return of Treasuries over the reference basket, $r^{F}_{c,q}$ the realised annual return on the basket itself, and bars denoting the means over the same M returns. Each realised return is dated by the first quarter q of its holding period, which runs from the end of quarter $q-1$ to the end of quarter $q+P-1$, so the most recent return in the window, dated $t-P+1$, is completed at the end of quarter t, and $V_{c,t}$ and $C_{c,t}$ use only returns completed by the end of t. Unlike the expected excess return, the realised returns include capital gains, and both legs, net of the same credit-default-swap spreads, are translated into investor c's currency, so that they reflect movements of both the dollar and the basket's component currencies against that currency. The window is long enough to deliver stable second moments and short enough to retain the time-series variation the estimation exploits. Because a full window must be available in the first estimation quarter, realised returns are constructed back to well before the start of the holdings sample. Figure A3 and Figure A4 in Appendix C plot the two series by investor country over the estimation sample.

Residualisation controls. The second step of the instrument construction projects each donor cell's portfolio share on a constant and the controls reported in Panel C of Table 1: the quarterly realised return of U.S. Treasuries relative to the members of the reference government bond basket, the VIX equity-market implied-volatility index, the quarterly growth rate of marketable Treasury debt, and the quarterly change in the Effective Federal Funds Rate.

Only the first varies across cells within a quarter, due to the different composition of donor cells; the other three are common to all cells at a given date, so they contribute cross-cell variation only through the heterogeneous coefficients each donor's own regression assigns them.

Sources for all variables above are listed in Appendix B.5.

Summary statistics. Table 1 reports, on the estimation sample, summary statistics for every object that enters the estimated first-order condition and the Diffuse GIV instrument construction procedure. Three features are of particular relevance for the estimation. First, the level of the portfolio share in Panel A differs sharply across countries: the euro area, the United Kingdom and China each allocate a little over a third of their foreign sovereign portfolio to Treasuries, Japan and Canada around two thirds, and the U.S. sectors four fifths. The curvature term $\psi_{1,c,i}/s_{c,i,t}$ is therefore evaluated in very different regions across cells, and a given curvature parameter translates into a much larger contribution to the premium for a low-share cell than for a high-share one. Second, the within-country time-series dispersion of the share, which together with the instrumented return is what identifies the parameters, is itself heterogeneous: it is an order of magnitude larger for the United Kingdom than for Canada, so the cells are not equally informative and the precision of the estimates should be expected to vary accordingly. Third, Panel B shows that the covariance $C_{c,t}$ between the Treasury excess return and the reference-basket return is negative on average for every one of the six countries. Through the $-\gamma_{c,i}C_{c,t}$ term in (13), a negative covariance raises the optimal Treasury share for any level of the excess return, because Treasuries hedge the investor's foreign sovereign benchmark, mainly through dollar appreciation when foreign sovereign bond yields are low. Therefore, the realised covariance captures the safe-haven properties of Treasuries and the dollar (Maggiori, 2017, Gourinchas and Rey, 2022), to the extent that low government bond yields reflect an economic downturn, for example through the central bank lowering the policy rate in response to a recession. This property is then reflected in a negative risk premium, distinct from the model's notion of Treasury premium, which instead encompasses motives for holding Treasuries *beyond* their risk-return profile.

Table 1. Summary statistics: estimation sample

	N	Mean	SD	p10	Median	p90
Panel A: Foreign portfolio share in U.S. Treasuries (%)						
Euro area	49	36.8	3.9	32.5	36.7	41.5
Japan	49	61.1	4.9	55.5	60.2	68.4
United Kingdom	49	36.1	11.3	20.0	38.6	48.3
Canada	49	67.2	2.3	64.4	67.4	70.2
China	49	36.4	4.4	30.3	37.6	41.3
United States	49	80.1	6.6	71.4	80.9	89.2
Panel B: Return moments by country						
Excess return, $E_{c,t}$ (% p.a.)						
Euro area	49	0.72	0.57	0.17	0.50	1.57
Japan	49	0.64	0.65	−0.15	0.61	1.58
United Kingdom	49	0.93	0.61	0.29	0.80	1.84
Canada	49	0.88	0.62	0.19	0.73	1.81
China	49	0.86	0.61	0.18	0.72	1.79
United States	49	0.86	0.61	0.18	0.72	1.79
Variance, $V_{c,t}$ ($\times 10^4$)						
Euro area	49	41.8	12.9	24.1	43.1	57.0
Japan	49	74.7	12.1	56.4	77.6	88.2
United Kingdom	49	49.4	10.1	37.0	48.5	66.2
Canada	49	48.2	10.6	34.9	47.6	64.5
China	49	43.6	7.9	33.7	43.7	53.9
United States	49	43.1	7.6	33.6	43.0	54.9
Covariance, $C_{c,t}$ ($\times 10^4$)						
Euro area	49	−7.7	23.6	−46.6	−9.6	21.9
Japan	49	−54.8	19.6	−79.5	−58.3	−29.1
United Kingdom	49	−13.7	13.2	−31.4	−12.1	3.9
Canada	49	−12.1	15.4	−35.7	−5.8	1.7
China	49	−27.6	15.5	−40.2	−31.7	−0.8
United States	49	−40.3	14.7	−59.7	−38.0	−20.6
Panel C: Residualisation controls						
Relative return, UST vs. reference basket (% p.a.)	294	3.29	15.52	−19.10	3.17	24.35
VIX (index)	49	17.91	5.30	12.83	16.47	25.42
Marketable Treasury debt growth, face value (%)	49	2.00	2.28	0.25	1.91	3.11
Change in effective federal funds rate (pp)	49	0.08	0.42	−0.33	0.01	0.33

Notes: Quarterly data, Q4 2013–Q4 2025 (49 quarters), the estimation sample on which the residualisation controls, the portfolio share, the excess return, the variance and the covariance are jointly observed. Panel A: portfolio share, aggregated to the country level as the ratio of summed Treasury holdings to summed holdings of sovereign debt issued outside the investor's own jurisdiction; the United States row aggregates its three modelled private sectors, and China and Canada are single total-economy cells. Panel B: excess return $E_{c,t}$ at an annual rate, with both legs in U.S. dollars; variance $V_{c,t}$ and covariance $C_{c,t}$ of realised annual total returns. Panel C: the residualisation controls of the Diffuse GIV instrument; the relative return is summarised over the distribution of donor sets. *Sources:* Author's calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 foreign exchange rates and H.15 selected interest rates releases, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

5 Estimated parameters

This section reports the GMM estimates of the parameter vector $\theta_{c,i} = (\psi_{0,c,i}, \psi_{1,c,i}, \gamma_{c,i})$ for the 17 modelled cells: the 14 foreign cells spanning the euro area, Japan, and the United Kingdom by sector, together with the China and Canada country aggregates, and the 3 domestic U.S. sector cells. Two bounds are maintained throughout: the concavity restriction $\psi_{1,c,i} \geq 0$ of Section 2.1 and non-negative risk aversion $\gamma_{c,i} \geq 0$. Therefore, all estimates constrained to zero by these bounds are reported without any confidence interval.

Figure 3 displays the convenience parameters as a dot-and-whisker chart that arranges the level parameter $\psi_{0,c,i}$ and the curvature parameter $\psi_{1,c,i}$ side by side across cells, and Figure 4 displays the estimated risk aversion on the same cell ordering. Table A3 in Appendix D.1 collects all three parameters with their 90% confidence intervals by country and sector.

Table A4 in Appendix D.2 reports estimation diagnostics. Instruments are strong on average, with a median Olea and Pflueger (2013) effective F-statistic of 16.79 for $x_{c,i,t}$ and of 22.85 for $1/s_{c,i,t}$. However, 7 cells have an effective F-statistic below the rule-of-thumb threshold of 10 for at least one of the two variables, and 1 for both. Appendix F.2 shows that performing the experiments of Section 6 without the cells affected by weak instruments leaves the sign and size of the model-implied Treasury premium, and China's position as its largest driver, unchanged. Table A4 also reports Hansen (1982) overidentification tests, which fail to reject the overidentifying restrictions in all cells at any conventional significance level.

Overall, parameter estimates are precise for a sample of only 49 time-series observations per cell. 16 out of 25 Treasury preference parameters not constrained to zero are statistically significant at the 90% level, as are all 11 non-constrained risk aversion parameters. Nevertheless, Appendix F.1 reports the results of the Section 6 experiments setting all parameters that are not significant at the 90% level to zero, showing that the same results survive, including the higher sensitivity of the equilibrium return to supply in the presence of Treasury preferences.

5.1 Risk aversion

Estimates of the risk aversion parameter $\gamma_{c,i}$ away from the zero bound range from 1.41 to 4.56, with 6 out of 17 cells bounded to zero.

The range of $\gamma_{c,i}$ sits comfortably within the boundaries of the literature on the estimation of risk aversion from portfolio choices. The seminal paper Friend and Blume (1975) finds a relative risk aversion coefficient of around 2, while more recent panel-data evidence from Chiappori and Paiella (2011) reports an average coefficient of around 4. However, these studies all draw their estimates from multi-asset portfolios and, to my knowledge, no comparable benchmark for risk aversion exists in the context of sovereign portfolio choice.

Parameters are quite concentrated for foreign investors, with no large differences either across countries or across sectors. On the other hand, U.S. non-bank investors display a higher risk aversion than any of the foreign investors. The higher risk aversion of U.S. investors is at odds with explanations of the exorbitant privilege that rely on higher U.S. risk tolerance (Gourinchas and Rey, 2022), and points instead to the financial-development channel of Maggiori (2017), in which the asymmetry arises from intermediary balance sheets rather than preferences.

The risk aversion estimates for all British sectors, China, and U.S. banks are bounded to zero, so their portfolio choice problem collapses to the risk-neutral limit described in Appendix A.4: their first-order condition reduces to the convenience block alone, and in all of these cells except

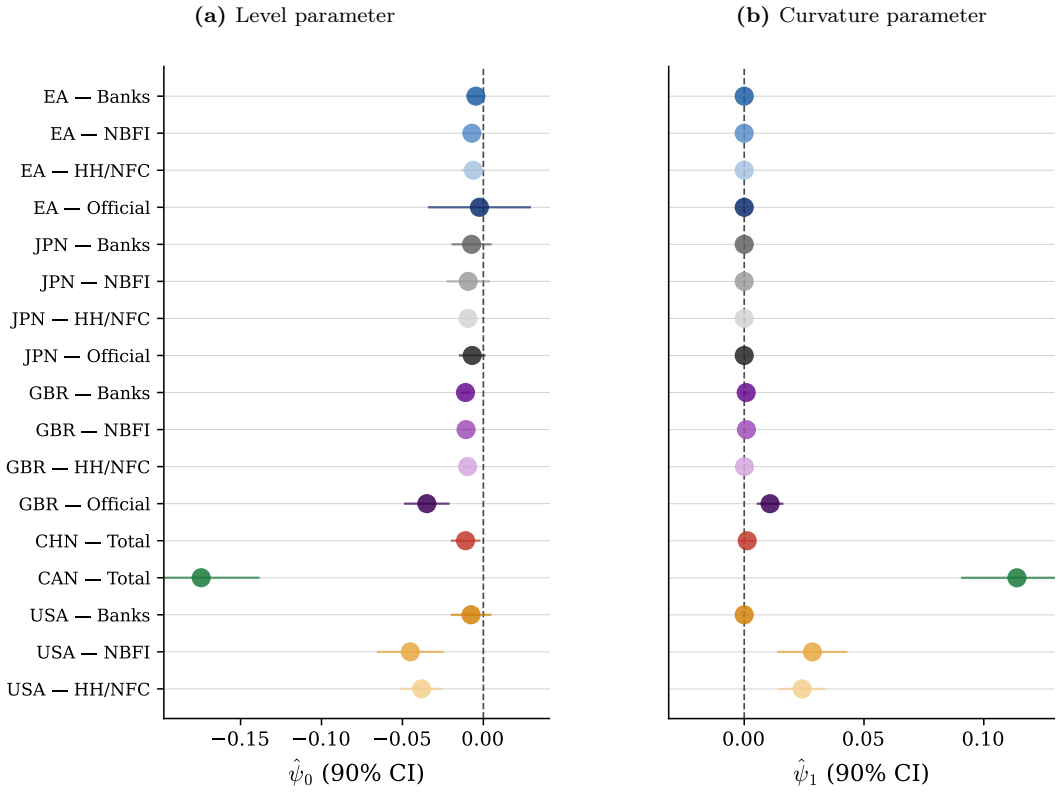


Figure 3. Estimated convenience parameters by country and sector

Notes: Point estimates of the level parameter $\hat{\psi}_{0,c,i}$ and the curvature parameter $\hat{\psi}_{1,c,i}$ with 90% confidence intervals for the 17 cells, estimated cell by cell by two-step GMM, Q4 2013–Q4 2025. Confidence intervals are based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987). Estimates at the bound $\psi_{1,c,i} = 0$ are plotted at zero without a confidence interval. *Sources:* Author's estimates based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

U.S. banks the whole allocation between Treasuries and the non-U.S. sovereign benchmark is explained by $\psi_{0,c,i}$ and $\psi_{1,c,i}$. For U.S. banks, whose $\psi_{1,c,i}$ is also zero, the first-order condition contains no share term and does not pin down the share (case (iii) of Appendix A.4). This result does not imply that these investors are risk-neutral in a general sense, but merely that they behave *as if* risk-neutral within the narrow confines of a sovereign portfolio choice problem with Treasury preferences.

Overall, the estimated risk aversion parameters should be interpreted jointly with, and conditional on, the Treasury preference parameters: a smaller $\gamma_{c,i}$ assigns less of the observed Treasury portfolio share to the hedging and variance terms, leaving more of it to be accounted for by convenience.

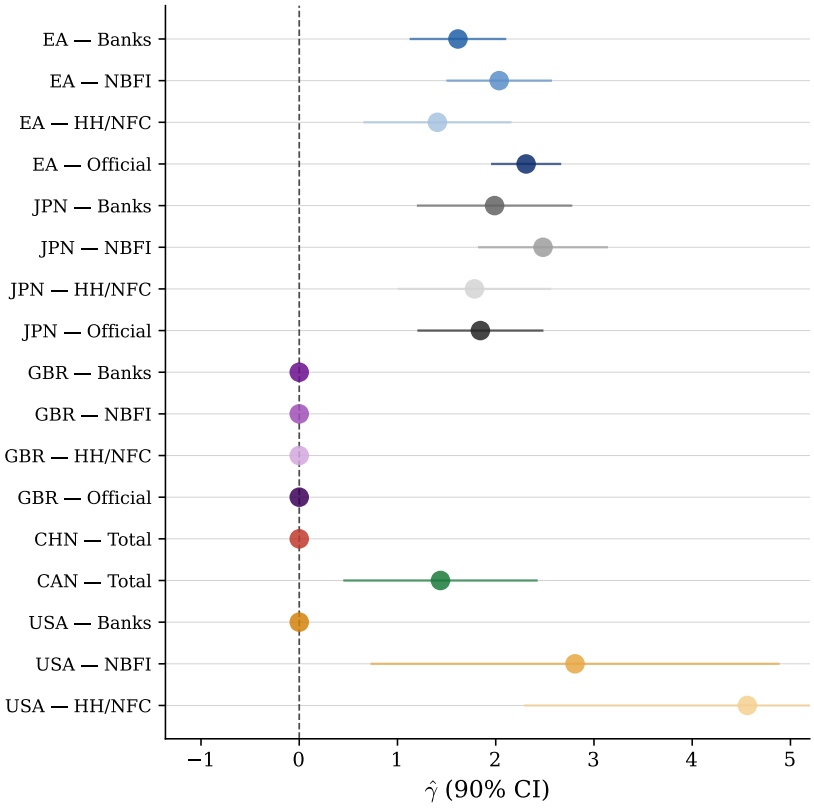


Figure 4. Estimated risk aversion by country and sector

Notes: Point estimates of risk aversion $\hat{\gamma}_{c,i}$ with 90% confidence intervals for the 17 cells, estimated cell by cell by two-step GMM, Q4 2013–Q4 2025. Confidence intervals are based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987). Estimates at the bound $\gamma_{c,i} = 0$ are plotted at zero without a confidence interval. *Sources:* Author's estimates based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

5.2 Treasury preference parameters

Level parameter. The level parameter $\psi_{0,c,i}$ is negative in every cell, ranging from -0.1743 for the Canadian aggregate to -0.0024 for the euro-area official sector (Figure 3 and Table A3); taken alone, the level term raises each cell's required excess return by $-\psi_{0,c,i}$, whatever its portfolio share.

Curvature parameter. The curvature parameter $\psi_{1,c,i}$ rests at the concavity bound $\psi_{1,c,i} = 0$ in every euro-area and every Japanese cell and for U.S. banks, that is in 9 of the 17 cells and in 8 of the 14 foreign cells. Across the 8 cells off the bound, it ranges from 0.0001 for the British households and non-financial corporations to 0.1138 for the Canadian aggregate (Figure 3 and Table A3).

Because $\psi_{1,c,i}$ is the only preference parameter that enters the slope of Treasury demand (Proposition 2), in the cells at the zero bound Treasury preferences shift demand without affecting its

elasticity at interior solutions. However, they can still affect elasticities in equilibrium as $\psi_{1,c,i}$ determines how often each cell's optimal portfolio share sits at a corner solution.

Treasury preferences and required excess returns. Before turning to equilibrium, it is useful to gauge how much the estimated level and curvature preference parameters move the Treasury excess return that each investor, taken in isolation, requires. Wherever both parameters are away from zero, the two channels work against each other, and a cell's marginal convenience reflects the balance between them at its portfolio share. Rearranging the first-order condition (4) gives $E_{c,t} = \gamma_{c,i}(V_{c,t}s_{c,i,t} + C_{c,t}) - (\psi_{0,c,i} + \psi_{1,c,i}/s_{c,i,t})$. I evaluate the preference term $\hat{\psi}_{0,c,i} + \hat{\psi}_{1,c,i}/\bar{s}_{c,i}$ at each cell's time-series average observed Treasury share $\bar{s}_{c,i}$ over its estimation sample, holding the risk terms fixed and using point estimates, including $\hat{\psi}_{1,c,i} = 0$ where the bound binds. A positive term lowers the cell's required excess return and a negative term raises it. Two caveats apply to this simple exercise. The calculation is partial-equilibrium and moves one investor at a time, so it is not the investor's contribution to the equilibrium Treasury premium, which is the object of the counterfactual experiments in Section 6.4. Furthermore, it is evaluated at the observed $1/\bar{s}_{c,i}$, which differs from the average of the equilibrium values of $1/s_{c,i,t}$.

The term is negative in all cells: at their average holdings, Treasury preferences raise every investor's required Treasury excess return, by between 24 basis points per annum for the euro-area official sector and 98 basis points per annum for the U.S. households and non-financial corporations, with a median of 76 basis points per annum (Figure 5). The increase ranges from 24 to 95 basis points per annum across foreign cells and from 76 to 98 across the U.S. sectors. The sign is due to $\hat{\psi}_{0,c,i} < 0$: where the curvature term is positive, it offsets only part of the level term. Canada illustrates the offset: its level and curvature estimates are the largest in the panel in absolute value, yet they net to a preference term of -50 basis points per annum, a smaller increase in the required return than in the median cell. The British official sector and the two U.S. non-bank sectors show the same offset on a smaller scale, with point estimates of the total effect that are small differences between much larger level and curvature components; in the other interior cells, curvature offsets a smaller part of the level term. At the shares they actually hold, investors require compensation beyond risk to hold Treasuries rather than the non-U.S. sovereign benchmark, a sign of "Treasury inconvenience".

The positive effect of Treasury preferences on the required return of foreign investors echoes the post-crisis disappearance of the Treasury premium documented by Du et al. (2018a), who find that the yield advantage of Treasuries over currency-hedged G10 sovereigns turned negative at medium and long maturities after 2015; and its subsequent fall into deeper negative territory reported in Du et al. (2026) and Fleckenstein and Longstaff (2024). It is also consistent with the security-level evidence in Tabova and Warnock (2026) that foreign holders, especially the official sector, do not accept below-market returns on Treasuries. Two mechanisms rationalise the sign. For private foreign investors, the persistently negative cross-currency basis since 2014 raised the cost of hedging dollar exposure, so that hedged Treasury returns no longer exceeded those on home-currency sovereigns (Du et al., 2018b, Du and Huber, 2026). For the foreign official sector, the steady diversification of reserves away from the dollar is partly attributable to geopolitical considerations, with sanctions risk lowering the convenience of dollar assets for non-aligned reserve managers (Eichengreen et al., 2019, Arslanalp et al., 2023, Bianchi and Sosa-Padilla, 2025).

Domestic investors display Treasury inconvenience too: for every domestic sector, Treasury preferences raise rather than lower the required return. Although surprising at first sight, the sign matches post-crisis evidence on negative swap spreads, dealer balance sheet costs and the cash-

futures basis, all of which imply that the marginal domestic holder of a cash Treasury requires compensation rather than paying a convenience yield. For non-bank financial institutions, pension funds and insurers have shifted to duration hedging through swaps, which removes the incentive to hold Treasuries outright irrespective of their risk-return profile (Klingler and Sundaresan, 2019, Jermann, 2020), while asset managers obtain duration synthetically through futures rather than cash bonds (Barth et al., 2024), and security brokers and dealers, which the Financial Accounts classify among non-bank financial institutions, became net long holders that require compensation for warehousing Treasuries under the leverage ratio (Du et al., 2023). For banks, the Basel III leverage ratio made balance sheet space costly even for safe, liquid assets (Duffie, 2018, Boyarchenko et al., 2018). The estimates for U.S. banks, which in the Financial Accounts comprise private depository institutions and exclude security brokers and dealers, pair zero risk aversion and zero curvature with a negative level parameter alone, consistent with a risk-neutral intermediary that pays a roughly constant balance-sheet cost per dollar of Treasuries, as a leverage ratio levied per dollar of assets implies at a given shadow price; the level estimate is imprecise, however, and a shadow cost that rises with balance-sheet use would make banks' demand less elastic than the model implies. For the household sector, which in the Financial Accounts includes domestic hedge funds as a residual, the expansion of the cash-futures basis trade implies that levered funds hold cash Treasuries only when these trade cheap relative to futures (Barth and Kahn, 2025).

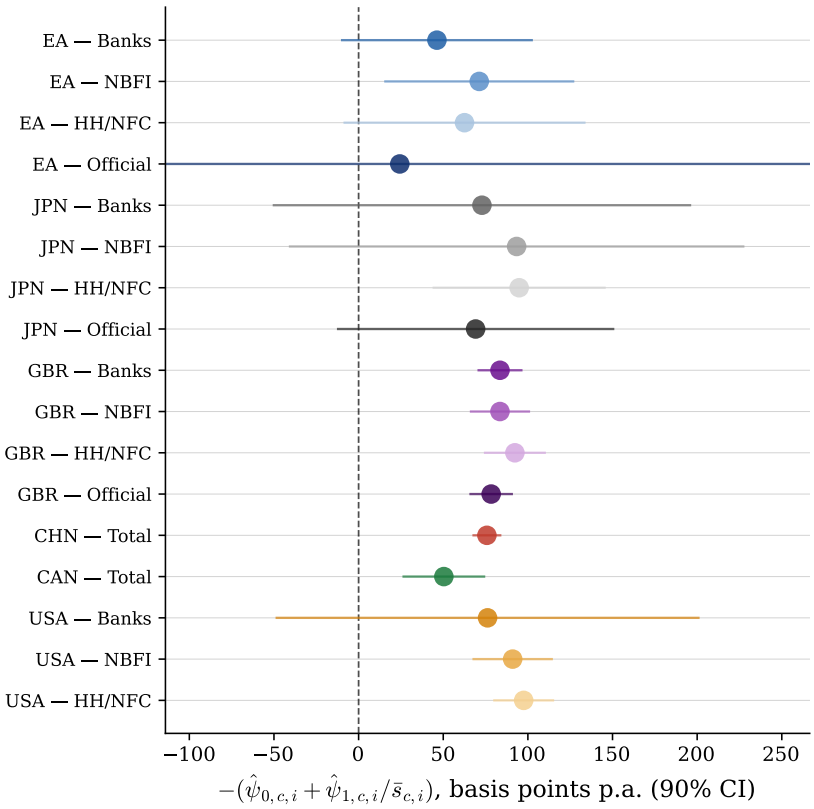


Figure 5. Effect of estimated Treasury preferences on required Treasury excess returns

Notes: Point estimates of $-(\hat{\psi}_{0,c,i} + \hat{\psi}_{1,c,i}/\bar{s}_{c,i})$, in basis points per annum, with 90% confidence intervals. The term is evaluated at the cell's time-series average observed Treasury share $\bar{s}_{c,i}$ over the estimation sample, Q4 2013–Q4 2025, holding the risk terms of the first-order condition (4) fixed. Confidence intervals are computed by the delta method from the GMM covariance of $(\hat{\psi}_{0,c,i}, \hat{\psi}_{1,c,i})$, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987), treating $\bar{s}_{c,i}$ as fixed; where $\hat{\psi}_{1,c,i}$ is at the bound $\psi_{1,c,i} = 0$, the interval reflects the uncertainty in $\hat{\psi}_{0,c,i}$ only. *Sources:* Author's estimates based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

6 Model experiments

I use the estimated model to quantify how Treasury preferences shape investors' portfolios and the equilibrium Treasury return, and to attribute the model-implied Treasury premium to individual countries and sectors. Every experiment compares the model at the estimated preference parameters $(\hat{\psi}_{0,c,i}, \hat{\psi}_{1,c,i})$ with a counterfactual in which both parameters are set to zero, $\psi_{0,c,i} = \psi_{1,c,i} = 0$, for a single cell, a group of cells or all cells at once. In the counterfactual, investors value Treasuries only for their risk-return profile. Everything else is held fixed: estimated risk aversion $\hat{\gamma}_{c,i}$, the return moments $V_{c,t}$ and $C_{c,t}$, portfolio wealth, Treasury supply S_t and the holdings of the inelastic residual investor H_t^{Res}. Each experiment is computed quarter by quarter over the estimation sample, Q4 2013–Q4 2025, and then averaged over quarters.

The experiments are of two kinds. The portfolio experiments of Sections 6.1 and 6.2 evaluate each cell's optimal share (5) and its return sensitivity at the cell's observed excess return, one cell at a time, so they describe how Treasury preferences shape an investor's demand schedule at given prices. The return experiments of Sections 6.3 and 6.4 are full general-equilibrium counterfactuals: for every quarter and every configuration of preference parameters, I re-solve the market-clearing condition (9) for the equilibrium Treasury return, so that all 17 cells re-optimise in response to the change in the return. The model-implied Treasury premium TP_t of equation (12) is the difference between the equilibrium return without and with Treasury preferences.

Sections 6.1–6.4 report the experiments on portfolio shares, on their sensitivity to excess returns, on the excess return, and on the country and sector decomposition of the Treasury premium. Appendix E reports two further equilibrium experiments, on how other holders absorb a cell's change in position and on the sensitivity of the equilibrium return to Treasury supply, which build on the model extensions of Appendix A.6. Appendix F repeats every experiment setting statistically insignificant parameters to zero, and the equilibrium experiments excluding cells with weak instruments.

6.1 Portfolio shares

Treasury preferences lower the optimal Treasury share of every investor. Figure 6 compares, for each cell, the average optimal share at the estimated parameters with the average share in the counterfactual without Treasury preferences. The counterfactual share is higher in all 17 cells, by between 12 percentage points for U.S. households and non-financial corporations and 57 percentage points for British households and non-financial corporations.

The sign follows directly from the estimates. The level parameter $\hat{\psi}_{0,c,i}$ is negative in all 17 cells, so the level term penalises every unit of Treasuries at any excess return, and where the curvature parameter is positive it offsets only part of that penalty (Section 5.2). At the returns they actually face, investors therefore hold fewer Treasuries than the risk-return trade-off alone would warrant.

Two features explain the size of the gaps. First, Treasuries offer a particularly good risk-return trade-off, with a positive expected excess return and a negative covariance providing insurance, so mean-variance demand at observed excess returns alone would place most of the foreign sovereign portfolio in Treasuries, and the counterfactual shares cluster near the no-short-selling ceiling. The prevalence of optimal portfolio shares at 1 in the counterfactual shows that motives beyond risk-return optimisation are essential to account for the observed U.S. Treasury shares in sovereign bond portfolios. Second, for the risk-neutral cells the counterfactual share is a corner

by construction, so the gap measures the distance to that corner rather than a smooth portfolio reallocation.

These results run against the view that convenience bolsters foreign demand for Treasuries. At the estimated preferences, Treasury holdings are *depressed*, not inflated, relative to their risk-return benchmark. This result is the portfolio counterpart of the post-crisis erosion of the Treasury premium documented in Du et al. (2018a) and Du et al. (2026), and of the loss of Treasury convenience studied in Jiang et al. (2026).

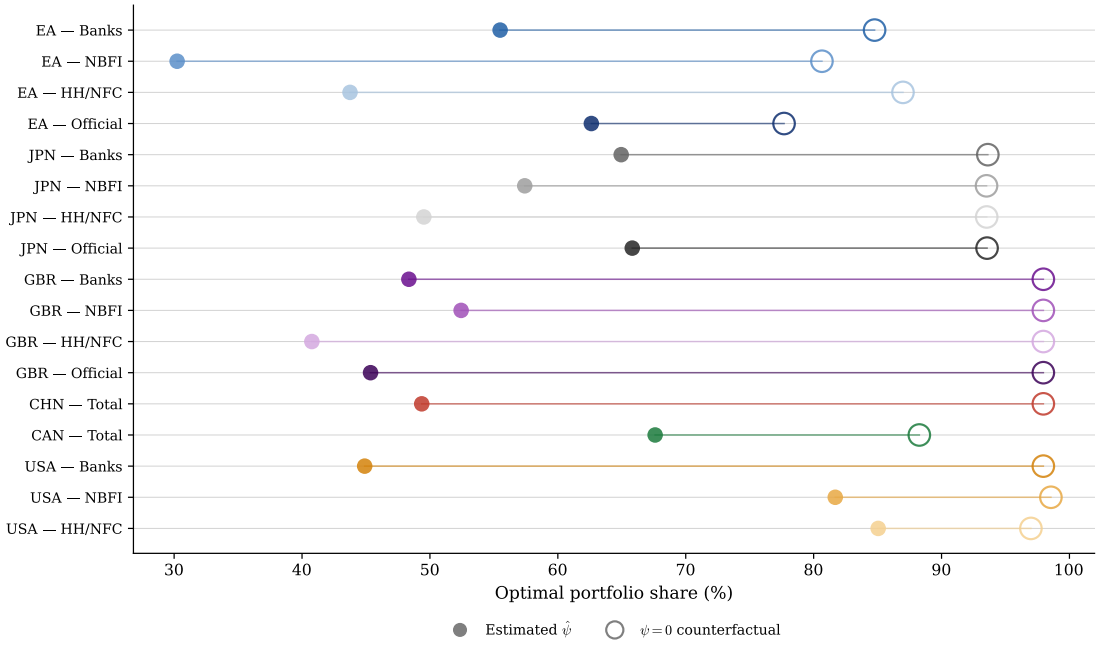


Figure 6. Portfolio share experiment

Notes: Time-series average of the optimal Treasury portfolio share $s^*_{c,i,t}$, in percent, at the estimated preference parameters $(\hat{\psi}_{0,c,i}, \hat{\psi}_{1,c,i})$ (filled markers) and in the counterfactual $\psi_{0,c,i} = \psi_{1,c,i} = 0$ (open markers), Q4 2013–Q4 2025. Shares are evaluated quarter by quarter, one cell at a time, from equation (5) at the cell's observed excess return $E_{c,t}$, variance $V_{c,t}$ and covariance $C_{c,t}$, with estimated risk aversion $\hat{\gamma}_{c,i}$, and are bounded between zero and one. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution of Appendix A.4. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

6.2 Portfolio sensitivity

The second experiment measures how strongly portfolio shares respond to excess returns with and without Treasury preferences. For each cell and quarter, I compute the derivative $\mathrm{d}s_{c,i,t}/\mathrm{d}E_{c,t}$ of Proposition 2 at the share that is optimal under each set of parameters. The derivative is zero in any quarter in which the share sits at the lower or upper bound of $[0, 1]$, since a small change in the excess return then leaves the share unchanged. I then average the derivative over all quarters.

Portfolio shares are more sensitive to excess returns at the estimated parameters than without

Treasury preferences in every cell except Canada, U.S. banks and U.S. households and non-financial corporations (Figure 7). This pattern reflects three distinct mechanisms.

For the risk-neutral British and Chinese cells, the counterfactual sensitivity is zero in every quarter, because without Treasury preferences their demand is a step function: perfectly elastic at the return at which they are indifferent between the two assets, and at a corner otherwise. At the estimated parameters their demand has a finite slope, which comes entirely from the curvature of Treasury preferences. For these investors, diminishing marginal convenience is the only force that makes Treasury demand respond smoothly to returns. U.S. banks are the limiting case: with both $\hat{\gamma}$ and $\hat{\psi}_1$ at zero, their sensitivity is zero with and without preferences as their optimal portfolio share always sits at a corner: 1 if the observed excess return is higher than their required return, and 0 otherwise.

For the cells whose curvature parameter is zero and whose risk aversion is positive, all euro-area cells and all Japanese cells, Proposition 2 implies that Treasury preferences leave the slope $1/(\gamma_{c,i}V_{c,t})$ unchanged in any quarter in which the share is interior. The measured sensitivities still differ because of a composition effect. Removing the negative level parameter raises the desired share. Some quarters in which the share was at zero at the estimates become interior, but many more reach the upper bound of one, and each constrained quarter contributes a zero to the average. Across these cells, the number of constrained quarters rises in the counterfactual. Setting preferences to zero thus lowers the average sensitivity although $\psi_{1,c,i}$ is unchanged. Since the interior slope varies with the return variance $V_{c,t}$, the size of the effect depends on which quarters become constrained as well as on how many.

The direct dampening effect of curvature predicted by Proposition 2 operates only in the cells where both risk aversion and curvature are estimated away from zero: Canada, U.S. non-bank financial institutions and U.S. households and non-financial corporations. It is clearest for Canada, the cell with the largest curvature parameter, whose sensitivity is lower at the estimates than in the counterfactual. For U.S. non-bank financial institutions the same composition effect as above dominates, and for U.S. households and non-financial corporations the two forces roughly offset. Among the British cells, the official sector, whose curvature parameter is the largest in the country, has by far the lowest sensitivity at the estimates.

These results offer a complementary perspective on the recent literature that estimates U.S. Treasury demand elasticities for different sectors.

First, the implied semi-elasticities, in units of percent change in portfolio shares over 1% change in expected excess returns, range from 5.8 for Canada to 373.3 for British households and non-financial corporations, up to an order of magnitude larger than the comparable estimates in the literature (Fang et al., 2025, Chaudhary et al., 2026, Eren et al., 2026). This discrepancy is due to the different object of analysis: the sensitivity of portfolio *shares* to yield *differentials* here, as opposed to the sensitivity of holdings to yields in the literature. The two objects are related but not equivalent: the portfolio share can move due to changes in the benchmark holdings too, and yield differentials are less volatile than raw yields, potentially leading to larger elasticities. The two approaches are complementary, focusing more on reallocation in the case of this paper as opposed to overall demand for government debt in the literature.

Second, the foreign official sector is the least elastic overall, except for Japan, consistent with the results in Eren et al. (2026) and Tabova and Warnock (2026). Likewise, NBFIs and the HH/NFC sector — which includes hedge funds — are the most elastic, in line with the higher demand elasticity for non-banks in Fang et al. (2025) and for the household sector in Chaudhary et al. (2026). However, as Cavaleri (2025) emphasises, elasticities alone cannot separate risk

aversion from convenience. This experiment does so explicitly by holding risk aversion fixed while switching off Treasury preferences. Shutting down Treasury preferences results in very similar elasticities across sectors within countries, while cross-country differences persist. Therefore, the documented cross-sectoral differences in the elasticity of government bond demand cannot be explained by differences in risk aversion alone. This result is not only driven by corner solutions: it holds even for the euro area and Japan, where all sectors sit away from zero in both the equilibrium and counterfactual.

Third, all foreign investors, except for Canada, display a higher elasticity at the estimated parameters than the two unconstrained domestic U.S. sectors, NBFIs and HH/NFC. This finding challenges the notion that foreign investors are particularly inelastic buyers of U.S. Treasuries (Krishnamurthy and Lustig, 2019, Jiang et al., 2021, 2024b). It also qualifies the argument in Tabova and Warnock (2026) that price sensitivities differ more between private and official investors than between domestic and foreign investors. My estimates show that both margins matter for differences in the demand for Treasuries *relative* to other government bonds.

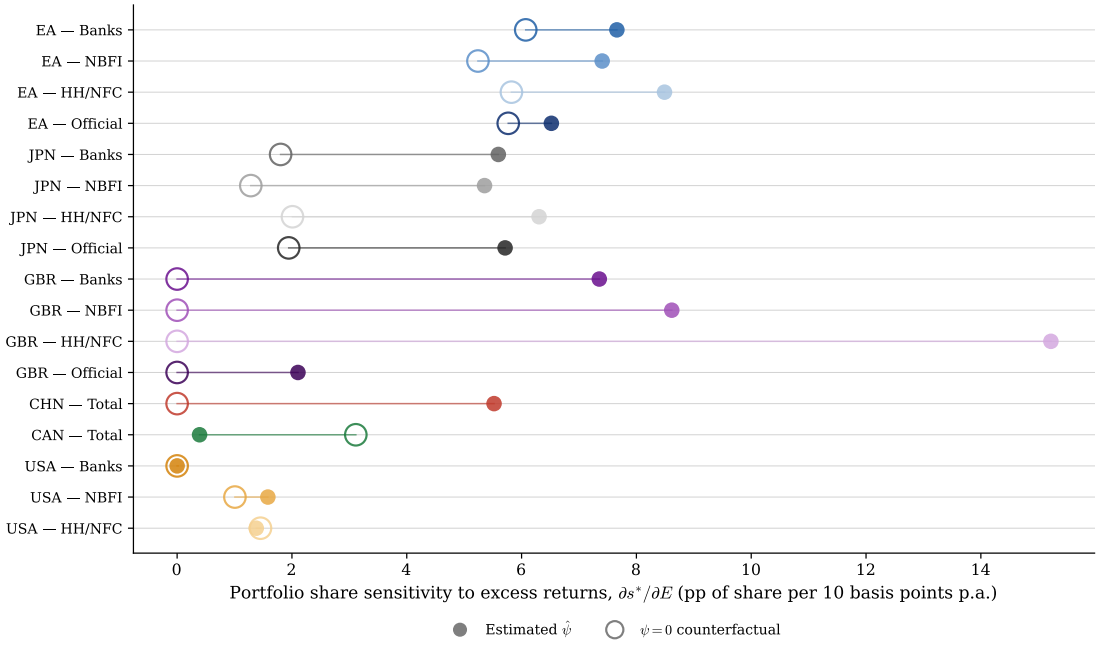


Figure 7. Portfolio sensitivity experiment

Notes: Time-series average of the sensitivity of the optimal Treasury portfolio share to the excess return, $\mathrm{d}s^*_{c,i,t}/\mathrm{d}E_{c,t}$ (Proposition 2), in percentage points of portfolio share per ten basis points per annum of excess return, at the estimated preference parameters (filled markers) and in the counterfactual $\psi_{0,c,i} = \psi_{1,c,i} = 0$ (open markers), Q4 2013–Q4 2025. The derivative is evaluated quarter by quarter, one cell at a time, at the share that is optimal under each set of parameters and at the cell's observed $(E_{c,t}, V_{c,t}, C_{c,t})$, and is set to zero in quarters in which that share is at zero or one. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution of Appendix A.4; where their curvature is also zero, their sensitivity is zero in every quarter. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

6.3 Excess returns

The third experiment asks how much of the equilibrium excess return on Treasuries is due to Treasury preferences. I measure the equilibrium excess return $E_t^* = R_{US,t+1}^* - \bar{R}_t^F$ against a common benchmark $\bar{R}_t^F$, the average of the cells' benchmark returns weighted by the number of cells per country. Since each cell's own benchmark enters the market-clearing condition, the common benchmark is used for reporting only. The excess return splits into two parts: the contribution of Treasury preferences, $-TP_t$, and the excess return $R_{US,t+1}^*\big|_{\psi=0} - \bar{R}_t^F$ that would prevail without them, which I refer to as the risk component.

Treasury preferences raise the equilibrium Treasury return. The model-implied Treasury premium TP_t is negative in every quarter, averaging -78 basis points per annum and ranging between -92 and -50 basis points across quarters. Treasury preferences account for more than the entire model-implied equilibrium excess return (Figure 8), which averages 70 basis points per annum. The risk component averages -8 basis points per annum, reflecting the hedging properties of Treasuries. Therefore, the model prices the hedging properties of Treasuries, which offer positive excess returns and a negative covariance with the reference basket. The positive excess return that clears the market is compensation for Treasury inconvenience.

The mechanism is the aggregate counterpart of the portfolio experiment. With a negative level parameter in every cell, Treasury preferences shift the whole demand schedule down, and the return must rise until investors absorb the outstanding supply.

A negative Treasury premium matches the post-crisis evidence on Treasury yields relative to other advanced-economy sovereigns. Du et al. (2018a) find that the medium- and long-maturity Treasury premium disappeared after the global financial crisis, dropping to -8 basis points on average *vis à vis* G10 currencies between 2010 and 2016. A refinement and update of this measure in Du et al. (2026) reports its further fall into negative territory, to about -25 basis points on average. Differencing domestic swap spreads across countries, which dispenses with the FX swap leg altogether, points in the same direction: over 2021–2025, ten-year Treasuries yielded about 26 basis points *above* the dollar swap rate while G10 government bonds traded roughly 19 basis points below their own swap rates (Du et al., 2026), implying a relative premium of about -45 basis points. Likewise, the cross-country measures of convenience yields with respect to domestic option-derived risk-free rates in Diamond and Van Tassel (2026) imply a relative convenience yield of about -30 basis points for the United States relative to the country with the highest domestic convenience yield in the sample, Australia. Therefore, my measure of the Treasury premium is on the same order of magnitude as the spread-based proxies commonly calculated by the literature, albeit larger than any of them.

The model-implied premium differs from these spread-based proxies in three respects. It is inferred from quantities as well as prices, accounting for the endogenous reaction of all investors to changes in Treasury preferences. In addition, it is calculated on an unhedged basis, rather than the usual hedged basis adopted in the literature, netting out the currency risk premium separately priced by the model. Finally, it captures by definition all incentives or disincentives to hold Treasuries beyond risk-return trade-offs, while spread-based measures may measure only aspects of the Treasury premium like liquidity (Longstaff, 2004, Krishnamurthy and Vissing-Jorgensen, 2012, Nagel, 2016), collateral value (Duffie, 1996, D'Amico et al., 2018), or hedging properties (Acharya and Laarits, 2023). These differences highlight the value of adopting an equilibrium framework in providing an alternative and complementary view of Treasury convenience.

6.4 Who drives the Treasury premium?

Attributing TP_t to individual cells in an additive manner requires a more complex exercise than simply switching off Treasury preferences cell-by-cell, because the equilibrium return is a nonlinear function of all cells' preferences. By Propositions 5 and 6, the effect of one cell's preferences on the return is proportional to its absorption share $a_{j,t}$, which depends on the demand slopes of all other cells, which in turn depend on their own preferences and on the equilibrium return. Switching off cells one after the other therefore yields contributions that depend on the order in which cells are switched off. Instead, I attribute the premium with the Shapley value (Shapley, 1953), which assigns to each cell its marginal effect on TP_t averaged over all possible orderings of the 17 cells. By construction, cell contributions are independent of any ordering and sum exactly to TP_t, so country and sector contributions are simple sums of cell contributions.

Size, elasticity and preferences. A cell's contribution combines three distinct forces: the size of its Treasury position, the elasticity of its demand, which depends on $\gamma_{c,i}$ and $\psi_{1,c,i}$, and the strength of its Treasury preferences, $\psi_{0,c,i}$ and $\psi_{1,c,i}$. The marginal effect of a cell's preferences on the equilibrium return is its absorption share $a_{j,t}$, its dollar demand slope $W_{j,t}/D_{j,t}$ relative to the market's, scaled by the size of the preference change. The absorption share is itself position size weighted by elasticity. Comparing each cell's share of modelled Treasury holdings, its absorption share and its share of the premium therefore indicates which force drives its contribution.

Size alone explains little. China holds 11% of the Treasuries held by the 17 cells on average, but accounts for 38% of the premium, 3.34 times its holdings weight. The reason is elasticity. Because China's risk aversion is estimated at zero, its curvature parameter is the only source of curvature in its demand: Treasury preferences give China's demand its finite slope, and without them it would be perfectly elastic at its benchmark, and perfectly inelastic elsewhere. China's preferences thus act through the elasticity of its demand as well as through its level, and the same holds for the British cells. China's demand elasticity is large and sharply drops when removing Treasury preferences, as shown in Section 6.2. As a result, its absorption share, 27%, is the second largest of any cell, after that of U.S. banks, discussed below. U.S. non-bank financial institutions show the opposite pattern. They are the largest modelled holder, with 43% of holdings, but their demand elasticity is barely affected by their Treasury preferences and is small overall relative to their size, with an absorption share of only 4%. Therefore, their contribution, 16% of the premium, is only 0.37 times their holdings weight.

The strength of Treasury preferences and the overall level of elasticity act jointly. U.S. households and non-financial corporations show the largest effect of Treasury preferences on the required return of any investor taken in isolation, 98 basis points per annum (Section 5.2). Yet their contribution is only 0.21 times their holdings weight, because their demand is relatively inelastic, owing to the highest risk-aversion estimate in the panel, 4.56. Thus, they have an absorption share of only 1%. The euro-area official sector, whose preferences raise its required return the least and whose absorption share, 1%, is below its 2% of holdings, contributes only 0.23 times its holdings weight. The general-equilibrium content of the exercise lies in this contrast with the partial-equilibrium effects of Section 5.2: who drives the premium depends on who is marginal, not on who values Treasuries the least in isolation.

U.S. banks are the limiting case of the same mechanism. With both risk aversion and curvature at zero, their demand is perfectly elastic at their indifference return even at the estimated parameters (Appendix A.4). Whenever they hold an interior share, they absorb every marginal change in supply at an unchanged return and set the equilibrium return, which their level parameter

then determines, while other investors' preferences move only the quantity they hold. Their absorption share, 45% on average and the largest of any cell, is therefore the limit of elasticity rather than of size, which is why it is large relative to both their 10% of holdings and their 16% of the premium.

Country and sector contributions. On average over the sample, every country and every one of the 17 cells contributes with the same sign: each investor's preferences raise the equilibrium Treasury return. Across countries, the contributions are highly concentrated (Figure 9). China accounts for −29 basis points per annum, 38% of the premium, and the three U.S. sectors for −27 basis points, 34%. Japan follows at −9 basis points, and the United Kingdom and the euro area contribute −6 and −6 basis points. Canada's contribution is negligible.

Summing cell contributions within each sector across the euro area, Japan, the United Kingdom and the United States gives the sector decomposition. Non-bank financial institutions contribute the most, −23 basis points per annum or 30% of the premium, followed by banks with −16 basis points (21%), the official sector with −6 basis points (8%), and households and non-financial corporations with −3 basis points (4%). The China and Canada aggregates together contribute −29 basis points (38%). Within countries (Figure 10), the U.S. contribution comes almost entirely from banks and non-bank financial institutions, −12 and −12 basis points. The official sector is the largest Japanese contributor, at −5 basis points, while non-bank financial institutions lead in the euro area and the United Kingdom, and take second place in Japan.

The official cells of the euro area, Japan and the United Kingdom together contribute 8% of the premium, somewhat less than their 9% share of holdings. This calculation, however, understates the role of official investors. China enters the estimation as a single country aggregate, but on average 83% of its Treasury holdings are official over Q4 2020–Q2 2025, the only window with a sector split (Section 3.2 and Figure 2, panel (d)). Attributing China's contribution to its official sector in this proportion, official investors account for 39% of the premium, the largest share of any sector.

The concentration of the premium in a few elastic holders is consistent with the observation of Krishnamurthy and Ma (2025) and Subacchi and van den Noord (2026) that the equilibrium convenience yield reflects the incentives of marginal investors rather than the sum of all investors' valuations. The decomposition identifies those investors as mainly domestic and foreign NBFIs, and the foreign official sector.

The prominence of China, and hence of official demand, challenges the argument in Jiang et al. (2024b) that foreign investors, and official investors in particular, earn low returns on their Treasury purchases, which the authors interpret as the behaviour of price-inelastic buyers of safe dollar assets. In the model, by contrast, China's weight comes from the elasticity of its demand, and its preferences raise rather than lower the equilibrium return. The leading role of foreign official demand in driving the Treasury premium negative is related to the steady diversification of official reserves away from the dollar (Arslanalp et al., 2022, Chinn et al., 2025), and lends credence to the ongoing concerns of Treasuries losing their global safe asset status (Boissay and Huang, 2025, Hassan et al., 2025, Acharya and Laarits, 2026).

The large contribution of NBFIs, magnified by their sharp increase in Treasury holdings over the last two decades, shows the vulnerability of the funding costs of the U.S. government to "flighty", returns-driven investors who are quick to shed Treasuries in times of crisis, as evidenced in the sell-off in the wake of the COVID pandemic (Duffie, 2020, Vissing-Jorgensen, 2021, He et al.,

2022, Kashyap et al., 2025).[7]

[7] In contrast, the small contribution from the HH/NFC sector, which includes hedge funds, implies that hedge-fund-driven episodes like the partial unwind of leveraged Treasury positions in the aftermath of the April 2025 U.S. tariff announcement had little impact on Treasury convenience on average in the sample period, consistent with the transitory effect measured on Treasury yields (Dyer and Fleming, 2026).

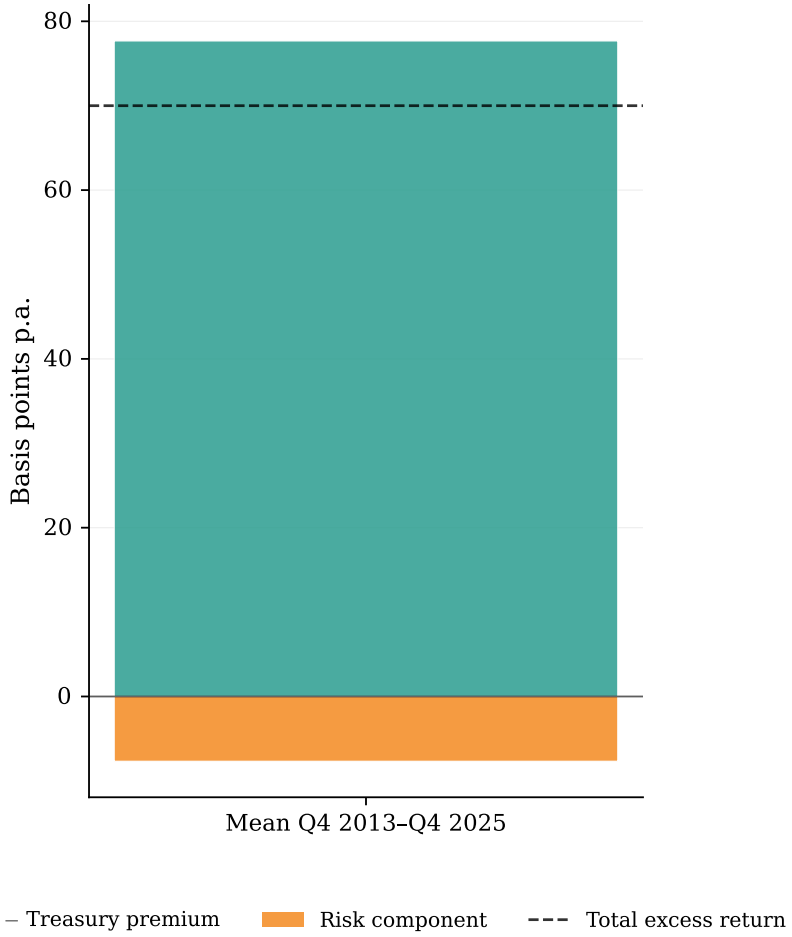


Figure 8. Excess return decomposition

Notes: Time-series averages, in basis points per annum, Q4 2013–Q4 2025. The dashed line is the model-implied equilibrium excess return of U.S. Treasuries, $E_t^* = R_{US,t+1}^* - \bar{R}_t^F$, where $\bar{R}_t^F$ averages the cells' benchmark returns with weights equal to each country's number of cells. It is split into the Treasury-premium component $-TP_t = R_{US,t+1}^* - R_{US,t+1}^*\big|_{\psi=0}$, positive when Treasury preferences raise the equilibrium return, and the risk component $R_{US,t+1}^*\big|_{\psi=0} - \bar{R}_t^F$. Both equilibrium returns solve the market-clearing condition (9) quarter by quarter, with each cell's own benchmark return; $\bar{R}_t^F$ is used for reporting only. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

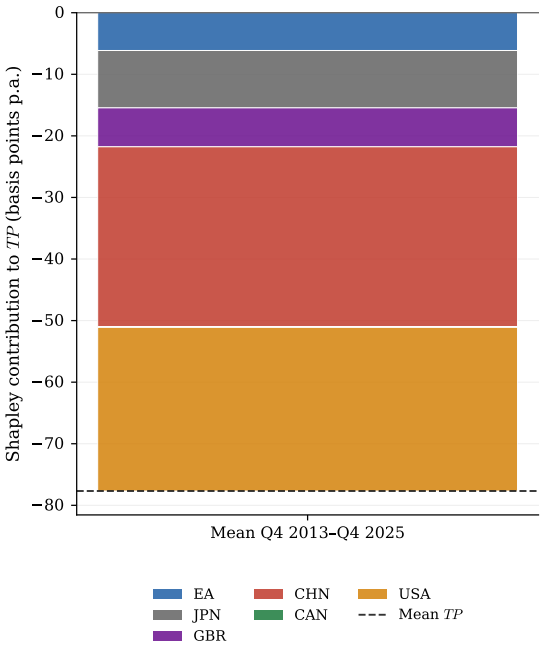


Figure 9. Treasury premium decomposition by country

Notes: Shapley contributions of each country's cells to the model-implied Treasury premium $TP_t = R^*_{US,t+1}\big|_{\psi=0} - R^*_{US,t+1}$, time-series averages in basis points per annum, Q4 2013–Q4 2025. The dashed line marks the average TP_t; a negative contribution means that the country's preferences raise the equilibrium Treasury return. Shapley values are computed quarter by quarter by exact enumeration of all 2^{17} coalitions of cells, re-solving the market-clearing condition (9) for each coalition; cell contributions sum to TP_t, and a country's contribution is the sum of its cells' contributions. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

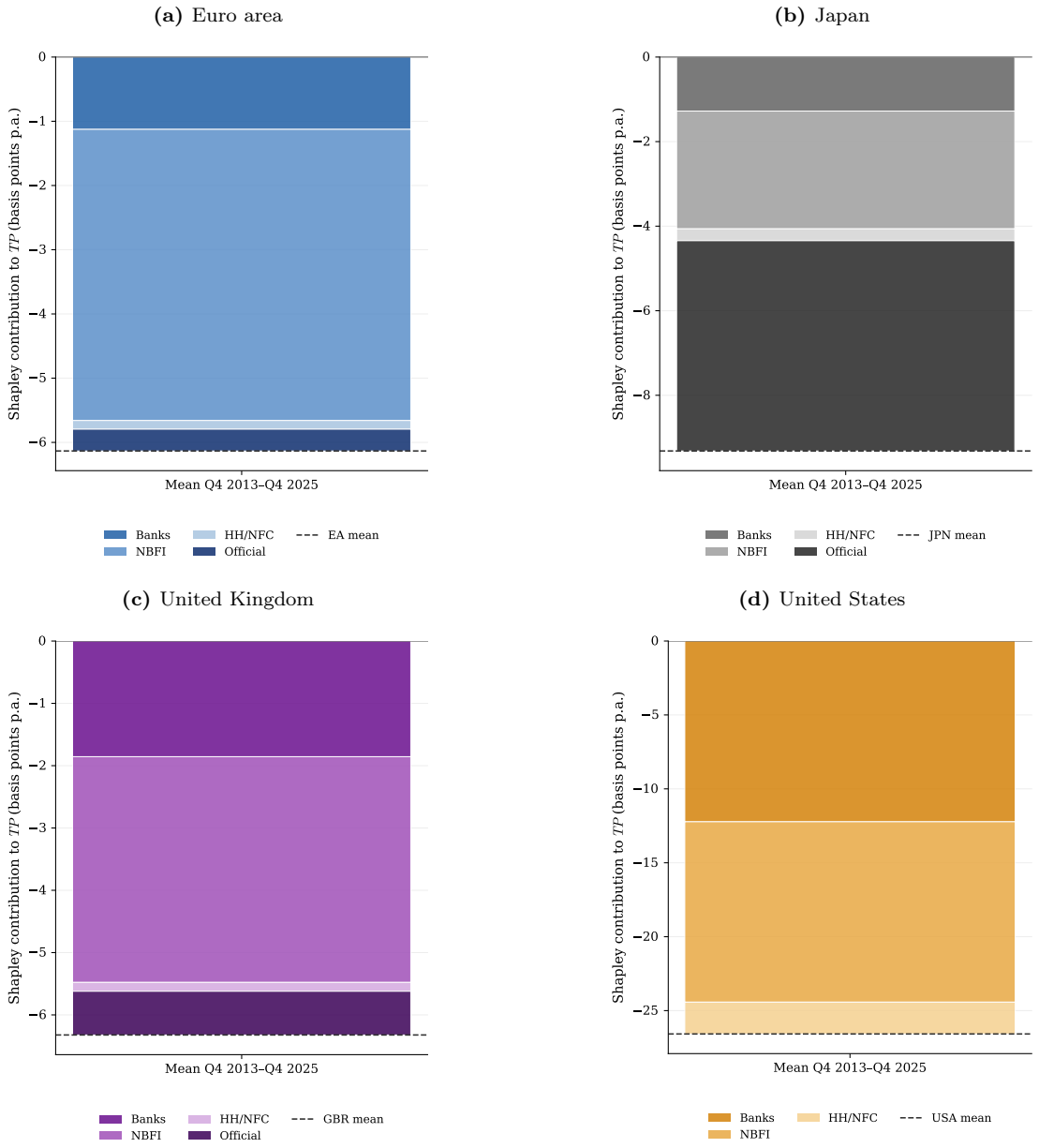


Figure 10. Treasury premium decomposition by sector

Notes: Shapley contributions of each sector's cell to the model-implied Treasury premium $TP_t = R^*_{US,t+1}\big|_{\psi=0} - R^*_{US,t+1}$, time-series averages in basis points per annum, Q4 2013–Q4 2025, for the four countries with a sector split; negative values raise the equilibrium return. The dashed line marks the country total. Vertical scales differ across panels. Shapley values are computed quarter by quarter by exact enumeration of all 2^{17} coalitions of cells, re-solving the market-clearing condition (9) for each coalition; cell contributions sum to TP_t. China, Canada and the U.S. official sector are not shown; cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4).
Sources: Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

7 Conclusion

This paper introduces a measure of the U.S. Treasury premium built on the structural parameters of a portfolio-choice model with Treasury preferences, estimated sector by sector for the largest holding countries. The premium is negative, close to −80 basis points, and most of it traces to non-bank financial institutions at home and abroad and to the foreign official sector, where China dominates.

The method has applications beyond the Treasury market. The estimation separates risk aversion from convenience preferences as drivers of Treasury demand, and thereby speaks to the emerging literature on the theoretical foundations of demand system asset pricing (Fuchs et al., 2023). The same approach can quantify other sources of taste-based heterogeneity in asset demand wherever a non-pecuniary term enters the investor's objective: preferences for green assets (Pástor et al., 2021, Baker et al., 2022), the liability-driven demand for duration of pension funds and insurers (Klingler and Sundaresan, 2019, Greenwood and Vissing-Jorgensen, 2018), regulatory incentives to hold particular securities (Corell et al., 2025), or home and currency bias (Coeurdacier and Rey, 2013, Maggiori et al., 2020).

The policy implications run along the three fault lines of the decomposition. The weight of the foreign official sector, and of China in particular, leaves U.S. debt sustainability exposed to geopolitical realignment. Reserve holdings have historically followed security alliances as much as pecuniary returns (Eichengreen et al., 2019), and the trade disputes of the second Trump administration, together with the broader fragmentation of the U.S.-led economic order (Clayton et al., 2024), raise the odds that official holders retreat from Treasuries for reasons unrelated to their financial merits.

The large negative contribution of NBFIs is concerning for a different reason. Their share of government debt has grown steadily since the Great Financial Crisis, and sovereign bond markets in advanced economies increasingly rely on price-sensitive investors to absorb new issuance (Fang et al., 2025, International Monetary Fund, 2025, Bank for International Settlements, 2026). The estimates quantify the cost of that reliance: for NBFIs, Treasury preferences do not compress yields but raise the return the sector requires.

Finally, the results caution against engineering domestic demand by regulation. A prime example is the GENIUS Act, which requires payment stablecoin issuers to back their liabilities with Treasury bills and cash-like assets and is expected to create a new, structurally captive source of Treasury demand (Ahmed and Aldasoro, 2025, Liang and Neiman, 2026). The estimates suggest instead that domestic investors respond to returns, and that their Treasury preferences raise rather than lower the compensation they require.

The U.S. government cannot count on a captive clientele of investors, whether domestic or foreign, to compress its funding costs.

References

Acharya, V. and Laarits, T. (2026). Tariff war shock and the convenience yield of US Treasuries a hedging perspective. CEPR Discussion Papers 20985, Centre for Economic Policy Research.

Acharya, V. V. and Laarits, T. (2023). When do Treasuries Earn the Convenience Yield? – A Hedging Perspective. NBER Working Paper 31863, National Bureau of Economic Research.

Ahmed, R. and Aldasoro, I. (2025). Stablecoins and safe asset prices. BIS Working Paper 1270, Bank for International Settlements.

Alstadsæter, A., Johannesen, N., and Zucman, G. (2018). Who owns the wealth in tax havens? Macro evidence and implications for global inequality. *Journal of Public Economics*, 162:89–100.

Arslanalp, S., Eichengreen, B., and Simpson-Bell, C. (2022). The stealth erosion of dollar dominance and the rise of nontraditional reserve currencies. *Journal of International Economics*, 138(C).

Arslanalp, S., Eichengreen, B., and Simpson-Bell, C. (2023). Gold as international reserves: A barbarous relic no more? *Journal of International Economics*, 145:103822.

Arslanalp, S. and Tsuda, T. (2014). Tracking global demand for advanced economy sovereign debt. *IMF Economic Review*, 62(3):430–464.

Baker, M., Bergstresser, D., Serafeim, G., and Wurgler, J. (2022). The pricing and ownership of US green bonds. *Annual Review of Financial Economics*, 14:415–437.

Bank for International Settlements (2026). High public debt and shifting financial markets: Challenges for central banks. Annual economic report 2026, chapter ii, Bank for International Settlements, Basel.

Bank for International Settlements, European Central Bank, and International Monetary Fund (2015). Handbook on securities statistics. Technical report, International Monetary Fund, Washington, D.C. ISBN 978-1-4983-8831-3.

Barth, D., Beltran, D. O., Hoops, M., Kahn, R. J., Liu, E., and Perozek, M. G. (2025). The cross-border trail of the Treasury basis trade. FEDS Notes 2025-10-15-2, Board of Governors of the Federal Reserve System (U.S.).

Barth, D. and Kahn, R. J. (2025). Hedge funds and the Treasury cash-futures basis trade. *Journal of Monetary Economics*, 155(C).

Barth, D., Kahn, R. J., Monin, P. J., and Sokolinskiy, O. (2024). Reaching for duration and leverage in the Treasury market. Finance and Economics Discussion Series 2024-039, Board of Governors of the Federal Reserve System.

Beck, R., Coppola, A., Lewis, A., Maggiori, M., Schmitz, M., and Schreger, J. (2026). The geography of capital allocation in the euro area. Global Capital Allocation Project.

Bergant, K., Fidora, M., and Schmitz, M. (2020). International capital flows at the security level: evidence from the ECB's asset purchase programme. Working Paper Series 2388, European Central Bank.

Bernanke, B. S. (2005). The global saving glut and the U.S. current account deficit. Remarks at the Sandridge Lecture, Virginia Association of Economists, Richmond, Virginia, March 10. Board of Governors of the Federal Reserve System.

Bernanke, B. S., Bertaut, C. C., Kamin, S. B., and DeMarco, L. P. (2011). International capital flows and the returns to safe assets in the United States, 2003-2007. International Finance Discussion Papers 1014, Board of Governors of the Federal Reserve System (U.S.).

Bertaut, C. C. and Judson, R. A. (2023). Measuring U.S. cross-border securities flows: New data and a guide for researchers. FEDS Notes 2023-10-02, Board of Governors of the Federal Reserve System (U.S.).

Bianchi, J. and Sosa-Padilla, C. (2025). International sanctions and dollar dominance. *Economic Journal*, 135(672):2567–2577.

Bodenstein, M., Cuba-Borda, P. A., Gornemann, N. M., Presno, I., Prestipino, A., Queraltó, A., and Raffo, A. (2023). Global flight to safety, business cycles, and the dollar. Working Papers 799, Federal Reserve Bank of Minneapolis.

Boissay, F. and Huang, W. (2025). The safe haven properties of US Treasuries and portfolio flows of global investors. BIS Quarterly Review, Box, Bank for International Settlements.

Boyarchenko, N., Gupta, P., Steele, N., and Yen, J. (2018). Negative swap spreads. *Federal Reserve Bank of New York Economic Policy Review*, 24(2):1–14.

Bretscher, L., Schmid, L., Sen, I., and Sharma, V. (2026). Institutional Corporate Bond Pricing. *The Review of Financial Studies*, 39(3):605–660.

Cavaleri, F. (2025). The convenience yield and the demand for U.S. Treasury securities. Working paper.

Chaudhary, M., Fu, J. Z., and Zhou, H. (2026). Anatomy of the Treasury market: Who moves yields? Working paper, SSRN. First draft November 2024; this draft April 19, 2026.

Chiappori, P.-A. and Paiella, M. (2011). Relative risk aversion is constant: Evidence from panel data. *Journal of the European Economic Association*, 9(6):1021–1052.

Chinn, M. D., Frankel, J. A., and Ito, H. (2025). Reserves, sanctions and tariffs in a time of uncertainty. Working Paper 34177, National Bureau of Economic Research.

Choi, J., Dang, D., Kirpalani, R., and Perez, D. J. (2024). Exorbitant privilege and the sustainability of US public debt. *AEA Papers and Proceedings*, 114:143–47.

Clayton, C., Maggiori, M., and Schreger, J. (2024). A theory of economic coercion and fragmentation. NBER Working Paper 33309, National Bureau of Economic Research.

Coeurdacier, N. and Rey, H. (2013). Home bias in open economy financial macroeconomics. *Journal of Economic Literature*, 51(1):63–115.

Cordes, L. and Ferris, E. E. S. (2024). Who buys Treasuries when the Fed reduces its holdings. FEDS Notes 2024-06-14-3, Board of Governors of the Federal Reserve System (U.S.).

Corell, F., Holm-Hadulla, F., Leombroni, M., Mota, L., and Papoutsi, M. (2025). Drivers of convenience yields. Working paper.

D'Amico, S., Fan, R., and Kitsul, Y. (2018). The scarcity value of Treasury collateral: Repo-market effects of security-specific supply and demand factors. *Journal of Financial and Quantitative Analysis*, 53(5):2103–2129.

Diamond, W. and Van Tassel, P. (2026). Risk-free rates and convenience yields around the world. *Journal of Finance*, 81(4):2063–2108.

Doerr, S., Eren, S. E., and Malamud, S. (2023). Money market funds and the pricing of near-money assets. BIS Working Papers 1096, Bank for International Settlements.

Du, W., Hebert, B., and Li, W. (2023). Intermediary balance sheets and the treasury yield curve. *Journal of Financial Economics*, 150(3):103722.

Du, W. and Huber, A. W. (2026). Dollar asset holdings and hedging around the globe. *The Review of Financial Studies*. Advance article, hhag067.

Du, W., Im, J., and Schreger, J. (2018a). The U.S. Treasury Premium. *Journal of International Economics*, 112:167–181.

Du, W., Keerati, R., and Schreger, J. (2026). Decoupling dollar and Treasury privilege. *IMF Economic Review*.

Du, W., Tepper, A., and Verdelhan, A. (2018b). Deviations from covered interest rate parity. *Journal of Finance*, 73(3):915–957.

Duffie, D. (1996). Special repo rates. *Journal of Finance*, 51(2):493–526.

Duffie, D. (2018). Post-crisis bank regulations and financial market liquidity. Paolo Baffi Lecture on Money and Finance, Banca d'Italia, Rome.

Duffie, D. (2020). Still the world's safe haven? Redesigning the U.S. Treasury market after the COVID-19 crisis. Working Paper 62, Hutchins Center on Fiscal and Monetary Policy, Brookings Institution.

Dyer, H. and Fleming, M. J. (2026). Treasury market liquidity since April 2025. Liberty Street Economics, Federal Reserve Bank of New York.

Eichengreen, B., Mehl, A., Chiţu, L., and Beck, T. (2019). Mars or Mercury? the geopolitics of international currency choice. *Economic Policy*, 34(98):315–363.

Engel, C. and Wu, S. P. Y. (2023). Liquidity and Exchange Rates: An Empirical Investigation. *The Review of Economic Studies*, 90(5):2395–2438.

Eren, E., Schrimpf, A., and Xia, F. D. (2026). The demand for government debt. *Management Science*.

Fama, E. F. (1984). Forward and spot exchange rates. *Journal of Monetary Economics*, 14(3):319–338.

Fang, X., Hardy, B., and Lewis, K. (2025). Who holds sovereign debt and why it matters. *The Review of Financial Studies*, 38(8):2326–2361.

Fleckenstein, M. and Longstaff, F. A. (2024). Treasury richness. *Journal of Finance*, 79(4):2797–2844.

Florez-Orrego, S., Maggiori, M., Schreger, J., Sun, Z., and Tinda, S. (2024). Global capital allocation. *Annual Review of Economics*, 16(1):623–53.

Friend, I. and Blume, M. E. (1975). The demand for risky assets. *American Economic Review*, 65(5):900–922.

Fuchs, W. M., Fukuda, S., and Neuhann, D. (2023). Demand-System Asset Pricing: Theoretical Foundations. *SSRN Electronic Journal*.

Gabaix, X. and Koijen, R. S. J. (2021). In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis. NBER Working Paper 28967, National Bureau of Economic Research.

Gabaix, X. and Koijen, R. S. J. (2024). Granular Instrumental Variables. *Journal of Political Economy*, 132(7):2274–2303.

Gabaix, X., Koijen, R. S. J., Mainardi, F., Oh, S., and Yogo, M. (2024). Asset Demand of U.S. Households. NBER Working Paper 32001, National Bureau of Economic Research.

Georgiadis, G., Müller, G. J., and Schumann, B. (2023). Dollar trinity and the global financial cycle. Discussion Papers of DIW Berlin 2058, DIW Berlin, German Institute for Economic Research.

Gourinchas, P.-O. and Rey, H. (2022). Exorbitant privilege and exorbitant duty. CEPR Discussion Paper 16944, Centre for Economic Policy Research.

Graziano, M., Lombardo, M. P., and Phillot, M. (2026). Fiscal funding, treasury convenience and the real exchange rate. *SSRN Electronic Journal*.

Greenwood, R. M. and Vissing-Jorgensen, A. (2018). The Impact of Pensions and Insurance on Global Yield Curves. *SSRN Electronic Journal*.

Haddad, V., Huebner, P., and Loualiche, E. (2025). How Competitive is the Stock Market? Theory, Evidence from Portfolios, and Implications for the Rise of Passive Investing. *American Economic Review*, 115(3):975–1018.

Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4):1029–1054.

Hassan, T. A., Mertens, T. M., Wang, J., and Zhang, T. (2025). Trade war and the dollar anchor. *Brookings Papers on Economic Activity*, Fall:65–130.

He, Z., Nagel, S., and Song, Z. (2022). Treasury inconvenience yields during the COVID-19 crisis. *Journal of Financial Economics*, 143(1):57–79.

International Monetary Fund (2025). Global financial stability report: Shifting ground beneath the calm. Technical report, International Monetary Fund, Washington, DC.

Ito, H. and McCauley, R. (2020). Currency composition of foreign exchange reserves. *Journal of International Money and Finance*, 102(C).

Jansen, K. A. E., Li, W., and Schmid, L. (2026). Granular Treasury demand with arbitrageurs. NBER Working Paper 33243, National Bureau of Economic Research. First version December 2024; revised June 2026.

Jermann, U. (2020). Negative swap spreads and limited arbitrage. *The Review of Financial Studies*, 33(1):212–238.

Jiang, Z., Krishnamurthy, A., and Lustig, H. (2021). Foreign Safe Asset Demand and the Dollar Exchange Rate. *The Journal of Finance*, 76(3):1049–1089.

Jiang, Z., Krishnamurthy, A., and Lustig, H. (2024a). Dollar safety and the global financial cycle. *The Review of Economic Studies*, 91(5):2878–2915.

Jiang, Z., Krishnamurthy, A., and Lustig, H. (2024b). The rest of the world's dollar-weighted return on U.S. Treasurys. *IMF Economic Review*, 72(4):1320–1346.

Jiang, Z., Krishnamurthy, A., Lustig, H., and Sun, J. (2024c). Convenience yields and exchange rate puzzles. NBER Working Papers 32092, National Bureau of Economic Research, Inc.

Jiang, Z., Lustig, H., Van Nieuwerburgh, S., and Xiaolan, M. Z. (2024d). The U.S. Public Debt Valuation Puzzle. *Econometrica*, 92(4):1309–1347.

Jiang, Z., Lustig, H., Van Nieuwerburgh, S., and Xiaolan, M. Z. (2025). Bond Convenience Yields in the Eurozone Currency Union. NBER Working Paper 34307, National Bureau of Economic Research.

Jiang, Z., Richmond, R. J., and Zhang, T. (2024e). A portfolio approach to global imbalances. *Journal of Finance*, 79(3):2025–2076.

Jiang, Z., Richmond, R. J., and Zhang, T. (2026). Convenience lost. *Journal of Financial Economics*, 184.

Judson, R. A. and Kim, N. (2026). Measuring cross-border securities positions: Explaining asymmetries between U.S. Treasury TIC and IMF PIP (formerly CPIS) data. FEDS Notes 2026-02-13-1, Board of Governors of the Federal Reserve System (U.S.).

Kashyap, A. K., Stein, J. C., Wallen, J. L., and Younger, J. (2025). Treasury market dysfunction and the role of the central bank. *Brookings Papers on Economic Activity*, 56(1):221–296.

Klingler, S. and Sundaresan, S. (2019). An explanation of negative swap spreads: Demand for duration from underfunded pension plans. *The Journal of Finance*, 74(2):675–710.

Klingler, S. and Sundaresan, S. (2023). Diminishing treasury convenience premiums: Effects of dealers' excess demand and balance sheet constraints. *Journal of Monetary Economics*, 135(C):55–69.

Koijen, R. S., Koulischer, F., Nguyen, B., and Yogo, M. (2021). Inspecting the mechanism of quantitative easing in the Euro Area. *Journal of Financial Economics*, 140(1):1–20.

Koijen, R. S. J. and Yogo, M. (2019). A Demand System Approach to Asset Pricing. *Journal of Political Economy*, 127(4):1475–1515.

Koijen, R. S. J. and Yogo, M. (2026). Exchange Rates and Asset Prices in a Global Demand System. *Journal of Political Economy*, 134(8):2441–2487.

Krishnamurthy, A. (2002). The bond/old-bond spread. *Journal of Financial Economics*, 66(2-3):463–506.

Krishnamurthy, A. and Lustig, H. N. (2019). Mind the gap in sovereign debt markets: The U.S. Treasury basis and the dollar risk factor. In *2019 Jackson Hole Economic Symposium, Stanford University Graduate School of Business Research Paper*. SSRN Working Paper 3443231.

Krishnamurthy, A. and Ma, Y. (2025). The demand and supply of convenience assets. *Annual Review of Financial Economics*, 17:225–242.

Krishnamurthy, A. and Vissing-Jorgensen, A. (2012). The Aggregate Demand for Treasury Debt. *Journal of Political Economy*, 120(2):233–267.

Laarits, T., Ross, C., and Ross, S. (2026). Aggregate collateral demand. Working paper.

Liang, N. and Neiman, B. (2026). Stablecoins after GENIUS: Private money, public debt, and the global dollar. Hutchins center working paper, Brookings Institution, Hutchins Center on Fiscal and Monetary Policy.

Longstaff, F. A. (2004). The Flight-to-Liquidity Premium in U.S. Treasury Bond Prices. *The Journal of Business*, 77(3):511–526.

Maggiori, M. (2017). Financial intermediation, international risk sharing, and reserve currencies. *American Economic Review*, 107(10):3038–3071.

Maggiori, M., Neiman, B., and Schreger, J. (2020). International currencies and capital allocation. *Journal of Political Economy*, 128(6):2019 – 2066.

Nagel, S. (2016). The Liquidity Premium of Near-Money Assets*. *The Quarterly Journal of Economics*, 131(4):1927–1971.

Nenova, T. (2025). Global or Regional Safe Assets: Evidence from Bond Substitution Patterns. BIS Working Papers 1254, Bank for International Settlements.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Olea, J. L. M. and Pflueger, C. (2013). A robust test for weak instruments. *Journal of Business & Economic Statistics*, 31(3):358–369.

Pástor, Ľ., Stambaugh, R. F., and Taylor, L. A. (2021). Sustainable investing in equilibrium. *Journal of Financial Economics*, 142(2):550–571.

Setser, B. W. (2023). China isn't shifting away from the dollar or dollar bonds. Council on Foreign Relations, Follow the Money blog.

Setser, B. W. (2026). Finding China in the U.S. TIC data. Council on Foreign Relations, Follow the Money blog.

Shapley, L. S. (1953). A value for n-person games. In Kuhn, H. W. and Tucker, A. W., editors, *Contributions to the Theory of Games, Volume II*, pages 307–317. Princeton University Press, Princeton.

Subacchi, P. and van den Noord, P. J. (2026). Should Europe continue to fund U.S. federal debt? *De Economist*, 174(3):227–251.

Tabova, A. and Warnock, F. (2026). Preferred habitats and timing in the world's safe asset. *Journal of International Economics*, 161(C):104233.

Valchev, R. (2020). Bond Convenience Yields and Exchange Rate Dynamics. *American Economic Journal: Macroeconomics*, 12(2):124–166.

Vissing-Jorgensen, A. (2021). The Treasury market in spring 2020 and the response of the Federal Reserve. *Journal of Monetary Economics*, 124:19–47.

Appendix

Who Drives the U.S. Treasury Premium?

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A Model derivations and extensions

A.1 Mean-variance utility derivation

Consider an investor $j \in \mathcal{J}$ choosing the portfolio share of U.S. Treasuries $s_{j,t}$ with gross return $R_{US,t+1}$. Foreign cells $j = (c, i)$ also hold the foreign sovereign portfolio excluding the United States and country c, with gross return $R^F_{c,t+1}$ defined in (3); the three U.S. domestic sector cells—banks, NBFI, and households and non-financial corporations—hold the foreign sovereign portfolio excluding only the United States, with return $R^F_{US,t+1}$. All returns are unhedged and measured in the investor's currency. Denote the relevant benchmark return generically by $R^F_{j,t+1}$. Terminal wealth per unit of portfolio value $W_{j,t}$ is the share-weighted average of the two assets' gross returns, $r^{port}_{j,t+1} \coloneqq W_{j,t+1}/W_{j,t} = s_{j,t}(R_{US,t+1} - R^F_{j,t+1}) + R^F_{j,t+1}$. The investor derives utility from terminal wealth and directly from holding Treasuries through a convenience term $h_j(s_{j,t})$. The investor's utility is

$$U(r^{port}_{j,t+1}, s_{j,t}) = -e^{-\gamma_j (r^{port}_{j,t+1} + h_j(s_{j,t}))},$$

where h_j is increasing and concave in $s_{j,t}$. This utility function preserves the desirable properties of standard exponential utility: it is increasing and concave in $r^{port}_{j,t+1}$, and it displays constant absolute risk aversion with coefficient γ_j.

By taking the first and second derivatives with respect to $s_{j,t}$,

$$\begin{aligned} U'(s_{j,t}) &= \gamma_j h_j'(s_{j,t}) e^{-\gamma_j (r^{port}_{j,t+1} + h_j(s_{j,t}))} > 0 \\ U''(s_{j,t}) &= \gamma_j e^{-\gamma_j (r^{port}_{j,t+1} + h_j(s_{j,t}))} \left(h_j''(s_{j,t}) - \gamma_j (h_j'(s_{j,t}))^2 \right) < 0. \end{aligned}$$

The marginal utility of holding Treasuries is therefore declining, so investors require a higher monetary return to absorb more Treasuries in equilibrium. This mechanism is widely used in the literature to link the outstanding amount of U.S. government debt with the equilibrium convenience yield (Krishnamurthy and Vissing-Jorgensen (2012) and Engel and Wu (2023), among others).

Assume that $R_{US,t+1}$ and $R^F_{j,t+1}$ are conditionally jointly normal, so that $r^{port}_{j,t+1} \sim N(\mu_W, \sigma^2_W)$ with $\mu_W = \mathbb{E}_t[r^{port}_{j,t+1}]$ and $\sigma^2_W = \mathbb{V}_t[r^{port}_{j,t+1}]$. Then expected utility is

$$\begin{aligned} \mathbb{E}_t\left[-e^{-\gamma_j (r^{port}_{j,t+1} + h_j(s_{j,t}))} \right] &= \int_{-\infty}^{\infty} -e^{-\gamma_j (r + h_j(s_{j,t}))} \frac{1}{\sigma_W \sqrt{2\pi}} e^{-\frac{(r-\mu_W)^2}{2\sigma^2_W}} dr \\ &= e^{-\gamma_j h_j(s_{j,t})} \int_{-\infty}^{\infty} -e^{-\gamma_j r} \frac{1}{\sigma_W \sqrt{2\pi}} e^{-\frac{(r-\mu_W)^2}{2\sigma^2_W}} dr. \end{aligned}$$

Following the standard mean-variance derivation by completing the square under the integral,

$$\begin{aligned} \mathbb{E}_t\left[-e^{-\gamma_j (r^{port}_{j,t+1} + h_j(s_{j,t}))} \right] &= -e^{-\gamma_j \left(\mu_W - \frac{\gamma_j}{2}\sigma^2_W + h_j(s_{j,t}) \right)} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sigma_W \sqrt{2\pi}} e^{-\frac{(r-\mu_W+\gamma_j\sigma^2_W)^2}{2\sigma^2_W}} dr}_{=1} \\ &= -e^{-\gamma_j \left(\mu_W - \frac{\gamma_j}{2}\sigma^2_W + h_j(s_{j,t}) \right)}. \end{aligned}$$

It follows that

$$\max_{s_{j,t}} \mathbb{E}_t\left[-e^{-\gamma_j (r^{port}_{j,t+1} + h_j(s_{j,t}))} \right] = \max_{s_{j,t}} \mu_W - \frac{\gamma_j}{2}\sigma^2_W + h_j(s_{j,t}).$$

Maximising expected utility with an exponential utility function in the terminal portfolio return and U.S. Treasuries reduces to standard mean-variance preferences with an additive term for U.S. Treasury holdings, which is increasing and concave in $s_{j,t}$.

A.2 Optimal portfolio share

Proof of Proposition 1.

Proof. Starting from the FOC (4) (suppressing the cell index j within this proof for readability, so that γ, E, V, C and the two ψ's all denote a single cell's objects),

$$E - \gamma V s - \gamma C + \psi_0 + \frac{\psi_1}{s} = 0.$$

Multiply by s:

$$sE - \gamma V s^2 - \gamma C s + \psi_0 s + \psi_1 = 0.$$

Rearrange grouping terms in powers of s:

$$-\gamma V s^2 + (E - \gamma C + \psi_0)s + \psi_1 = 0.$$

Multiply by -1 to put into standard quadratic form:

$$\gamma V s^2 - (E - \gamma C + \psi_0)s - \psi_1 = 0.$$

Let $a = \gamma V$, $b = -(E - \gamma C + \psi_0)$, $c = -\psi_1$. The two roots are

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{E - \gamma C + \psi_0 \pm \sqrt{(E - \gamma C + \psi_0)^2 + 4\gamma V \psi_1}}{2\gamma V}.$$

If $\psi_1 > 0$ the discriminant exceeds $(E - \gamma C + \psi_0)^2$, so the square root term exceeds the absolute value of the linear term. The "minus" root is therefore non-positive, and is strictly negative for $\psi_1 > 0$. Because $s > 0$ is required by the domain of the log convenience term and by the no-short-selling constraint, I discard the minus root and keep the plus root:

$$s^*_{j,t} \;=\; \frac{E_{j,t} - \gamma_j C_{j,t} + \psi_{0,j} \;+\; \sqrt{(E_{j,t} - \gamma_j C_{j,t} + \psi_{0,j})^2 + 4\gamma_j V_{j,t} \psi_{1,j}}}{2\gamma_j V_{j,t}}.$$

□

A.3 Derivatives of $s^*_{j,t}$

A.3.1 Proposition 2

Proof. Define

$$f_j(E_{j,t}, s_{j,t}) \;\equiv\; E_{j,t} - \gamma_j V_{j,t}\, s_{j,t} - \gamma_j C_{j,t} + \psi_{0,j} + \frac{\psi_{1,j}}{s_{j,t}}.$$

The FOC (4) implies $f_j(E_{j,t}, s_{j,t}(E_{j,t})) \equiv 0$.

Compute partial derivatives:

$$f_E = \frac{\partial f_j}{\partial E_{j,t}} = 1, \qquad f_s = \frac{\partial f_j}{\partial s_{j,t}} = -\gamma_j V_{j,t} - \frac{\psi_{1,j}}{s_{j,t}^2}.$$

Differentiate $f_j(E_{j,t}, s_{j,t}(E_{j,t})) = 0$ with respect to $E_{j,t}$ by the implicit function theorem and solve:

$$0 = f_E + f_s \frac{\mathrm{d}s_{j,t}}{\mathrm{d}E_{j,t}} \quad \Rightarrow \quad \frac{\mathrm{d}s_{j,t}}{\mathrm{d}E_{j,t}} = -\frac{f_E}{f_s} = \frac{1}{\gamma_j V_{j,t} + \psi_{1,j}/s_{j,t}^2}.$$

□

A.3.2 Proposition 3

Proof. Differentiate $f_j(E_{j,t}, s_{j,t}) = 0$ with respect to $\psi_{0,j}$ holding $R_{US,t+1}$, and hence $E_{j,t}$, fixed:

$$0 = f_{\psi_0} + f_s \frac{\partial s_{j,t}}{\partial \psi_{0,j}},$$

where $f_{\psi_0} = \partial f_j / \partial \psi_{0,j} = 1$. Hence

$$\frac{\partial s_{j,t}}{\partial \psi_{0,j}}\Big|_R = -\frac{f_{\psi_0}}{f_s} = \frac{1}{\gamma_j V_{j,t} + \psi_{1,j}/s_{j,t}^2}.$$

□

A.3.3 Proposition 4

Proof. Differentiate $f_j(E_{j,t}, s_{j,t}) = 0$ with respect to $\psi_{1,j}$ holding $R_{US,t+1}$, and hence $E_{j,t}$, fixed:

$$0 = f_{\psi_1} + f_s \frac{\partial s_{j,t}}{\partial \psi_{1,j}}, \qquad f_{\psi_1} = \frac{\partial f_j}{\partial \psi_{1,j}} = \frac{1}{s_{j,t}}.$$

Thus

$$\frac{\partial s_{j,t}}{\partial \psi_{1,j}}\Big|_R = -\frac{f_{\psi_1}}{f_s} = \frac{1/s_{j,t}}{\gamma_j V_{j,t} + \psi_{1,j}/s_{j,t}^2}.$$

□

A.4 The risk-neutral limit ($\gamma = 0$)

This section characterises the model when risk aversion is zero, the boundary of the admissible parameter space when γ is estimated. Throughout, the cell index j and time subscripts are suppressed where no ambiguity arises, so that γ in this subsection denotes a single cell's risk aversion γ_j, and R^F denotes the cell's benchmark return as in Appendix A.1.

Preferences. By Appendix A.1, for any $\gamma > 0$ the investor's problem is equivalent to maximising the certainty equivalent $\mu_W - \frac{\gamma}{2}\sigma_W^2 + h(s)$. As $\gamma \to 0$ the variance term vanishes and the objective converges to

$$\max_{s\in[0,1]} \mathbb{E}_t[r_{t+1}^{port}] + h(s) \;=\; \max_{s\in[0,1]} \underbrace{(E + \psi_0)}_{=:\,a}\, s \;+\; \psi_1 \log s \;+\; \mathbb{E}_t[R_{t+1}^F], \tag{18}$$

where $E = \mathbb{E}_t[R_{US,t+1} - R_{t+1}^F]$ and the last term is a constant.

Optimal portfolio share. The objective (18) is strictly concave on $(0, 1]$ if and only if $\psi_1 > 0$, since its second derivative is $-\psi_1/s^2$. Three cases arise.

(i) $\psi_1 > 0$ and $a < 0$. The first-order condition $a + \psi_1/s = 0$ has the unique solution

$$s^* = -\frac{\psi_1}{a} = \frac{\psi_1}{-(E + \psi_0)}, \tag{19}$$

which is interior whenever $\psi_1 < -a$; if $\psi_1 \geq -a$ the objective is increasing on $(0, 1]$ and $s^* = 1$.

(ii) $\psi_1 > 0$ and $a \geq 0$. The objective is strictly increasing on $(0, 1]$ and $s^* = 1$.

(iii) $\psi_1 = 0$. The objective is linear, and $s^* = 1$ if $a > 0$, $s^* = 0$ if $a < 0$, with any $s \in [0, 1]$ optimal if $a = 0$. This is the standard risk-neutral corner solution.

A risk-neutral investor holds an interior share of Treasuries only if their expected excess return net of the level convenience is negative, $E + \psi_0 < 0$: the concave term $\psi_1 \log s$ then makes the first units of Treasuries very valuable while the linear term penalises further accumulation.

The perfectly elastic limit. The step demand of case (iii) represents the limit case of $\psi_{1,j}$ driving elasticity in the risk-neutral limit. As $\gamma_j \to 0$ with $\psi_{1,j} = 0$, the denominator $D_{j,t} = \gamma_j V_{j,t}$ tends to zero, so cell j's demand becomes perfectly elastic at its indifference excess return, $E_{j,t} = -\psi_{0,j}$.

Derivatives of the optimal share. In the interior case, the cell-level demand-elasticity denominator becomes $D = \psi_1/s^2 > 0$, so Propositions 2–4 specialise to

$$\frac{\mathrm{d}s}{\mathrm{d}E} = \frac{\partial s}{\partial \psi_0}\Big|_R = \frac{s^2}{\psi_1}, \qquad \frac{\partial s}{\partial \psi_1}\Big|_R = \frac{s}{\psi_1}, \tag{20}$$

as can be verified directly from (19). Two features distinguish the risk-neutral demand from its risk-averse counterpart. First, the response of the share to the excess return is not bounded by $1/(\gamma V)$ but scales with s^2/ψ_1: cells with large Treasury shares or weak curvature are highly elastic, and the elasticity tends to infinity as $\psi_1 \to 0$, recovering the corner solutions of case (iii). Second, risk aversion and convenience curvature are substitutes in generating a finite demand elasticity: for any $\gamma > 0$ the elasticity $1/(\gamma V + \psi_1/s^2)$ can be matched by a risk-neutral investor with curvature $\tilde{\psi}_1 = \psi_1 + \gamma V s^2$. This substitutability explains why, in the first-order condition (4), a low estimate of γ_j shifts the burden of explaining a downward-sloping Treasury demand onto $\psi_{1,j}$, so that a boundary estimate of γ_j does not necessarily imply corner solutions, but rather a demand elasticity governed entirely by convenience curvature.

Equilibrium. With all cells interior, the aggregate demand slope and absorption shares of Section 2.4 become

$$\mathcal{A}_t = \sum_{j \in \mathcal{J}} \frac{W_{j,t}\, s_{j,t}^2}{\psi_{1,j}}, \qquad a_{j,t} = \frac{W_{j,t}\, s_{j,t}^2/\psi_{1,j}}{\mathcal{A}_t}, \tag{21}$$

and the equilibrium comparative statics follow by substitution:

$$\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}\psi_{0,j}} = -a_{j,t}, \qquad \frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}\psi_{1,j}} = -\frac{a_{j,t}}{s_{j,t}}, \qquad \frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}S_t} = \frac{1}{\mathcal{A}_t},$$

and Proposition 7 holds with $D_{k,t} = \psi_{1,k}/s_{k,t}^2$. The curvature of dollar demand is $\kappa_{j,t} = 2W_{j,t}s_{j,t}^3/\psi_{1,j}^2$, and the cross derivative of Proposition 10 reduces to

$$\mathcal{A}_t^{-2}\left[-W_{j,t}s_{j,t}^2/\psi_{1,j}^2 + a_{j,t}\mathcal{K}_t/s_{j,t}\right].$$

Its direct component is unambiguously negative, consistent with the necessary condition $\psi_{1,j}/s_{j,t}^2 > \gamma_j V_{j,t}$ being trivially met at $\gamma = 0$, so that in the risk-neutral model stronger curvature lowers the sensitivity of the equilibrium return to supply through the direct channel, the risk-aversion channel of Proposition 10 being absent, and only the general-equilibrium feedback $a_{j,t}\mathcal{K}_t/s_{j,t}$ can offset it.

Existence and uniqueness of the equilibrium still hold in the risk-neutral limit, with one qualification. Since each cell's share is capped at one, modelled dollar demand $\sum_j W_{j,t}s_{j,t}^*$ is bounded above by $\sum_j W_{j,t}$, and the market-clearing function G_t is non-decreasing rather than strictly increasing in $R_{US,t+1}$ over regions where all cells are at the corner. As $R_{US,t+1} \to -\infty$, $a \to -\infty$ for every cell and $s_{j,t}^* = \psi_{1,j}/|a| \to 0$, so $G_t \to H_t^{\text{Res}} - S_t < 0$; as $R_{US,t+1}$ increases, every cell eventually reaches $s_{j,t}^* = 1$ and $G_t \to \sum_j W_{j,t} + H_t^{\text{Res}} - S_t$. A solution exists if $\sum_j W_{j,t} + H_t^{\text{Res}} > S_t$, which is guaranteed whenever the observed portfolio shares are interior, since then $\sum_j W_{j,t} > \sum_j W_{j,t}s_{j,t} = S_t - H_t^{\text{Res}}$. At any such solution at least one cell is interior, so G_t is strictly increasing in a neighbourhood of the equilibrium and the solution is unique.

If any cell displays a step demand as described in case (iii) above, its absorption share tends to one and every other cell's to zero. If such a cell holds an interior share in equilibrium, it absorbs every marginal change in demand or supply at an unchanged return: the equilibrium return equals its indifference return, which its level parameter determines, the derivatives of the return with respect to other cells' parameters and to supply are zero, and other cells' preferences move only the quantity it holds.

A.5 Comparative statics of the equilibrium return

Throughout this subsection, define the aggregate dollar-demand slope

$$A(R_{US,t+1}) \equiv \sum_{j\in\mathcal{J}} \frac{W_{j,t}}{\gamma_j V_{j,t} + \psi_{1,j}/s_{j,t}^2} = \sum_{j\in\mathcal{J}} \frac{W_{j,t}}{D_{j,t}} = \mathcal{A}_t,$$

the aggregate modelled-demand precision sum at the equilibrium. The market-clearing condition $G_t(R_{US,t+1}) = \sum_j W_{j,t}s_{j,t}^* + H_t^{\text{Res}} - S_t = 0$ defines $R_{US,t+1}^*$ implicitly. With $H_{j,t}^* = W_{j,t}s_{j,t}^*$ and $E_{j,t} = R_{US,t+1} - \mathbb{E}_t[R_{j,t+1}^F]$ for each cell, the partial derivative of G_t with respect to $R_{US,t+1}$ is

$$\frac{\partial G_t}{\partial R_{US,t+1}} = \sum_j W_{j,t}\frac{\partial s_{j,t}^*}{\partial E_{j,t}} = \sum_j \frac{W_{j,t}}{D_{j,t}} = \mathcal{A}_t > 0,$$

which delivers the strict monotonicity used in all comparative statics below. Throughout, $\mathrm{d}(\cdot)/\mathrm{d}\theta$ denotes the total derivative of an equilibrium object with respect to a parameter θ, obtained by differentiating $G_t(R_{US,t+1}^*(\theta);\theta) = 0$ in equilibrium, while $\partial(\cdot)/\partial\theta|_R$ denotes the partial derivative at a fixed return. For the return itself, the implicit function theorem gives

$$\frac{\mathrm{d}R_{US,t+1}^*}{\mathrm{d}\theta} = -\frac{\partial G_t/\partial\theta|_R}{\partial G_t/\partial R_{US,t+1}} = -\frac{\partial G_t/\partial\theta|_R}{\mathcal{A}_t},$$

where the numerator is the excess dollar demand created by the parameter change at the initial return and the denominator sums the re-optimisation of every cell in $\mathcal{J}$. The response of cells other than the one whose parameter changes is therefore embedded in $\mathcal{A}_t$, and the derivative is a general-equilibrium object. Finally, recall the absorption share $a_{j,t} \equiv (W_{j,t}/D_{j,t})/\mathcal{A}_t$, with $\sum_j a_{j,t} = 1$.

A.5.1 Proposition 5

Proof. By Proposition 3, at fixed $R_{US,t+1}$ (and hence at fixed $E_{j,t}$), $\partial s^*_{j,t}/\partial \psi_{0,j}|_R = 1/D_{j,t}$, while cells $j' \neq j$ are unaffected at a fixed return. Hence

$$\frac{\partial G_t}{\partial \psi_{0,j}}\Big|_R = \frac{W_{j,t}}{D_{j,t}}.$$

By the implicit function theorem,

$$\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}\psi_{0,j}} = -\frac{\partial G_t/\partial \psi_{0,j}|_R}{\partial G_t/\partial R_{US,t+1}} = -\frac{W_{j,t}/D_{j,t}}{\mathcal{A}_t} = -a_{j,t}.$$

The derivative is total: the denominator $\mathcal{A}_t = \sum_{k\in\mathcal{J}} W_{k,t}/D_{k,t}$ collects the response of every cell's dollar demand to the induced change in the return, so the re-optimisation of cells $k \neq j$ is fully accounted for. Since $\mathcal{A}_t > 0$ and $W_{j,t} > 0$, if $D_{j,t} > 0$ for all j then $\mathrm{d}R^*_{US,t+1}/\mathrm{d}\psi_{0,j} \leq 0$. □

A.5.2 Proposition 6

Proof. By Proposition 4, at fixed $R_{US,t+1}$, $\partial s^*_{j,t}/\partial \psi_{1,j}|_R = (1/s_{j,t})/D_{j,t}$, while cells $j' \neq j$ are unaffected at a fixed return. Hence

$$\frac{\partial G_t}{\partial \psi_{1,j}}\Big|_R = \frac{W_{j,t}}{s_{j,t}\, D_{j,t}}.$$

By the implicit function theorem,

$$\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}\psi_{1,j}} = -\frac{\partial G_t/\partial \psi_{1,j}|_R}{\partial G_t/\partial R_{US,t+1}} = -\frac{W_{j,t}/(s_{j,t}\, D_{j,t})}{\mathcal{A}_t} = -\frac{a_{j,t}}{s_{j,t}}.$$

As in the proof of Proposition 5, the derivative is total because $\mathcal{A}_t$ embeds the return response of all cells. Since $\mathcal{A}_t > 0$ and $s_{j,t} > 0$, if $D_{j,t} > 0$ for all j then $\mathrm{d}R^*_{US,t+1}/\mathrm{d}\psi_{1,j} \leq 0$. □

A.6 Absorption of convenience shocks and supply sensitivity

This subsection characterises two further equilibrium objects of the model of Section 2: how a change in one cell's Treasury preferences is redistributed across all holders, and how Treasury preferences shape the sensitivity of the equilibrium return to Treasury supply. It uses the demand-slope denominator $D_{j,t}$, the aggregate slope $\mathcal{A}_t$ and the absorption shares $a_{j,t}$ defined in Section 2.4, and the same regularity condition $D_{j,t} > 0$ for all j. Proofs are in Appendix A.7. Appendix E reports the corresponding counterfactual experiments.

Absorption of convenience shocks across holders. Propositions 5–6 describe the price response to a change in one cell's convenience preferences. The same shock also redistributes Treasury holdings across all cells: the fall in the equilibrium return crowds out part of the demand of every other holder, and part of the additional demand of the shocked cell itself. The following proposition characterises this quantity response.

Proposition 7. *Let $k \in \mathcal{J}$ denote an arbitrary cell and $j \in \mathcal{J}$ the cell whose convenience parameters change. The total derivatives of the equilibrium portfolio shares with respect to the level parameter $\psi_{0,j}$ are*

$$\frac{\mathrm{d}s_{k,t}}{\mathrm{d}\psi_{0,j}} = \begin{cases} \dfrac{1}{D_{j,t}}\left(1 - a_{j,t}\right) & k = j, \\ -\dfrac{1}{D_{k,t}}\, a_{j,t} & k \neq j, \end{cases} \tag{22}$$

and with respect to the curvature parameter $\psi_{1,j}$

$$\frac{\mathrm{d}s_{k,t}}{\mathrm{d}\psi_{1,j}} = \begin{cases} \dfrac{1}{s_{j,t}D_{j,t}}\left(1 - a_{j,t}\right) & k = j, \\ -\dfrac{1}{D_{k,t}}\,\dfrac{a_{j,t}}{s_{j,t}} & k \neq j. \end{cases} \tag{23}$$

In dollar holdings $H^*_{k,t} = W_{k,t}s^*_{k,t}$, both shocks are absorbed according to the absorption shares:

$$\begin{aligned} \frac{\mathrm{d}H^*_{k,t}}{\mathrm{d}\psi_{n,j}} &= \Delta^n_{j,t}\left(\mathbf{1}\{k = j\} - a_{k,t}\right), \qquad n \in \{0, 1\}, \\ \Delta^0_{j,t} &\equiv \frac{W_{j,t}}{D_{j,t}}, \qquad \Delta^1_{j,t} \equiv \frac{W_{j,t}}{s_{j,t}D_{j,t}}, \end{aligned} \tag{24}$$

where $\Delta^n_{j,t}$ is the fixed-return excess dollar demand generated by the shock. Consequently $\sum_{k\in\mathcal{J}} \mathrm{d}H^*_{k,t}/\mathrm{d}\psi_{n,j} = 0$: market clearing is preserved. If $D_{k,t} > 0$ for all k, the shocked cell's share rises and every other cell's share falls.

Proof. In Appendix A.7.1. □

The proposition delivers a clean decomposition of who absorbs a convenience shock. At the initial return, a marginal increase in cell j's convenience preferences generates excess demand $\Delta^n_{j,t}$. The equilibrium return falls by $\Delta^n_{j,t}/\mathcal{A}_t$ and every cell k, including j, reduces its dollar demand by the fraction $a_{k,t}$ of that excess demand. Cell j therefore retains only a fraction $1 - a_{j,t} = \sum_{k\neq j} a_{k,t}$ of the increase in holdings it would have chosen at the initial return; the remainder is captured by the other holders in proportion to their absorption shares.

Two features of the result are worth stressing. First, the absorption shares $a_{k,t}$ are properties of the receiving cell alone and are common to both parameters and to all shocked cells: the same distribution $\{a_{k,t}\}$ governs the redistribution of any convenience shock, and only the scale $\Delta^n_{j,t}$ depends on its origin. Second, the fixed-return partial derivatives of Propositions 3–4 overstate the equilibrium response of the shocked cell's own share by the factor $1/(1 - a_{j,t})$, with the discrepancy increasing in the cell's weight in aggregate demand sensitivity. For a cell with small absorption share, general-equilibrium feedback on its own portfolio is negligible and nearly the full quantity adjustment falls on other holders; for a dominant cell, a large part of the shock is dissipated through the return rather than through its holdings. This distinction matters for the interpretation of the counterfactual experiments of Section 6 and Appendix E, where the relevant question is not only how much a sector's convenience preferences depress the return, but also which holders would step in to absorb the Treasuries that sector releases when those preferences weaken.

Treasury supply and the equilibrium return. The presence of convenience preferences affects not only the equilibrium interest rate itself, but also its relationship with debt supply. Due to the different roles played by the level shifter and curvature parameters, the net effect is not an unambiguous reduction in the sensitivity of excess returns to debt supply, instead depending on the interplay of demand level and elasticity effects.

Proposition 8. *The total derivative of the equilibrium U.S. Treasury return with respect to U.S. Treasury supply S_t, holding the level of dollar holdings of the residual bucket H_t^{Res} fixed as S_t varies is*

$$\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}S_t} = \frac{1}{\mathcal{A}_t}. \tag{25}$$

If $D_{j,t} > 0$ for all j, then $\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}S_t} \geq 0$.

Proof. In Appendix A.7.2. □

As in the standard mean-variance model, an increase in debt supply raises the equilibrium return to compensate risk-averse investors. Treasury preferences introduce an additional rationale: the diminishing marginal utility of Treasury convenience. As for the sensitivity of portfolio shares to excess returns, the partial effect of the curvature parameter $\psi_{1,j}$ plays the same role as risk aversion: it *increases* the responsiveness of returns to debt supply. To investigate the total effect of convenience preferences on the sensitivity of returns to supply, I now turn to second derivatives.

Treasury preferences and return sensitivity to supply. The supply sensitivity $\mathrm{d}R^*_{US,t+1}/\mathrm{d}S_t = 1/\mathcal{A}_t$ depends on the convenience parameters through two channels. The first is direct: a change in $\psi_{n,j}$ alters the elasticity of the shocked cell, $1/D_{j,t}$, both because $\psi_{1,j}$ enters $D_{j,t}$ and because the induced change in $s_{j,t}$ moves the curvature term $\psi_{1,j}/s^2_{j,t}$. The second is a general-equilibrium feedback: by Propositions 5–7, the shock lowers the equilibrium return and reduces the portfolio share of every other cell, which raises $D_{k,t}$ wherever $\psi_{1,k} > 0$ and makes aggregate demand less elastic. Two objects summarise these channels. Let

$$\kappa_{j,t} \equiv \frac{\mathrm{d}^2 H^*_{j,t}}{\mathrm{d}R^2_{US,t+1}} = \frac{2W_{j,t}\psi_{1,j}}{D^3_{j,t}\, s^3_{j,t}} \geq 0 \qquad \text{and}$$

$$\mathcal{K}_t \equiv \sum_{k\in\mathcal{J}} \kappa_{k,t} = \frac{\mathrm{d}\mathcal{A}_t}{\mathrm{d}R_{US,t+1}} \geq 0.$$

The first is the curvature of cell j's demand in dollar terms: the derivative, with respect to the U.S. Treasury return, of the cell's demand slope $W_{j,t}/D_{j,t} = \mathrm{d}H^*_{j,t}/\mathrm{d}R_{US,t+1}$, or equivalently the second derivative of its equilibrium dollar holdings. It measures how fast the cell's demand becomes more elastic as a higher return raises its share and lowers $\psi_{1,j}/s^2_{j,t}$; it is zero for cells with linear convenience preferences. The second is the aggregate curvature of modelled Treasury demand, the derivative of the aggregate slope $\mathcal{A}_t$ with respect to the return. The following two propositions give the total second cross derivatives of the return sensitivity to debt supply with respect to convenience parameters in equilibrium.

Proposition 9. *The total derivative of the return sensitivity to supply with respect to the level parameter $\psi_{0,j}$ is*

$$\frac{\mathrm{d}}{\mathrm{d}\psi_{0,j}}\left(\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}S_t}\right) = -\frac{1}{\mathcal{A}_t^2}\left(\kappa_{j,t} - a_{j,t}\,\mathcal{K}_t\right). \tag{26}$$

If $\mathcal{K}_t > 0$, the derivative is negative if and only if $\kappa_{j,t}/\mathcal{K}_t > a_{j,t}$.

Proof. In Appendix A.7.3. □

The two terms in (26) represent the two channels. The first term, $-\kappa_{j,t}/\mathcal{A}_t^2$, captures the direct effect and is negative whenever $\psi_{1,j} > 0$: the greater absorption capacity implied by $\psi_{0,j}$ dampens the reaction of returns to additional supply, but only for cells that derive increasing and marginally diminishing utility from holding Treasuries, as $\psi_{0,j}$ does not affect demand sensitivities otherwise. The second term, $a_{j,t}\mathcal{K}_t/\mathcal{A}_t^2$, captures the general-equilibrium effect and works in the opposite direction: because the shock lowers the return, it erodes the elasticity of every cell with concave preferences, so the market as a whole becomes marginally less able to absorb supply through quantities. The net sign is therefore an empirical matter. A cell's convenience level dampens supply sensitivity only if its share of aggregate demand curvature, $\kappa_{j,t}/\mathcal{K}_t$, exceeds its absorption share $a_{j,t}$, that is if the cell is relatively more important for how elasticity changes with the return than for the level of elasticity itself. For cells with small absorption shares the feedback is negligible and the direct effect prevails; for dominant cells it can reverse the sign.

Proposition 10. *The total derivative of the return sensitivity to supply with respect to the curvature parameter $\psi_{1,j}$ is*

$$\frac{\mathrm{d}}{\mathrm{d}\psi_{1,j}}\left(\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}S_t}\right) \;=\; \frac{1}{\mathcal{A}_t^2}\left[\frac{W_{j,t}}{D_{j,t}^2 s_{j,t}^2} - \frac{1}{s_{j,t}}\left(\kappa_{j,t} - a_{j,t}\,\mathcal{K}_t\right)\right]. \tag{27}$$

The derivative is negative if and only if

$$\frac{W_{j,t}}{D_{j,t}^2\, s_{j,t}}\left(\frac{2\psi_{1,j}}{D_{j,t}\, s_{j,t}^2} - 1\right) \;>\; a_{j,t}\,\mathcal{K}_t,$$

which requires $\psi_{1,j}/s_{j,t}^2 > \gamma_j V_{j,t}$.

Proof. In Appendix A.7.4. □

The curvature parameter adds a third force. Its first term in (27) is the direct effect on elasticity: a higher $\psi_{1,j}$ raises $D_{j,t}$ and, like risk aversion, *amplifies* the reaction of returns to supply. The second term is the share effect already at work for the level parameter: a higher $\psi_{1,j}$ raises $s_{j,t}$, lowers $\psi_{1,j}/s_{j,t}^2$ and dampens the reaction of returns. The direct component, obtained by setting $\mathcal{K}_t = 0$, is negative precisely when the curvature of Treasury preferences, $\psi_{1,j}/s_{j,t}^2$, exceeds variance-weighted risk aversion, $\gamma_j V_{j,t}$: if curvature prevails, investors accept lower returns for any level of debt supply and the dampening effect of convenience, mediated by the level shifter, dominates; if risk aversion prevails, the reduced absorption capacity forces the equilibrium to adjust to additional supply through prices rather than quantities, raising returns. The general-equilibrium feedback, $a_{j,t}\mathcal{K}_t/(s_{j,t}\mathcal{A}_t^2)$, adds to the amplifying side, so the condition for a dampening effect is stricter in equilibrium than the comparison of curvature and risk aversion alone. The relative strength of these forces is an empirical question, which Appendix E.2 answers at the estimated parameters.

A.7 Proofs

This subsection proves the propositions stated in Appendix A.6. The notation and the implicit-function argument are those of the proofs of Propositions 5 and 6 above.

A.7.1 Proposition 7

Proof. Fix the shocked cell j and let $n \in \{0,1\}$. For any cell $k \in \mathcal{J}$, the equilibrium share is $s^*_{k,t} = s^*_{k,t}\big(E_{k,t}(R^*_{US,t+1}); \psi_{0,k}, \psi_{1,k}\big)$ with $\partial E_{k,t}/\partial R_{US,t+1} = 1$. By the chain rule, the total derivative in equilibrium is

$$\frac{\mathrm{d}s_{k,t}}{\mathrm{d}\psi_{n,j}} \;=\; \frac{\partial s_{k,t}}{\partial \psi_{n,j}}\Big|_R \;+\; \frac{\mathrm{d}s_{k,t}}{\mathrm{d}E_{k,t}}\,\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}\psi_{n,j}}, \tag{28}$$

where the first term is non-zero only for $k = j$.

Level parameter ($n = 0$). By Propositions 2 and 3, $\mathrm{d}s_{k,t}/\mathrm{d}E_{k,t} = 1/D_{k,t}$ and $\partial s_{j,t}/\partial \psi_{0,j}|_R = 1/D_{j,t}$; by Proposition 5, $\mathrm{d}R^*_{US,t+1}/\mathrm{d}\psi_{0,j} = -a_{j,t}$. Substituting into (28),

$$\frac{\mathrm{d}s_{k,t}}{\mathrm{d}\psi_{0,j}} \;=\; \frac{\mathbf{1}\{k=j\}}{D_{j,t}} - \frac{a_{j,t}}{D_{k,t}},$$

which gives $\mathrm{d}s_{j,t}/\mathrm{d}\psi_{0,j} = (1-a_{j,t})/D_{j,t}$ for $k = j$ and $\mathrm{d}s_{k,t}/\mathrm{d}\psi_{0,j} = -a_{j,t}/D_{k,t}$ for $k \neq j$, as in (22).

Curvature parameter ($n = 1$). By Proposition 4, $\partial s_{j,t}/\partial \psi_{1,j}|_R = 1/(s_{j,t}D_{j,t})$, and by Proposition 6, $\mathrm{d}R^*_{US,t+1}/\mathrm{d}\psi_{1,j} = -a_{j,t}/s_{j,t}$. Substituting into (28),

$$\frac{\mathrm{d}s_{k,t}}{\mathrm{d}\psi_{1,j}} \;=\; \frac{\mathbf{1}\{k=j\}}{s_{j,t}D_{j,t}} - \frac{a_{j,t}}{s_{j,t}D_{k,t}},$$

which is (23).

Dollar holdings and market clearing. Multiplying by $W_{k,t}$, which is exogenous, and using $a_{k,t} = (W_{k,t}/D_{k,t})/\mathcal{A}_t$ and $W_{k,t} = W_{j,t}$ for $\mathbf{1}\{k=j\} = 1$,

$$\frac{\mathrm{d}H^*_{k,t}}{\mathrm{d}\psi_{0,j}} \;=\; \frac{W_{j,t}}{D_{j,t}}\,\mathbf{1}\{k=j\} - \frac{W_{k,t}}{D_{k,t}}\,\frac{W_{j,t}/D_{j,t}}{\mathcal{A}_t} \;=\; \frac{W_{j,t}}{D_{j,t}}\big(\mathbf{1}\{k=j\} - a_{k,t}\big) \;=\; \Delta^0_{j,t}\big(\mathbf{1}\{k=j\} - a_{k,t}\big),$$

and analogously $\mathrm{d}H^*_{k,t}/\mathrm{d}\psi_{1,j} = \Delta^1_{j,t}(\mathbf{1}\{k=j\} - a_{k,t})$ with $\Delta^1_{j,t} = W_{j,t}/(s_{j,t}D_{j,t})$, which is (24). Summing over $k \in \mathcal{J}$ and using $\sum_k a_{k,t} = 1$,

$$\sum_{k\in\mathcal{J}} \frac{\mathrm{d}H^*_{k,t}}{\mathrm{d}\psi_{n,j}} \;=\; \Delta^n_{j,t}\Big(1 - \sum_{k\in\mathcal{J}} a_{k,t}\Big) \;=\; 0,$$

so total modelled holdings are unchanged, consistent with market clearing (9) at fixed S_t and inelastic H^{Res}_t.

Signs. If $D_{k,t} > 0$ for all k, then $\mathcal{A}_t > 0$ and $a_{k,t} \in (0,1)$ for every k, so $1 - a_{j,t} > 0$. Hence $\mathrm{d}s_{j,t}/\mathrm{d}\psi_{n,j} > 0$ and $\mathrm{d}s_{k,t}/\mathrm{d}\psi_{n,j} < 0$ for $k \neq j$, $n \in \{0,1\}$. □

A.7.2 Proposition 8

Proof. Write market clearing in dollar-holdings form, $G_t = \sum_{j\in\mathcal{J}} W_{j,t}s^*_{j,t} + H^{\mathrm{Res}}_t - S_t$. The derivative captures the marginal cost of issuing one additional dollar of Treasuries that the modelled investors must absorb, with the level of the residual bucket H^{Res}_t held fixed as S_t varies. At fixed $R_{US,t+1}$ and fixed $\{s^*_{j,t}\}_j$, and with H^{Res}_t constant, $\partial G_t/\partial S_t|_R = -1$. By the implicit function theorem,

$$\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}S_t} \;=\; -\frac{\partial G_t/\partial S_t|_R}{\partial G_t/\partial R_{US,t+1}} \;=\; -\frac{-1}{\mathcal{A}_t} \;=\; \frac{1}{\mathcal{A}_t} \;>\; 0,$$

which is the form stated in Proposition 8. The derivative is total: the additional dollar of supply is absorbed by all cells in proportion to their absorption shares $a_{j,t}$, and the return rises by the amount required for this reallocation. Larger Treasury supply raises the equilibrium return as long as the regularity condition $D_{j,t} > 0$ holds for all j. □

A.7.3 Proposition 9

Proof. By Proposition 8, $\mathrm{d}R^*_{US,t+1}/\mathrm{d}S_t = 1/\mathcal{A}_t$ with $\mathcal{A}_t = \sum_{k\in\mathcal{J}} W_{k,t}/D_{k,t}$ and $D_{k,t} = \gamma_k V_{k,t} + \psi_{1,k}/s^2_{k,t}$, where every $s_{k,t}$ is evaluated at the equilibrium. Differentiating in equilibrium, with $W_{k,t}$ and $V_{k,t}$ exogenous,

$$\frac{\mathrm{d}}{\mathrm{d}\psi_{0,j}}\left(\frac{1}{\mathcal{A}_t}\right) = -\frac{1}{\mathcal{A}_t^2}\frac{\mathrm{d}\mathcal{A}_t}{\mathrm{d}\psi_{0,j}}, \qquad \frac{\mathrm{d}\mathcal{A}_t}{\mathrm{d}\psi_{0,j}} = -\sum_{k\in\mathcal{J}}\frac{W_{k,t}}{D^2_{k,t}}\frac{\mathrm{d}D_{k,t}}{\mathrm{d}\psi_{0,j}}. \tag{29}$$

No $D_{k,t}$ depends on $\psi_{0,j}$ directly; each depends on it only through the equilibrium share $s_{k,t}$, so

$$\frac{\mathrm{d}D_{k,t}}{\mathrm{d}\psi_{0,j}} = -\frac{2\psi_{1,k}}{s^3_{k,t}}\frac{\mathrm{d}s_{k,t}}{\mathrm{d}\psi_{0,j}} \quad\Longrightarrow\quad \frac{\mathrm{d}\mathcal{A}_t}{\mathrm{d}\psi_{0,j}} = \sum_{k\in\mathcal{J}}\frac{2W_{k,t}\psi_{1,k}}{D^2_{k,t}s^3_{k,t}}\frac{\mathrm{d}s_{k,t}}{\mathrm{d}\psi_{0,j}} = \sum_{k\in\mathcal{J}}\kappa_{k,t}D_{k,t}\frac{\mathrm{d}s_{k,t}}{\mathrm{d}\psi_{0,j}},$$

using $\kappa_{k,t} = 2W_{k,t}\psi_{1,k}/(D^3_{k,t}s^3_{k,t})$. The total share responses are given by Proposition 7: $\mathrm{d}s_{k,t}/\mathrm{d}\psi_{0,j} = \mathbf{1}\{k=j\}/D_{j,t} - a_{j,t}/D_{k,t}$. Substituting,

$$\frac{\mathrm{d}\mathcal{A}_t}{\mathrm{d}\psi_{0,j}} = \sum_{k\in\mathcal{J}}\kappa_{k,t}D_{k,t}\left(\frac{\mathbf{1}\{k=j\}}{D_{j,t}} - \frac{a_{j,t}}{D_{k,t}}\right) = \kappa_{j,t} - a_{j,t}\sum_{k\in\mathcal{J}}\kappa_{k,t} = \kappa_{j,t} - a_{j,t}\,\mathcal{K}_t.$$

The first term is the direct effect, arising from the shocked cell's own share response at the initial return; the second is the general-equilibrium feedback, arising from the return-induced share responses of all cells. Substituting into (29),

$$\frac{\mathrm{d}}{\mathrm{d}\psi_{0,j}}\left(\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}S_t}\right) = -\frac{1}{\mathcal{A}_t^2}\left(\kappa_{j,t} - a_{j,t}\,\mathcal{K}_t\right),$$

which is (26). Since $\mathcal{A}_t^2 > 0$, the expression is negative if and only if $\kappa_{j,t} > a_{j,t}\mathcal{K}_t$, i.e. $\kappa_{j,t}/\mathcal{K}_t > a_{j,t}$ whenever $\mathcal{K}_t > 0$. Finally, $\kappa_{k,t} = \mathrm{d}(W_{k,t}/D_{k,t})/\mathrm{d}R_{US,t+1}$ and $\mathcal{K}_t = \mathrm{d}\mathcal{A}_t/\mathrm{d}R_{US,t+1}$ follow from the same computation with $\mathrm{d}s_{k,t}/\mathrm{d}R_{US,t+1} = 1/D_{k,t}$ (Proposition 2) in place of $\mathrm{d}s_{k,t}/\mathrm{d}\psi_{0,j}$: $\mathrm{d}(W_{k,t}/D_{k,t})/\mathrm{d}R_{US,t+1} = -(W_{k,t}/D^2_{k,t})(-2\psi_{1,k}/s^3_{k,t})(1/D_{k,t}) = \kappa_{k,t}$; and $\mathcal{K}_t \geq 0$ from $\psi_{1,k} \geq 0$, $W_{k,t} > 0$, $D_{k,t} > 0$ and $s_{k,t} > 0$. □

A.7.4 Proposition 10

Proof. Proceed as in the proof of Proposition 9, with $\psi_{1,j}$ in place of $\psi_{0,j}$ in (29). The curvature parameter now enters $D_{j,t}$ directly as well as through the equilibrium shares:

$$\frac{\mathrm{d}D_{k,t}}{\mathrm{d}\psi_{1,j}} = \frac{\mathbf{1}\{k=j\}}{s^2_{j,t}} - \frac{2\psi_{1,k}}{s^3_{k,t}}\frac{\mathrm{d}s_{k,t}}{\mathrm{d}\psi_{1,j}},$$

so that

$$\frac{\mathrm{d}\mathcal{A}_t}{\mathrm{d}\psi_{1,j}} = -\frac{W_{j,t}}{D^2_{j,t}s^2_{j,t}} + \sum_{k\in\mathcal{J}}\kappa_{k,t}D_{k,t}\frac{\mathrm{d}s_{k,t}}{\mathrm{d}\psi_{1,j}}.$$

By Proposition 7, $\mathrm{d}s_{k,t}/\mathrm{d}\psi_{1,j} = \big(\mathbf{1}\{k=j\}/D_{j,t} - a_{j,t}/D_{k,t}\big)/s_{j,t}$. Hence

$$\frac{\mathrm{d}\mathcal{A}_t}{\mathrm{d}\psi_{1,j}} \;=\; -\frac{W_{j,t}}{D_{j,t}^2 s_{j,t}^2} + \frac{1}{s_{j,t}}\big(\kappa_{j,t} - a_{j,t}\,\mathcal{K}_t\big),$$

and therefore

$$\frac{\mathrm{d}}{\mathrm{d}\psi_{1,j}}\left(\frac{\mathrm{d}R^*_{US,t+1}}{\mathrm{d}S_t}\right) \;=\; -\frac{1}{\mathcal{A}_t^2}\,\frac{\mathrm{d}\mathcal{A}_t}{\mathrm{d}\psi_{1,j}} \;=\; \frac{1}{\mathcal{A}_t^2}\left[\frac{W_{j,t}}{D_{j,t}^2 s_{j,t}^2} - \frac{1}{s_{j,t}}\big(\kappa_{j,t} - a_{j,t}\,\mathcal{K}_t\big)\right],$$

which is (27). The first term is the direct effect of $\psi_{1,j}$ on the shocked cell's elasticity, the second the own-share effect at the initial return, and the third the general-equilibrium feedback.

Sign. Since $\mathcal{A}_t^2 > 0$, the derivative is negative if and only if $\kappa_{j,t}/s_{j,t} - W_{j,t}/(D_{j,t}^2 s_{j,t}^2) > a_{j,t}\mathcal{K}_t/s_{j,t}$. Using $\kappa_{j,t}/s_{j,t} = 2W_{j,t}\psi_{1,j}/(D_{j,t}^3 s_{j,t}^4)$, the left-hand side equals

$$\frac{W_{j,t}}{D_{j,t}^2 s_{j,t}^2}\left(\frac{2\psi_{1,j}}{D_{j,t}\,s_{j,t}^2} - 1\right),$$

so, multiplying through by $s_{j,t} > 0$, the derivative is negative if and only if

$$\frac{W_{j,t}}{D_{j,t}^2\,s_{j,t}}\left(\frac{2\psi_{1,j}}{D_{j,t}\,s_{j,t}^2} - 1\right) \;>\; a_{j,t}\,\mathcal{K}_t.$$

Because the right-hand side is non-negative, a necessary condition is that the bracket be positive, i.e. $D_{j,t} < 2\psi_{1,j}/s_{j,t}^2$, which is equivalent to $\psi_{1,j}/s_{j,t}^2 > \gamma_j V_{j,t}$. When $\mathcal{K}_t = 0$ this condition is also sufficient. □

B Dataset construction

This appendix documents the construction of the dataset summarised in Section 3. Appendix B.1 sets out the restatements applied to the country-level total Treasury holdings. Appendix B.2 describes, jurisdiction by jurisdiction, how those amounts are disaggregated into investor sectors, together with every maintained assumption the construction requires. Appendix B.3 describes the denominator of the portfolio share. Appendix B.4 collects the coverage caveats, and Appendix B.5 lists the provenance of all data used in the paper.

B.1 Country holdings and financial-centre restatements

The TIC system records foreign holdings of U.S. Treasury securities by the country of the immediate holder. Four mechanisms drive a wedge between that country and the country of the ultimate investor: round-tripping of U.S.-managed vehicles through offshore conduits; custody of sovereign reserves at the two European international central securities depositories, booked to Belgium and Luxembourg; investment-fund domiciliation in Luxembourg and Ireland on behalf of non-euro-area beneficial owners; and general custodial mediation through Hong Kong, Switzerland and the United Kingdom. I apply a sequence of restatements before any sector split. All of them operate on the TIC country totals; no adjustment is made to the resident-side sector sources except where stated explicitly. Figure A1 visualises the resulting flows for the final quarter of the sample, showing that the euro area is the largest donor and China the largest receiver.

Exclusion of round-trip offshore conduits. Four jurisdictions are excluded entirely because their TIC totals predominantly reflect U.S.-managed fund vehicles rather than genuine foreign portfolio investment: the Cayman Islands, which hosts U.S.-managed hedge fund and private equity vehicles; Bermuda, which hosts U.S.-managed reinsurance and hedge fund platforms; the British Virgin Islands, whose reported series shows abrupt jumps inconsistent with genuine foreign portfolio investment; and the Bahamas, whose offshore private banking positions display a volatile round-trip profile. None of these jurisdictions reports bilateral positions to the Bank for International Settlements (BIS), so the bilateral-banking allocation of Alstadsæter et al. (2018) used elsewhere in this appendix cannot be applied to them. While their holdings presumably reflect U.S. ultimate ownership to a large extent, they are nonetheless excluded without redistribution to any dataset country as a conservative assumption.

Luxembourg and Ireland: rest-of-world component of fund holdings. Luxembourg and Ireland are the domiciles of European investment funds whose beneficial owners include non-euro-area investors. Counting the rest-of-world share of those funds' positions inside the euro-area total would overstate genuine euro-area demand for Treasuries. In principle, the same correction would apply to funds domiciled in other jurisdictions that cater to foreign investors, such as the United Kingdom, but no public source provides a breakdown of fund share ownership by nationality for these jurisdictions. For the euro area, the look-through dataset of Beck et al. (2026) allows me to compute the rest-of-world share of debt holdings in Luxembourg- and Ireland-domiciled funds,

$$\alpha_t \;=\; \frac{\text{rest-of-world debt holdings in LUX and IRL funds}_t}{\text{total debt holdings in LUX and IRL funds}_t},$$

and subtract $\alpha_t \times (H^{\text{TIC}}_{\text{LUX},t} + H^{\text{TIC}}_{\text{IRL},t})$ from euro-area TIC total holdings. The removed component is dropped from the dataset rather than reallocated, because the country-level identities of the ultimate owners cannot be recovered from public data.

Luxembourg and Ireland: euro-area NBFI sector correction. The same look-through logic is carried through to the euro-area NBFI cell, which contains Treasury positions held by Luxembourg- and Ireland-domiciled non-money-market investment funds on behalf of owners resident both inside and outside the euro area. The non-euro-area component is netted out by scaling the NBFI cell,

$$H^{\text{NBFI,adj}}_{\text{EA},t} \;=\; \max\!\big(0,\; H^{\text{NBFI}}_{\text{EA},t}\,(1 - \alpha_t \beta_t)\big),$$

where α_t is the rest-of-world beneficial-owner share defined above and β_t is the share of the euro-area non-money-market investment-fund debt book domiciled in Luxembourg and Ireland. The correction is applied to the Treasury-holdings leg of the NBFI cell only, not to the denominator of the euro-area sector shares, and it is not mirrored by any further subtraction from the euro-area total: the rest-of-world share of Luxembourg and Irish positions has already been removed at the country level by the preceding restatement, and deducting it a second time would double-count. The adjustment leaves banks and HH/NFC untouched.

The share β_t is constructed from the ECB's sector-level securities holdings statistics as the Luxembourg-plus-Ireland share of euro-area non-money-market investment-fund holdings of debt securities. That source begins in 2021; for earlier quarters β_t is held at its sample mean, which is preferred to holding it at either endpoint because the series is close to flat over its observed range. Three scope departures are imposed by data availability, each relying on a maintained assumption. First, on instrument scope, β_t is built from holdings of all debt securities rather

than sovereign debt specifically; the maintained assumption is that the Luxembourg and Irish share of euro-area fund debt holdings is representative of their share of euro-area fund holdings of sovereign debt. Second, on holder composition, β_t isolates non-money-market investment funds rather than the full NBFI aggregate, which also includes insurance corporations, pension funds and other financial intermediaries; the maintained assumption is that Luxembourg and Irish NBFI activity is dominated by investment funds, consistent with those two countries hosting the overwhelming majority of euro-area fund assets under management. Third, on counterpart geography, β_t is computed on an all-counterparty basis rather than against U.S. issuers specifically, because the source does not publish that breakdown; the maintained assumption is that the Luxembourg and Irish share within all-counterparty debt is representative of their share within U.S. Treasuries. The correction removes USD 63.74 billion per quarter on average, or 20.2% of the euro-area NBFI cell, reflecting a substantial foreign component in the holdings of euro-area investment fund shares.

Custodial gaps and their reallocation. Two euro-area jurisdictions host international central securities depositories whose books are dominated by non-resident, largely official, clients: Belgium, home to Euroclear, and Luxembourg, home to Clearstream. Belgium's reported TIC total, approximately USD 477 billion in Q4 2025, is inflated accordingly. For Belgium I substitute the ultimate-ownership bilateral position in U.S. debt securities reported to the IMF for the TIC figure when constructing the euro-area total, and record the difference, clipped at zero, as the Euroclear custodial gap. For Luxembourg the analogous non-custodial fraction is estimated as the ratio of its ultimate-ownership bilateral position in U.S. debt securities to its reported holdings of U.S. debt; that fraction scales the Luxembourg total before the rest-of-world correction of the preceding paragraph is applied, and the complementary amount is recorded as the Clearstream custodial gap.

The two gaps are pooled and reallocated to the non-euro-area dataset countries in proportion to each country's share of total official reserve assets, computed over the universe of individually reporting economies outside the euro area and the United States. Reserve shares are a more suitable allocation key than bank-deposit shares because the clientele of both depositories is predominantly official. The dataset countries are a strict subset of that universe, so the reallocated shares sum to strictly less than one by construction: the part of the pool attributable to economies that are not dataset countries is dropped.

These restatements are applied only on the TIC side. The reason is a reporting asymmetry: TIC records positions by the location of the immediate holder, so a security custodied at Euroclear Belgium for a non-euro-area official investor is booked to Belgium, whereas the ECB's securities holdings statistics follow the look-through standard of sections 6.38–6.39 of the *Handbook on Securities Statistics* (Bank for International Settlements et al., 2015) and attribute custodied positions to the beneficial owner.

A potential limitation of this procedure is an overcorrection due to the different treatment of reserve holdings in the TIC and PIP datasets, as noted in Judson and Kim (2026). TIC holdings include foreign reserves, while PIP holdings do not per balance-of-payments conventions. Therefore, an uncorrected "custodial gap" obtained from subtracting the two series would also remove all reserve holdings. I sidestep this issue by subtracting from the TIC Treasury totals the estimate of Treasury holdings held as reserves for Belgium and Luxembourg, described in Appendix B.2.1. I use the same procedure when applying the financial centre correction to the United Kingdom.

Another limitation is the potential understatement of foreign investment fund Treasury holdings

in TIC, as it covers only funds with U.S. custodians. As a result, the estimated custodial gap for Belgium and Luxembourg is likely a lower bound.

Hong Kong bilateral redistribution. TIC attributes approximately USD 268 billion in Q4 2025 to Hong Kong, in part reflecting Chinese and other foreign investment routed through Hong Kong custodians and intermediaries. Following Alstadsæter et al. (2018), I redistribute the Hong Kong total to investor countries in proportion to bilateral cross-border banking positions,

$$\Delta H_t^{\text{HKG}\to k} \;=\; H_{\text{HKG},t}^{\text{TIC}} \times \frac{Q_{\text{HKG},k,t}}{\sum_{k'} Q_{\text{HKG},k',t}},$$

where $Q_{\text{HKG},k,t}$ denotes positions of banks located in Hong Kong vis-à-vis counterpart country k from the BIS Locational Banking Statistics database. Bilateral breakdowns are published only at the all-instruments level, so all-instrument rather than deposit liabilities serve as the weights. Because the all-instruments measure includes interbank and intragroup positions, it may modestly overstate the Chinese share given the large interbank exposures of mainland institutions. Increments accruing to the five foreign dataset regions and to the United States are added to their respective totals; the residual accruing to non-recipient countries is excluded. Singapore could benefit from a similar restatement, but its reported bilateral positions carry no counterpart breakdown at any instrument level, so it is excluded from the redistribution and its own TIC holdings (approximately USD 278 billion in Q4 2025) are dropped as a non-attributable residual.

Switzerland domestic-share split. Switzerland's TIC total mixes genuine Swiss-resident investment with assets managed by Swiss banks for foreign clients; it stands at approximately USD 294 billion in Q4 2025. I estimate the domestic fraction as the ratio of the ultimate-ownership Swiss bilateral position in U.S. debt securities to TIC Switzerland holdings of all U.S. debt,

$$\text{domestic share}_t \;=\; \frac{H_{\text{CHE},t}^{\text{IMF}}}{H_{\text{CHE},t}^{\text{TIC, all U.S. debt}}},$$

which averages 0.44 over the sample, and apply it to TIC Switzerland Treasury holdings. The foreign custodial residual is redistributed to the dataset countries and to the United States using the same bilateral-banking weights as for Hong Kong. Switzerland is not itself a dataset country, so its domestic total feeds only the restated totals of recipient countries.

United Kingdom domestic-share adjustment. The UK total is inflated by custodial holdings at Euroclear UK and at other UK custodian banks, as well as international investment funds and banks. I compute a domestic share on the same principle,

$$\text{domestic share}_t \;=\; \frac{H_{\text{GBR},t}^{\text{IMF}}}{H_{\text{GBR},t}^{\text{TIC, all NR U.S. debt}}},$$

which averages 0.65, and apply it to the restated UK non-reserve Treasury total. Unlike the Hong Kong and Swiss cases, the foreign custodial component is removed without redistribution: UK-resident positions blend custody, fund management and international banking activity in proportions that public data cannot separate.

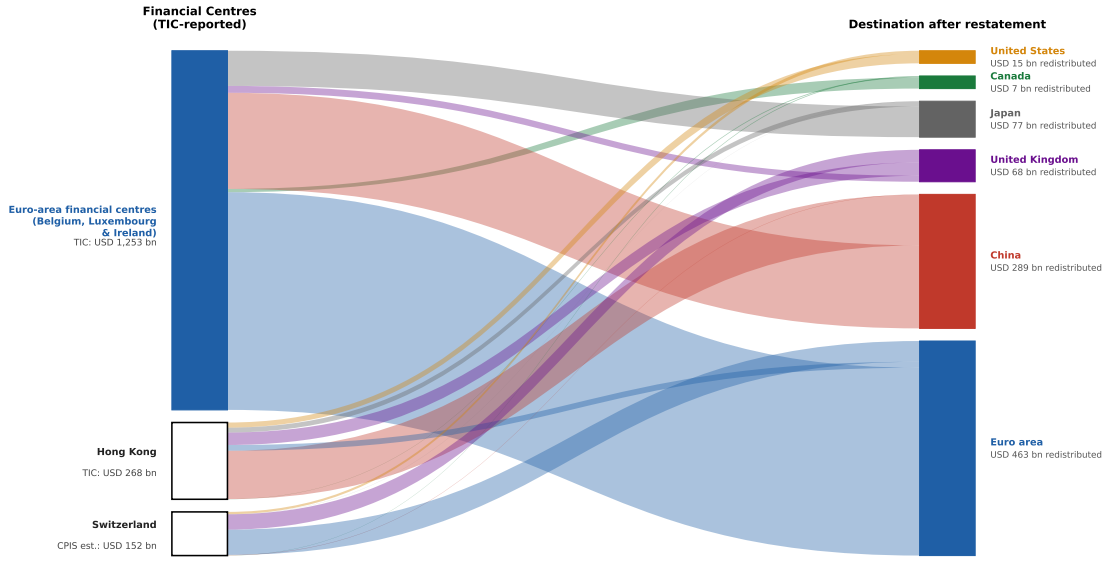


Figure A1. Financial-centre restatement flows, Q4 2025

Notes: Financial-centre restatement flows of U.S. Treasury holdings, in USD billions, Q4 2025. The left column holds the three restated and reallocated financial-centre blocks: the euro-area centres (Belgium, Luxembourg and Ireland) taken together, Hong Kong, and Switzerland. The right column holds the six dataset destinations, including the United States, which receives Hong Kong and Swiss custodial increments under the bilateral-banking allocation of Alstadsæter et al. (2018). Only reallocations are drawn. Amounts removed without reallocation (the rest-of-world share of Luxembourg and Irish fund holdings, the part of the pooled custodial gap accruing to economies outside the dataset, the non-attributable residual of the Hong Kong and Swiss books, and the UK custodial gap) are the difference between a left-hand block's height and the sum of its outgoing flows. The four offshore conduits, excluded outright, do not appear. *Sources:* Author's calculations based on U.S. Treasury (TIC); IMF (CPIS, PIP, IRFCL); BIS; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026).

B.2 Sector decomposition, country by country

Sector definitions. Four sectors are used throughout: banks, NBFIs, HH/NFC, and the official sector. The bank label is uniform across countries, but the sources differ in what they separately identify, and no single convention can be imposed on all of them. For the euro area, the bank sector corresponds to the ECB's monetary financial institutions (MFI) sector excluding the Eurosystem, which pools deposit-taking corporations with money market funds. This is not a modelling choice: over the SHS sub-period the ECB publishes only the pooled aggregate of deposit-takers and money market funds for these holdings and no separate money market fund series, so the euro area cannot be placed on a deposit-taker-only convention without breaking its own time series; the later SHSS sub-period is pooled to match it. Euro-area NBFI then comprises the remaining financial corporations, that is, investment funds other than money market funds, insurance corporations, pension funds and other financial intermediaries. For Japan, money market funds cannot be separately identified under any reporting combination available in the portfolio surveys, and the aggregate adding-up identity leaves no residual from which they could be recovered, so banks comprise deposit-taking corporations alone and money market funds remain inside NBFI. The same is true of the descriptive decompositions for China and Canada. For the United Kingdom and the United States the constraint does not bind—the IMF portfolio surveys report money market funds separately for the United Kingdom, and the Federal Reserve's Financial Accounts report them as a sector of their own—and in both cases they are placed in NBFI, so that as many jurisdictions as possible share one convention. The

euro area is therefore the only jurisdiction in which money market funds sit inside banks, and levels of bank holdings are not directly comparable between the euro area and the remaining countries. Because the structural estimation uses within-cell time-series variation and estimates a separate parameter pair for every cell, and because the euro-area numerator and denominator are on the same money market fund convention throughout, the remaining asymmetry bears on cross-country comparisons of the level of bank holdings and not on the parameter estimates. For the United States, banks are depository institutions alone, reflecting the distinct mapping of the Federal Reserve's Financial Accounts described below.

B.2.1 Euro area

The euro-area series is built from three components—a reserves component that spans the whole sample, and two non-reserve components covering the two halves of the sample.

Official reserves. The dollar value of euro-area official reserve holdings of U.S. Treasuries is constructed as a currency share applied to reported holdings of reserve securities. The currency share is built over a set of 12 reporting units: 11 euro-area national central banks whose published currency composition is available for at least half of the years from 1999 to 2020, together with the European Central Bank itself.[8] Annual currency shares are taken from Ito and McCauley (2020) through 2020 and extended to the end of the sample from the national central banks' own annual reports and financial statements, which supply annual observations. These are anchored at the fourth quarter and interpolated linearly within the year. The individual shares are aggregated into a euro-area dollar share by weighting each central bank by its own holdings of reserve securities reported to the IMF International Reserves and Foreign Currency Liquidity (IRFCL) database, with the set of banks expanding as countries accede to the euro area. Applying that share to the euro-area aggregate of reserve securities gives the reserves component of the official cell, and the same aggregate of reserve securities enters the denominator of the official sector's portfolio share.

Non-reserve sectors to Q1 2022. Over the first 34 quarters of the sample the ECB's Securities Holdings Statistics (SHS) report euro-area resident holdings by holder sector and by issuer country, so holdings of U.S. general-government debt securities are observed directly for each sector. Long-term and short-term maturities are summed explicitly, because the source publishes no combined-maturity total for debt securities. The portfolio share of each private sector is formed as that sector's holdings of U.S. general-government debt over its holdings of all non-euro-area general-government debt, a ratio that requires no currency conversion because both legs are denominated in euro. The official cell over this sub-period is the reserves component plus resident general-government holdings of U.S. general-government debt, with its portfolio share taken against the sum of the reserve-securities aggregate and general-government holdings of non-euro-area sovereign debt; the euro legs are converted at the quarterly average exchange rate before being added to the dollar-denominated reserves terms. The NBFI leg is corrected for the Luxembourg and Irish rest-of-world component as described above.

Non-reserve sectors from Q2 2022. From 2022 the ECB's successor dataset, Securities Holdings Statistics by Sector (SHSS) discontinues the holder-sector by issuer-country breakdown:

[8] These are the national central banks of Belgium, Germany, Estonia, Spain, Croatia, Italy, Latvia, Luxembourg, the Netherlands, Slovenia and Finland. Their own holdings of reserve securities reported to the IRFCL account on average for 71% of the reserve securities held by the twenty euro-area national central banks over the Q4 2013–Q4 2025 construction window; the ECB's own reserve securities are excluded from both legs of that coverage ratio.

for non-euro-area issuers the finest resolution is an aggregate rest-of-the-world, with no country code for the United States. The sector composition over the remaining 14 quarters is therefore rebuilt from the PIP database, following closely the methodology used for the United Kingdom and Japan. Sectoral holdings of U.S. debt securities are assembled country by country for the 14 euro-area members that report a usable holder-sector cross-section against the United States, and complemented for Luxembourg and Ireland—which do not—by applying their own reported sector composition of extra-national debt holdings to their total-economy position in U.S. debt. Because that composition is measured on all debt securities while the object of interest is Treasury holdings specifically, the raw composition is calibrated against the directly measured Treasury composition of the first sub-period over the 5 quarters in which both are observed, and the four calibrated shares are renormalised to sum to one in every quarter. The shares are applied to the non-reserve Treasury total, defined as the restated TIC euro-area total less the reserves component, so that the four sectors exhaust the TIC aggregate exactly over this sub-period. Portfolio shares are formed against each sector's holdings of extra-euro-area sovereign debt. The semi-annual PIP observations are interpolated linearly to quarterly frequency before being combined. The final two quarters, for which no survey observation exists to interpolate against, are extrapolated by a linear trend fitted on the trailing four quarters.

B.2.2 Japan

Japanese sector shares are provided by the PIP database, which reports bilateral holdings of total debt securities against the United States by deposit-taking corporations, other financial corporations, and other sectors. The reported amounts are semi-annual and are interpolated linearly to quarterly frequency. Because the survey indicator covers all U.S. debt rather than Treasuries alone, the private sectors are scaled against a matched TIC scope—long-term U.S. debt plus short-term Treasuries—before the resulting shares are applied to the restated TIC Treasury total. The official cell is the residual of the matched-scope total after the three private sectors, clamped at zero. It absorbs reserve holdings, the small reported general-government position, and any remaining gap between the TIC scope and reported non-reserve holdings. Japan is the one dataset country whose official cell is recovered residually rather than measured. The maintained assumption is that the sector allocation within all U.S. debt proxies the sector allocation within Treasuries specifically.

B.2.3 United Kingdom

UK sector holdings are built by combining directly observable reserve holdings, TIC totals restated for financial-centre corrections, and sectoral shares from PIP. The official cell is built directly as the dollar share of UK foreign-currency reserves, taken from the Bank of England's published currency breakdown, applied to holdings of reserve securities from the IRFCL database, plus the general-government position against the United States reported to PIP, interpolated to quarterly frequency.

The private sectors are then built on a non-custodial share. Let T_t^{BoE} denote the official reserve amount defined above. The non-custodial share is the ratio of reported UK total-economy holdings of U.S. debt to the TIC all-U.S.-debt total net of reserves,

$$\text{nc}_t \;=\; \frac{H_{\text{GBR},t}^{\text{IMF, all U.S. debt}}}{H_{\text{GBR},t}^{\text{TIC, all U.S. debt}} - T_t^{\text{BoE}}},$$

and the private Treasury total is $\text{nc}_t \times (H_{\text{GBR},t}^{\text{TIC, UST}} - T_t^{\text{BoE}})$ less the reported general-government position. The private total is then allocated across banks, NBFI and HH/NFC using weights

calculated from PIP data, with money market funds remaining in the non-bank aggregate. Direct reporting of the other-financial-corporations aggregate is available in 56% of survey periods; where it is missing, it is imputed as the sum of its reported components, an identity verified on the periods in which both are available. The UK total also carries the domestic-share adjustment described in Appendix B.1.

B.2.4 China

China enters the dataset as a single total-economy cell whose Treasury holdings are the restated TIC total. No holder-sector split enters estimation, because the only PIP indicator that offers a sector cross-section against the United States covers total portfolio investment—equity as well as debt—and no debt-specific indicator is disaggregated by holder sector for China. Imposing the total-portfolio composition on Treasury holdings would be a strong assumption: official holdings are disproportionately concentrated in Treasuries relative to equities and corporate bonds, so the implied official share would be biased downwards.

A descriptive sector decomposition is nonetheless constructed, for Figure 2 only. The three private cells are the reported PIP positions in total U.S. portfolio investment; the official cell is the difference between the unrestated TIC all-securities total and their sum, under the assumption that the Chinese official sector holdings of U.S. securities are solely in Treasuries. The resulting four shares, which sum to one by construction, are applied to the restated TIC Treasury total. This series is available only from Q4 2020, when the sector cross-section begins, and it is explicitly not used to build China's portfolio share or any estimation input. Chinese holdings are also affected by two of the restatements of Appendix B.1: a meaningful part of Chinese official holdings is custodied via Euroclear and so appears under Belgium, and a further part is attributed to Hong Kong.

B.2.5 Canada

Canada is the only dataset country whose Treasury holdings are observed entirely from national sources, and it too enters the estimation as a single total-economy cell. The resident leg is Canadian holdings of U.S. general-government debt securities from the Canadian international investment position statistics, available directly from Q4 2016 and chain-backcast on the growth rates of the predecessor series for the earlier quarters of the sample, converted to dollars at the quarterly average exchange rate. The official leg is the U.S.-dollar component of the Exchange Fund Account, constructed as the account's reported dollar share of convertible foreign currencies applied to Canada's reported holdings of reserve securities from IRFCL.

As with China, a descriptive four-sector decomposition is constructed for Figure 2 alone. It applies PIP sector shares—measured against the world rather than against the United States, because no bilateral sector cross-section against the United States is reported for Canada—to the resident leg, and adds the dollar reserve term to the official cell. It is available from Q2 2019, the first period in which all four sectors are simultaneously observed. The maintained assumption behind it is that Canada's sector allocation across all foreign debt securities proxies its allocation to U.S. Treasuries. That assumption is supported by the concentration of the Canadian foreign debt portfolio in U.S. instruments: over the 49 quarters of the sample, U.S. debt accounts on average for 67.1% of Canadian holdings of foreign debt securities, within a range of 62.2% to 70.6%. It is, in any case, an assumption that carries no weight in the estimation, since the Canadian estimation cell is the country aggregate.

B.2.6 United States

U.S. domestic holdings are taken from the Federal Reserve's Financial Accounts of the United States, whose instrument tables report holdings of Treasury securities by resident sector. Three sectors are modelled: banks; NBFI, comprising the remaining financial sectors; and households together with non-financial corporations. The U.S. official sector—the federal government together with the Federal Reserve, whose System Open Market Account holds several trillion dollars of Treasuries—is not modelled. Because official rows are excluded from the set of modelled cells, System Open Market Account holdings enter the inelastic residual of the market-clearing condition automatically and are not subtracted from supply separately. This treatment reflects the fact that those positions are determined by the conduct of monetary policy rather than by portfolio optimisation. The United States is also a recipient in the restatements of Appendix B.1, taking the increments allocated to it from Hong Kong and Switzerland under the bilateral-banking procedure, which are small relative to domestic holdings.

B.3 The portfolio-share denominator

The portfolio share is $s_{j,t} = H^{UST}_{j,t}/W_{j,t}$, where the numerator is the cell's Treasury holdings constructed above and the denominator $W_{j,t}$ is the cell's total holdings of sovereign debt issued outside its own jurisdiction. The denominator is measured, wherever the data permit, from the same resident-side source as the numerator, so that the two legs are consistent by construction and the share is bounded above by one mechanically.

Directly observed denominators. For the euro area, each sector's holdings of extra-euro-area sovereign debt are read directly from the ECB SHS or SHSS statistics, which is precisely the ratio already described in Appendix B.2.1. For the official cell, the aggregate of reserve securities is added to the general-government leg. For Canada, both legs come from the Canadian national accounts and the Exchange Fund Account: the denominator is resident holdings of all foreign general-government debt securities plus the dollar and non-dollar components of reserves. For the United States, the domestic leg is the sector's Treasury holdings from the Financial Accounts and the foreign leg is its reported holdings of foreign general-government debt, taken from the U.S. outbound portfolio survey over the full counterpart universe; the 24 counterparts with a reported sector cross-section are used directly, and the remaining counterparts, reported only in aggregate, are split using the pooled sector composition of the former.

Mirror-imputed denominators. No resident-side sovereign portfolio is observed for Japan, the United Kingdom or China. For these three, the denominator is imputed with a mirror procedure. Let $H_{j,k,t}$ denote the cell's reported bilateral portfolio holdings of debt securities issued in destination country k, and let $\omega_{k,t}$ denote the share of sovereign debt in the externally held debt of destination k, computed from the IMF International Investment Position (IIP) database. The non-reserve part of the denominator is $\sum_{k\in\mathcal{D}} \omega_{k,t} H_{j,k,t}$, summed over the set $\mathcal{D}$ of destinations with full coverage in the position statistics and excluding both the United States and the investor's own jurisdiction. For Japan and the United Kingdom this is done separately for each holder sector, using the holder-sector by counterpart cross-section from PIP. The official sector's denominator is instead its own reported holdings of reserve securities. For China the denominator is a single total-economy object, the sum of reserve securities and the mirror-imputed non-reserve term, with no sectoral decomposition. Semi-annual PIP positions are interpolated to quarterly frequency for each counterpart separately before being weighted and summed. Both Chinese legs require a back-cast before Q2 2015, when the bilateral survey and the reserve-securities

series begin. The non-reserve leg is back-cast by a single trend fitted on the weighted aggregate over the entire sample and anchored at its first observed value; the reserve leg is back-cast by applying the securities share of reserves—the ratio of reported holdings of reserve securities to total foreign-currency reserves, averaged over the entire sample of quarters in which both are observed—to the total foreign-currency reserves reported in the Chinese international investment position. The two legs are back-cast differently because the non-reserve aggregate trends cleanly ($R^2 = 0.904$ on the anchored fit) but has no contemporaneously observable counterpart to scale against, whereas the reserve leg barely trends ($R^2 = 0.007$) yet bears a near-constant ratio to total foreign-currency reserves—confined to $[0.990, 0.999]$ across the sample—which are observed back to 2013 and were falling steeply over precisely the back-cast window.

B.4 Coverage and caveats

The data construction procedure suffers from four main limitations. First, the construction of sectoral holdings relies on the assumption that the split of holdings across sectors observed for different sets of securities applies to U.S. Treasuries. This assumption is particularly strong when applied to a wide set of securities or destination countries, which motivates the exclusion of the Canadian and Chinese sectoral breakdown from estimation. Second, the financial-centre restatements suffer from data limitations, so they do not exhaust all sources of obfuscation of true ultimate ownership. Even where they are applied they rely on mild to moderately strong assumptions. Therefore, some residual bias introduced by offshore investing may remain. Third, China and Canada enter the estimation only as country aggregates; their sector decompositions are descriptive, cover shorter windows, and rest on maintained assumptions that the estimation does not use. Fourth, the mirror-imputed denominators for Japan, the United Kingdom and China rest on destination-country position statistics whose coverage is incomplete, so the destination set is restricted to fully covered economies and the denominators should be read as lower bounds on the true sovereign portfolio to the extent that holdings in excluded destinations are non-trivial.

B.5 Data sources

Tables A1 and A2 list the provenance of every dataset used in the paper, including the returns and control series described in Section 4.3.

Table A1. Data sources: holdings and portfolio shares

Data	Source	Role in the paper
Panel A: Holdings and portfolio shares		
Foreign holdings of U.S. Treasuries by country	U.S. Department of the Treasury, Treasury International Capital system	Country totals; base for the financial-centre restatements
Euro-area holdings of U.S. general-government debt by holder sector	European Central Bank, Securities Holdings Statistics	Euro-area sector numerators and share denominators to 2022 Q1
Euro-area holdings of extra-euro-area debt by holder sector	European Central Bank, Securities Holdings Statistics by Sector	Euro-area sector composition and denominator from 2022 Q2
Bilateral portfolio holdings by holder sector and counterpart economy	International Monetary Fund, Coordinated Portfolio Investment Survey and Portfolio Investment Positions	Sector composition outside the euro area; mirror-imputed denominators
Bilateral international investment positions	International Monetary Fund, International Investment Position statistics	Sovereign-debt shares used to convert portfolio positions into sovereign-debt positions
Official reserve assets held in securities	International Monetary Fund, International Reserves and Foreign Currency Liquidity	Official-sector numerators and denominators
Currency composition of official reserves, euro area	Ito and McCauley (2020), extended with euro-area national central banks' annual reports and financial statements	Dollar share of euro-area reserves
Currency composition of official reserves, United Kingdom	Bank of England, currency breakdown of UK foreign currency reserves	Dollar share of UK reserves
Canadian holdings of foreign and U.S. debt securities	Statistics Canada, international investment position statistics	Canadian numerator and denominator
Canadian official international reserves	Bank of Canada, Exchange Fund Account official international reserves	Canadian official reserve leg
Chinese international investment position	State Administration of Foreign Exchange of China	Total foreign-currency reserves, the base of the pre-2015 backcast of Chinese reserve securities
U.S. domestic holdings of Treasury and other debt securities by sector	Board of Governors of the Federal Reserve System, Financial Accounts of the United States	U.S. sector numerators and domestic leg of the U.S. denominator
Ultimate-ownership look-through of Luxembourg and Irish fund holdings	Beck et al. (2026)	Non-euro-area ownership share behind euro-area fund positions
Bilateral cross-border banking positions	Bank for International Settlements, locational banking statistics	Allocation key for the custodial-centre redistributions
Exchange rates used to convert holdings and denominators to U.S. dollars	Board of Governors of the Federal Reserve System, foreign exchange rates release (Canadian dollar); European Central Bank, euro foreign exchange reference rates (euro)	Quarterly-average conversion of Canadian and euro-area positions

Notes: Panel A: sources used to build the holdings panel and the portfolio-share denominator of Appendices B.2 and B.3, including the exchange rates at which Canadian and euro-area positions are converted to U.S. dollars.
Sources: Author's compilation of the providers listed in the table.

Table A2. Data sources: returns, Treasury supply and controls

Data	Source	Role in the paper
Panel B: Returns		
Sovereign bond index membership	FTSE Russell, World Government Bond Index, ground-rule update history and constituent factsheet	Reference-portfolio universe
Redemption yields, total returns and market values of index constituents	FTSE Russell, World Government Bond Index, obtained via LSEG Datastream	Reference-portfolio weights, yields and returns
Sovereign credit default swap spreads	Markit (S&P Global), obtained via LSEG Datastream	Credit-risk adjustment to sovereign returns
Bilateral spot exchange rates	Board of Governors of the Federal Reserve System, foreign exchange rates release; Organisation for Economic Co-operation and Development	Currency conversion of returns and of index market values
Panel C: Treasury supply and residualisation controls		
Market value of gross U.S. federal debt	Federal Reserve Bank of Dallas	Treasury supply in the market-clearing condition
One-quarter return of U.S. Treasuries relative to the investor's non-U.S. reference basket	Constructed from the Panel B series (FTSE Russell, World Government Bond Index; Markit credit default swap spreads; exchange rates)	Residualisation control
Equity-market implied volatility index (quarterly average)	CBOE Global Markets, VIX, obtained via FRED	Residualisation control
Quarterly growth of marketable U.S. Treasury debt held by the public, at par, quarter-end	U.S. Department of the Treasury, Monthly Statement of the Public Debt	Residualisation control
Quarterly change in the effective federal funds rate (quarterly average)	Board of Governors of the Federal Reserve System, Selected Interest Rates (H.15), obtained via FRED	Residualisation control

Notes: Panel B: sources used to build the excess returns and their second moments. World Government Bond Index membership, which determines each investor's reference basket, is taken from the index ground-rule update history and constituent factsheet; the constituents' redemption yields, total returns and market values are obtained via LSEG Datastream. Panel C: the Treasury supply series of the market-clearing condition (9) and the four controls of the residualisation regression (16) used to construct the Diffuse GIV instrument, labelled "Residualisation control" in the table. *Sources:* Author's compilation of the providers listed in the table.

C Estimation inputs time series

This appendix plots the four time-varying inputs to the first-order condition (13) that Section 4.3 describes in construction terms only: the expected excess return on U.S. Treasuries over each investor's non-U.S. sovereign benchmark, the conditional variance and covariance that enter the risk terms, and the portfolio share of Treasuries in each cell's foreign sovereign portfolio. The excess return, variance and covariance are shown over the 49-quarter estimation sample, and the portfolio share over the quarters in which it is observed, Q4 2013–Q4 2025.

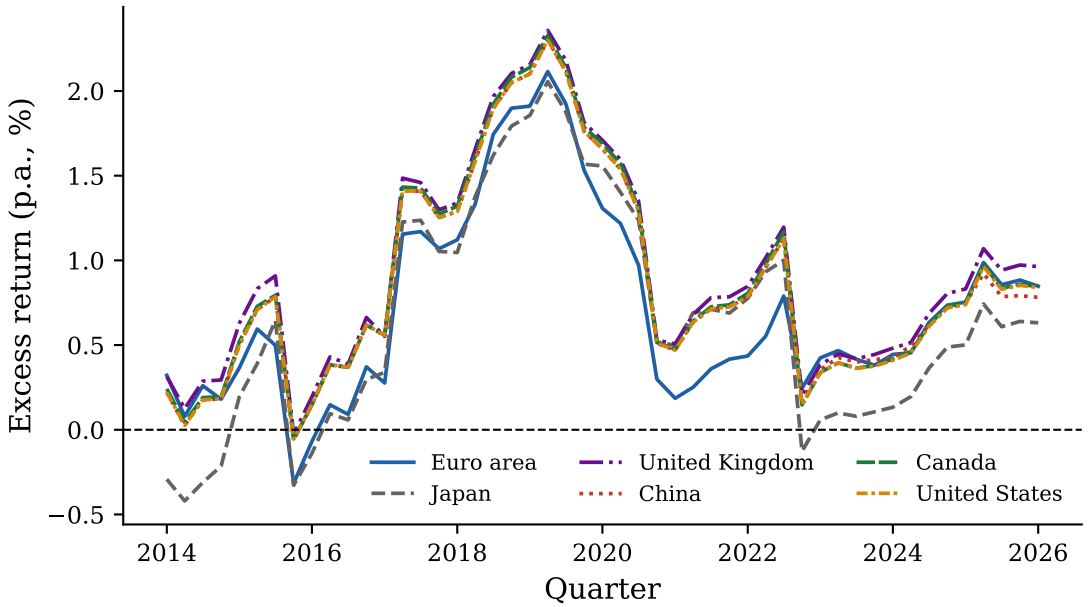


Figure A2. Expected excess return on U.S. Treasuries by investor country

Notes: Expected excess return $E_{c,t}$ of U.S. Treasuries over investor c's reference basket, the FTSE World Government Bond Index universe excluding the United States and investor c's own sovereign, quarterly, in percent at an annual rate, Q4 2013–Q4 2025. Baseline measurement basis: start-of-window yields to maturity, both legs net of five-year sovereign credit-default-swap spreads, under the random-walk treatment of the exchange rate, so that both legs are measured in U.S. dollars with no currency translation. The series is common to all cells resident in the same country. *Sources:* Author's calculations based on FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; Board of Governors of the Federal Reserve System (H.10, via FRED); OECD (exchange rates, via FRED).

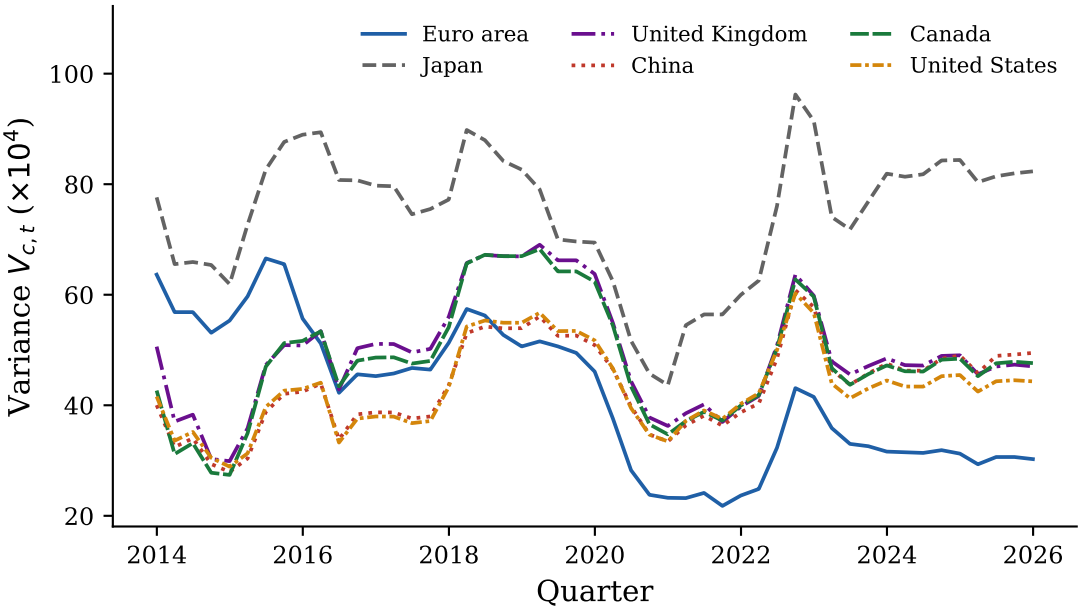


Figure A3. Conditional variance of the Treasury excess return by investor country

Notes: Conditional variance $V_{c,t}$ of the realised annual excess return of U.S. Treasuries over investor c's reference basket, the FTSE World Government Bond Index universe excluding the United States and investor c's own sovereign, quarterly, Q4 2013–Q4 2025. Variance of annual returns expressed as decimals, multiplied by 10^4 (equivalently, of returns expressed in percent). Sample variance (degrees-of-freedom correction of one) over a rolling window of twenty quarters of realised twelve-month total returns, including capital gains, with both legs net of five-year sovereign credit-default-swap spreads and translated into investor c's currency; the most recent return in the window is completed at the end of quarter t. The series is common to all cells resident in the same country. *Sources:* Author's calculations based on FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; Board of Governors of the Federal Reserve System (H.10, via FRED); OECD (exchange rates, via FRED).

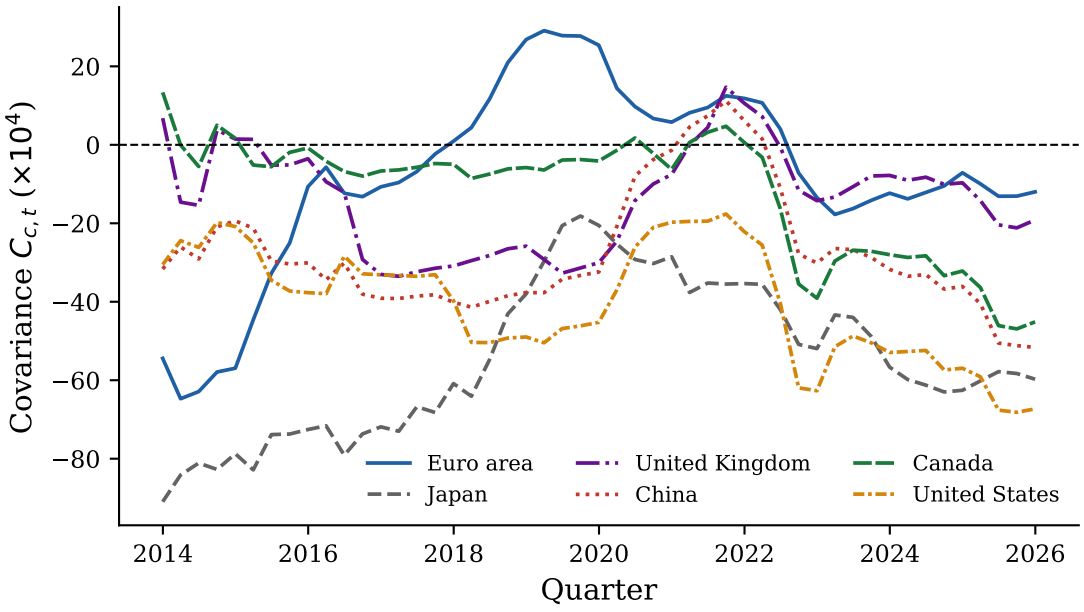


Figure A4. Conditional covariance of the Treasury excess return with the reference-basket return by investor country

Notes: Conditional covariance $C_{c,t}$ between the realised annual excess return of U.S. Treasuries over investor c's reference basket, the FTSE World Government Bond Index universe excluding the United States and investor c's own sovereign, and the realised annual return on the basket itself, quarterly, Q4 2013–Q4 2025. Covariance of annual returns expressed as decimals, multiplied by 10^4 (equivalently, of returns expressed in percent). Sample covariance (degrees-of-freedom correction of one) over a rolling window of twenty quarters of realised twelve-month total returns, including capital gains, with both legs net of five-year sovereign credit-default-swap spreads and translated into investor c's currency; the most recent return in the window is completed at the end of quarter t. The dashed horizontal line marks zero. The series is common to all cells resident in the same country. *Sources:* Author's calculations based on FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; Board of Governors of the Federal Reserve System (H.10, via FRED); OECD (exchange rates, via FRED).

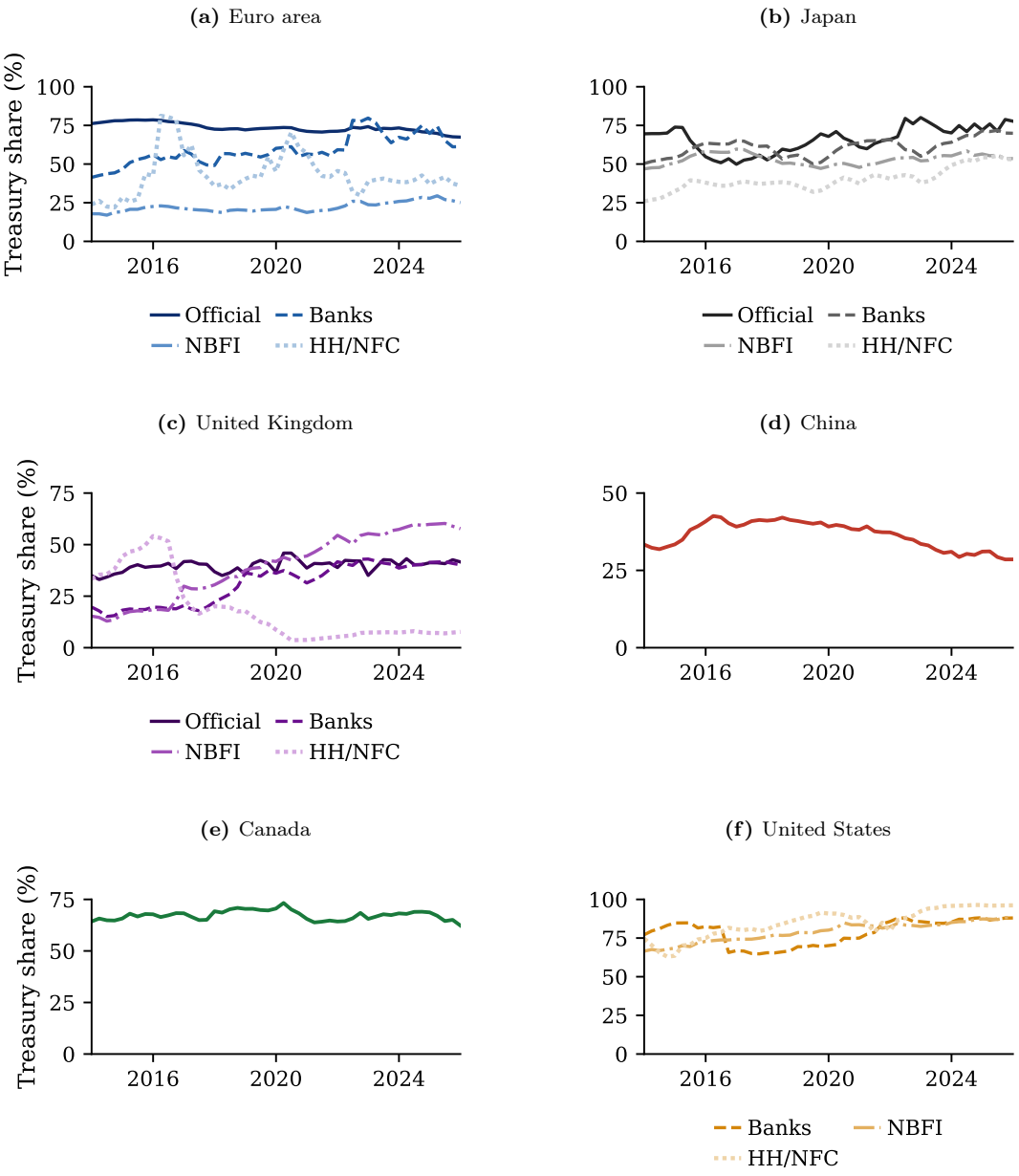


Figure A5. U.S. Treasury share of the foreign sovereign portfolio, by investor and sector

Notes: Portfolio share $s_{c,i,t}$, Treasury holdings divided by total holdings of sovereign debt issued outside the investor's own jurisdiction, quarterly, in percent, Q4 2013–Q4 2025. Panels (a), (b) and (c) show one line per modelled sector: official sector, banks, non-bank financial institutions, and households together with non-financial corporations. Panel (f) shows the three modelled private sectors; no U.S. official cell is modelled. China and Canada, panels (d) and (e), enter the estimation as single total-economy cells. *Sources:* Author's calculations based on U.S. Treasury (TIC); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026).

D Estimation

D.1 Estimated parameters and confidence intervals

This appendix collects the full set of cell-level point estimates discussed in Section 5. Table A3 reports, for each of the 17 modelled cells, the level parameter $\psi_{0,c,i}$, the curvature parameter $\psi_{1,c,i}$ and the risk-aversion parameter $\gamma_{c,i}$, each with its 90% confidence interval. The main text presents the same estimates graphically, in Figure 3 for the convenience parameters and Figure 4 for risk aversion.

D.2 Identification and specification diagnostics

First-stage instrument relevance is strong in most cells but weak in a minority of them. Table A4 reports, cell by cell, the Olea and Pflueger (2013) effective F-statistic for two dependent variables projected on the same instrument block: the regressor $x_{c,i,t} = V_{c,t}\, s_{c,i,t} + C_{c,t}$ of (14), whose instrumented variation identifies risk aversion $\gamma_{c,i}$, and $1/s_{c,i,t}$, whose instrumented variation identifies the curvature parameter $\psi_{1,c,i}$. Stock-Yogo critical values are not tabulated for instrument blocks as wide as those used here, so the effective F-statistic is assessed against the conventional rule-of-thumb benchmark of ten.

By that benchmark, relevance is adequate in most of the panel. 7 of the 17 cells fall below it for at least one of the two dependent variables: euro-area banks; euro-area households and non-financial corporations; British banks; British non-bank financial institutions; British households and non-financial corporations; British official sector; U.S. non-bank financial institutions. 4 cells fall below it for $x_{c,i,t}$ and 4 for $1/s_{c,i,t}$, and the number that fall below it for both is 1 (British official sector). All other cells clear the benchmark on both dependent variables (Table A4).

The Hansen (1982) overidentification test does not reject the moment restrictions in any of the 17 cells at the 5% or the 10% level, indicating that the overidentifying restriction is plausibly satisfied in every cell and corroborating the validity of the Diffuse GIV instrument.

Table A3. GMM parameter estimates by country and sector

Country		Banks	NBFI	HH/NFC	Official
Euro area	$\hat{\psi}_0$	−0.0046	−0.0071	−0.0063	−0.0024
		[−0.0103, 0.0010]	[−0.0127, −0.0015]	[−0.0134, 0.0009]	[−0.0342, 0.0294]
	$\hat{\psi}_1$	0†	0†	0†	0†
	$\hat{\gamma}$	1.6154	2.0353	1.4065	2.3095
		[1.1236, 2.1072]	[1.4983, 2.5724]	[0.6533, 2.1597]	[1.9523, 2.6667]
Japan	$\hat{\psi}_0$	−0.0073	−0.0093	−0.0095	−0.0069
		[−0.0196, 0.0051]	[−0.0228, 0.0041]	[−0.0146, −0.0044]	[−0.0151, 0.0013]
	$\hat{\psi}_1$	0†	0†	0†	0†
	$\hat{\gamma}$	1.9884	2.4817	1.7838	1.8440
		[1.1971, 2.7796]	[1.8206, 3.1428]	[1.0029, 2.5648]	[1.2022, 2.4857]
United Kingdom	$\hat{\psi}_0$	−0.0110	−0.0107	−0.0098	−0.0349
		[−0.0142, −0.0079]	[−0.0136, −0.0078]	[−0.0126, −0.0069]	[−0.0490, −0.0208]
	$\hat{\psi}_1$	0.0008	0.0009	0.0001	0.0108
		[0.0001, 0.0016]	[0.0004, 0.0014]	[−0.0001, 0.0003]	[0.0052, 0.0163]
	$\hat{\gamma}$	0†	0†	0†	0†
China	$\hat{\psi}_0$	−0.0110			
		[−0.0201, −0.0019]			
	$\hat{\psi}_1$	0.0012			
		[−0.0019, 0.0044]			
	$\hat{\gamma}$	0†			
Canada	$\hat{\psi}_0$	−0.1743			
		[−0.2104, −0.1383]			
	$\hat{\psi}_1$	0.1138			
		[0.0905, 0.1372]			
	$\hat{\gamma}$	1.4377			
		[0.4482, 2.4271]			
United States	$\hat{\psi}_0$	−0.0076	−0.0451	−0.0382	—
		[−0.0201, 0.0049]	[−0.0657, −0.0245]	[−0.0510, −0.0255]	
	$\hat{\psi}_1$	0†	0.0284	0.0242	—
			[0.0138, 0.0431]	[0.0143, 0.0340]	
	$\hat{\gamma}$	0†	2.8073	4.5613	—
			[0.7249, 4.8897]	[2.2880, 6.8346]	
N (quarters)		49			

Notes: Point estimates of the level parameter $\hat{\psi}_0$, the curvature parameter $\hat{\psi}_1$ and risk aversion $\hat{\gamma}$ for the 17 cells, estimated cell by cell by two-step GMM under the bounds $\psi_1 \geq 0$ and $\gamma \geq 0$ (Section 4.1), Q4 2013–Q4 2025 (49 quarters per cell). 90% confidence intervals in brackets, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987). Instruments: a constant and the Diffuse GIV block of Section 4.2, the principal components of the de-meaned demand residuals of donor cells resident in other countries and belonging to other sectors; the block has between 9 and 15 columns depending on the cell, so each cell uses between 10 and 16 moment conditions. † marks $\hat{\psi}_1$ or $\hat{\gamma}$ at its lower bound of zero, shown without a confidence interval. *Sources:* Author's estimates based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

Table A4. Estimation diagnostics by cell

Country	Sector	$\hat{F}^{OP}$ (dep. var. $x = Vs + C$)	$\hat{F}^{OP}$ (dep. var. $1/s$)	L	J p-value	N
Euro area	Banks	18.95	7.37	10	0.474	49
Euro area	NBFI	72.87	25.20	10	0.291	49
Euro area	HH/NFC	54.86	6.95	10	0.355	49
Euro area	Official	36.04	52.84	11	0.725	49
Japan	Banks	13.49	29.71	10	0.510	49
Japan	NBFI	13.96	19.00	10	0.505	49
Japan	HH/NFC	18.61	32.31	10	0.492	49
Japan	Official	32.94	12.87	11	0.397	49
United Kingdom	Banks	2.79	20.21	10	0.426	49
United Kingdom	NBFI	7.18	26.79	10	0.407	49
United Kingdom	HH/NFC	35.75	8.83	10	0.358	49
United Kingdom	Official	7.03	5.33	11	0.522	49
China	Total	15.77	238.28	16	0.851	49
Canada	Total	137.53	33.64	16	0.833	49
United States	Banks	16.79	19.94	11	0.501	49
United States	NBFI	6.78	39.54	11	0.575	49
United States	HH/NFC	13.90	22.85	11	0.477	49

Notes: Cell-level diagnostics for the two-step GMM estimates, Q4 2013–Q4 2025. $\hat{F}^{OP}$: Olea and Pflueger (2013) effective first-stage F-statistic for two dependent variables projected on the same instrument block, $x = Vs + C$, the regressor on which γ loads in the first-order condition, and the reciprocal share $1/s$, the regressor on which ψ_1 loads. L: number of moment conditions in the cell's GMM system, $L = k_{\text{excl}} + 1$ (the k_{excl} excluded instruments of the donor-residual block plus the constant). J p-value: Hansen overidentification test, with L minus the number of estimated parameters degrees of freedom, computed with the Newey–West (HAC) weighting matrix with four lags (Newey and West, 1987). Weak-instrument benchmark: rule of thumb $\hat{F}^{OP} \geq 10$. *Sources:* Author's estimates based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

E Additional model experiments

This appendix reports two general-equilibrium experiments that build on the model extensions of Appendix A.6. The conventions are those of Section 6: the counterfactual sets $\psi_{0,c,i} = \psi_{1,c,i} = 0$, every other parameter and input is held at its estimated or observed value, the market-clearing condition (9) is re-solved quarter by quarter, and results are averaged over Q4 2013–Q4 2025.

E.1 Absorption of convenience shocks

The first experiment asks who takes the other side when one cell's Treasury preferences are switched off. For each quarter and each cell j, I set $\psi_{0,j} = \psi_{1,j} = 0$ for cell j alone, re-solve the equilibrium, and record the change in every cell's dollar holdings. Because supply and the residual investor's holdings are fixed, the other cells' changes sum exactly to minus cell j's own change. I express each counterparty's change as a percentage of cell j's own change and average over quarters, weighting each quarter by the size of cell j's own change. Proposition 7 provides an analytical benchmark: for a marginal change in preferences, counterparty k takes the fraction $a_{k,t}/(1 - a_{j,t})$ of cell j's change. Therefore, each counterparty's response is proportional to its absorption share $a_{k,t} = (W_{k,t}/D_{k,t})/\mathcal{A}_t$, which increases in its wealth $W_{k,t}$ and in the elasticity of its demand, $1/D_{k,t}$, which is higher the lower its risk aversion, the variance of its excess return and the curvature of its Treasury preferences at its current share, $D_{k,t} = \gamma_k V_{k,t} + \psi_{1,k}/s_{k,t}^2$. Quarters in which the shocked cell sits at the same bound of $[0, 1]$ with and without its preferences involve no change in positions and are excluded.

Switching off a cell's Treasury preferences *raises* its Treasury holdings for all cells. The largest change in position is that of China, whose holdings rise by USD 1698.5 billion on average. Its largest counterparty is U.S. banks, which takes 32.5% of the change. The prominence of banks is due to their estimated preferences: they act as perfectly-elastic, risk-neutral intermediaries charging a fixed "penalty" to hold Treasuries. By country, the cells of the United States take 49.5%, and by sector, banks take 42.7%. Figure A6 shows the contribution of each counterparty to switching off Treasury preferences in each cell. Averaged across shocked cells, U.S. banks take 43.2% of a cell's change and China 27.7%: the same few holders absorb any convenience shock. They do so because absorption requires both size and elasticity: a holder takes up a large part of the change only if it is large and its demand responds strongly to the return, so large but inelastic holders and elastic but small ones absorb little.

On average, a cell's own absorption share is 5.9%, so general-equilibrium feedback on the shocked cell's own position is small, and nearly all of the adjustment falls on other holders.

The experiment complements the literature that estimates which investors absorb changes in the supply of government debt from their demand elasticities (Fang et al., 2025, Koijen and Yogo, 2026, Eren et al., 2026, Jansen et al., 2026).

In contrast with Fang et al. (2025), I find that domestic NBFIs play a limited role in absorbing excess supply in the context of the U.S. Treasury market, with an average contribution of 5.8%. The key difference is the type of shock: here I consider a loss of Treasury preferences for a specific investor, rather than an increase in total debt supply. The responses of absorbing investors then follow from a counterfactual on the estimated structural preferences and from market clearing rather than from reduced-form estimates, and the counterparties to a shock to each investor are identified directly.

The prominence of U.S. banks and China among the counterparties reflects the elasticity ranking of Section 6.4: U.S. banks, risk-neutral perfectly elastic at their indifference return, and China, a

risk-neutral large holder with a positive, but small, demand curvature, have the largest absorption shares, 44.9 and 26.8%, and absorb most of any shock. Risk neutrality, however, does not drive the result entirely: the British household and official cells, also risk neutral, absorb little.

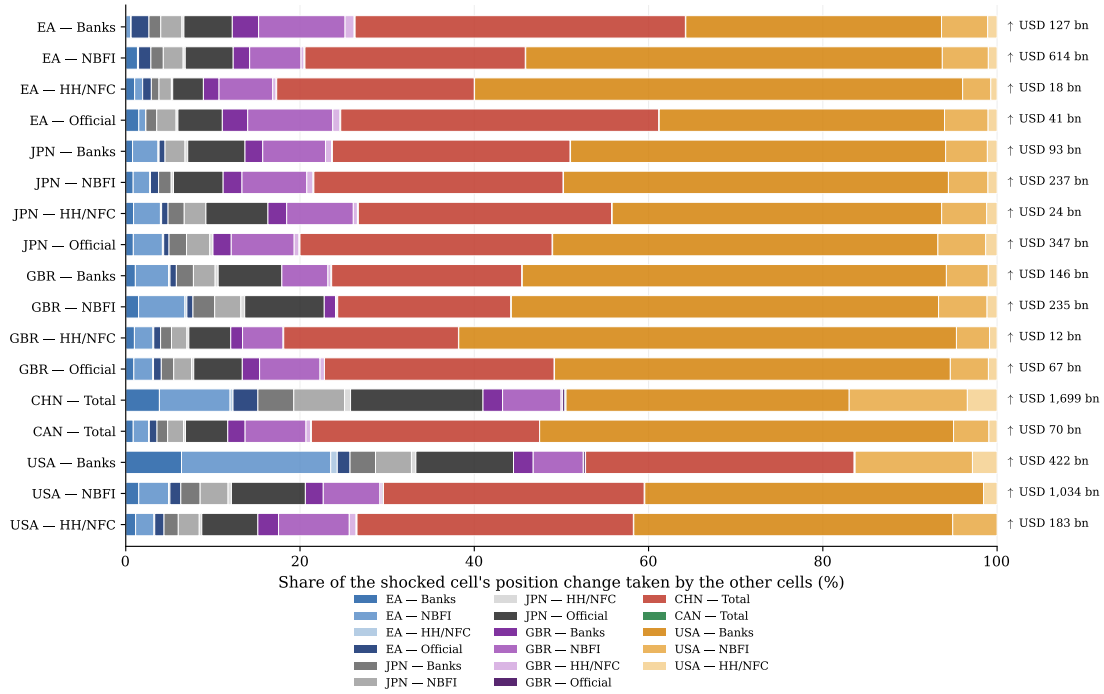


Figure A6. Counterparties to a cell's change in position

Notes: Each bar refers to the cell on the vertical axis, whose Treasury preferences alone are set to zero, Q4 2013–Q4 2025. Segments show the share of that cell's change in Treasury holdings taken by each other cell in the re-solved equilibrium, in percent, averaged over quarters with weights equal to the absolute size of the shocked cell's own change; segments sum to one hundred. The label on the right gives the direction and the average size of the shocked cell's own change, in USD billions. Quarters in which the shocked cell is at the same bound of $[0, 1]$ with and without its preferences are excluded (60 of 833 cell-quarters). Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

E.2 Supply sensitivity

The second experiment asks whether Treasury preferences make the equilibrium return more or less sensitive to Treasury supply, which is ambiguous by Propositions 9 and 10. By Proposition 8, the supply sensitivity is the inverse of the aggregate demand slope. I report it as the change in the equilibrium return, in basis points per annum, for a one percent increase in Treasury supply, evaluated at the equilibrium of each coalition of cells with active preferences. A perfectly elastic investor is the limiting case: when a risk-neutral cell without curvature holds an interior share, it absorbs additional supply at an unchanged return, and the supply sensitivity is zero (Appendix A.4).

Averaged over all quarters, Treasury preferences *raise* the sensitivity of the equilibrium return to supply. A one percent increase in supply raises the equilibrium return by 1.3 basis points per

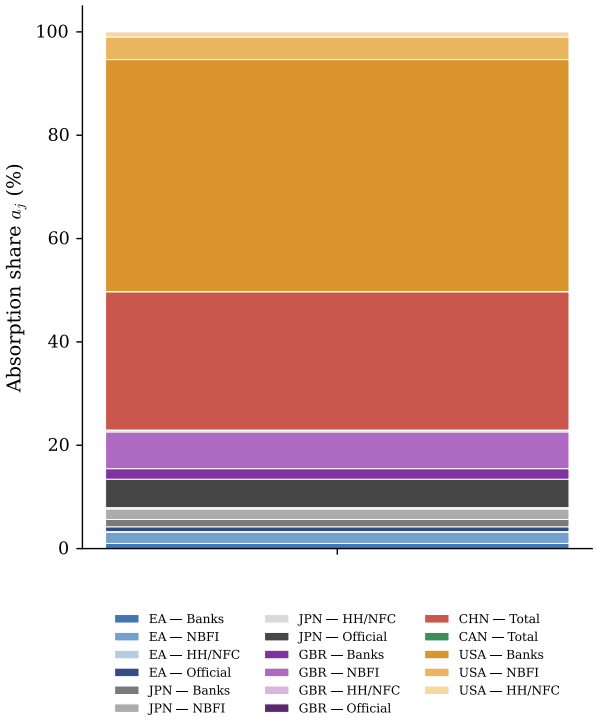


Figure A7. Absorption shares

Notes: Time-series averages of the absorption shares $a_{j,t} = (W_{j,t}/D_{j,t})/\mathcal{A}_t$ of Section 2.4, in percent, at the baseline equilibrium, Q4 2013–Q4 2025. In any quarter, a cell whose share is at a bound of $[0, 1]$ has an absorption share of zero, and a cell with a step demand that holds an interior share has an absorption share of one hundred percent. Cells with $\tilde{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

annum on average with estimated preferences and by 0.6 basis points without them, a difference of 0.7 basis points. Treasury preferences give the risk-neutral cells with positive curvature, China and the British cells, a finite elasticity and thereby raise it; U.S. banks, whose curvature is also zero, remain perfectly elastic with or without preferences, but their level parameter still affects the frequency at which they sit at a corner. The Shapley decomposition over all quarters attributes most of the difference to China, with a contribution of 0.78 basis points; the contributions of all other countries are small (Figure A8). This reflects China's elasticity channel: its curvature parameter is what gives it a finite slope, and without Treasury preferences its demand is perfectly elastic, which magnifies its contribution to the higher debt supply sensitivity.

The contributions do not all have the same sign: the preferences of some cells raise the supply sensitivity, while those of others lower it, as for the Japanese and U.S. cells taken together (-0.05 and -0.09 basis points; Figure A9 shows the contributions by sector). Sectors within a country can also pull in opposite directions: in the euro area, the preferences of banks raise the supply sensitivity, while those of NBFIs lower it (Figure A9a). Propositions 9 and 10 give the intuition. A cell's preferences amplify the sensitivity through its own elasticity: curvature makes its demand less elastic, and its negative level parameter lowers its Treasury share, which raises its marginal

curvature $\psi_{1,j}/s_{j,t}^2$ further. China is the leading example: without preferences it is perfectly elastic, and with them it must be compensated by a higher return for each additional Treasury it holds. A cell's preferences dampen the sensitivity through general-equilibrium feedback: because every estimated level parameter is negative, they raise the equilibrium return, which raises the Treasury shares of the cells with positive curvature and makes their demand more elastic. The Japanese official sector is an example: its preferences have no curvature, so they act on the supply sensitivity only through this feedback, which lowers it. U.S. banks lower it for a related reason: their preferences change the quarters in which they, perfectly elastic, hold an interior share and absorb supply at an unchanged return. The euro-area cells have no curvature either, so their preferences cannot amplify the sensitivity through their own elasticity at an interior share; they act only by shifting Treasury shares, which moves the equilibrium return and the quarters in which each cell sits at a corner, and the net of these effects differs in sign across sectors.

These results qualify the evidence that the special status of Treasuries lets the United States absorb additional debt with smaller increases in returns (Jiang et al., 2024e). In the model, Treasury preferences have an ambiguous impact on the sensitivity of returns to debt supply regardless of their direction, due to the balance of their direct and general equilibrium effects on elasticities. On net their effect is an *increase* in sensitivity, albeit modest.

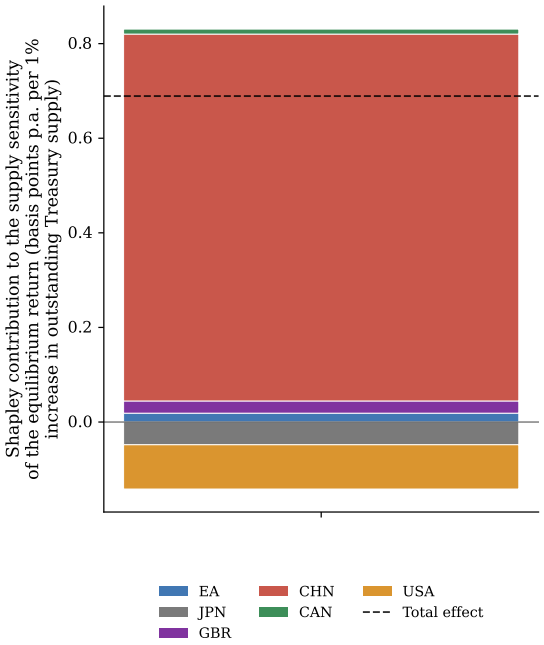


Figure A8. Supply sensitivity decomposition by country

Notes: Shapley contributions of each country's cells to the difference between the supply sensitivity of the equilibrium return, $\mathrm{d}R^*_{US,t+1}/\mathrm{d}\log S_t$, with and without Treasury preferences, in basis points per annum per one percent increase in Treasury supply, time-series averages over Q4 2013–Q4 2025. The dashed line is the total difference; a positive value means that Treasury preferences raise the sensitivity. Positive and negative contributions are stacked on opposite sides of zero, so the dashed line equals their net sum rather than the top of the bar. Shapley values are computed quarter by quarter by exact enumeration of all 2^{17} coalitions of cells, re-solving the market-clearing condition (9) for each coalition. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

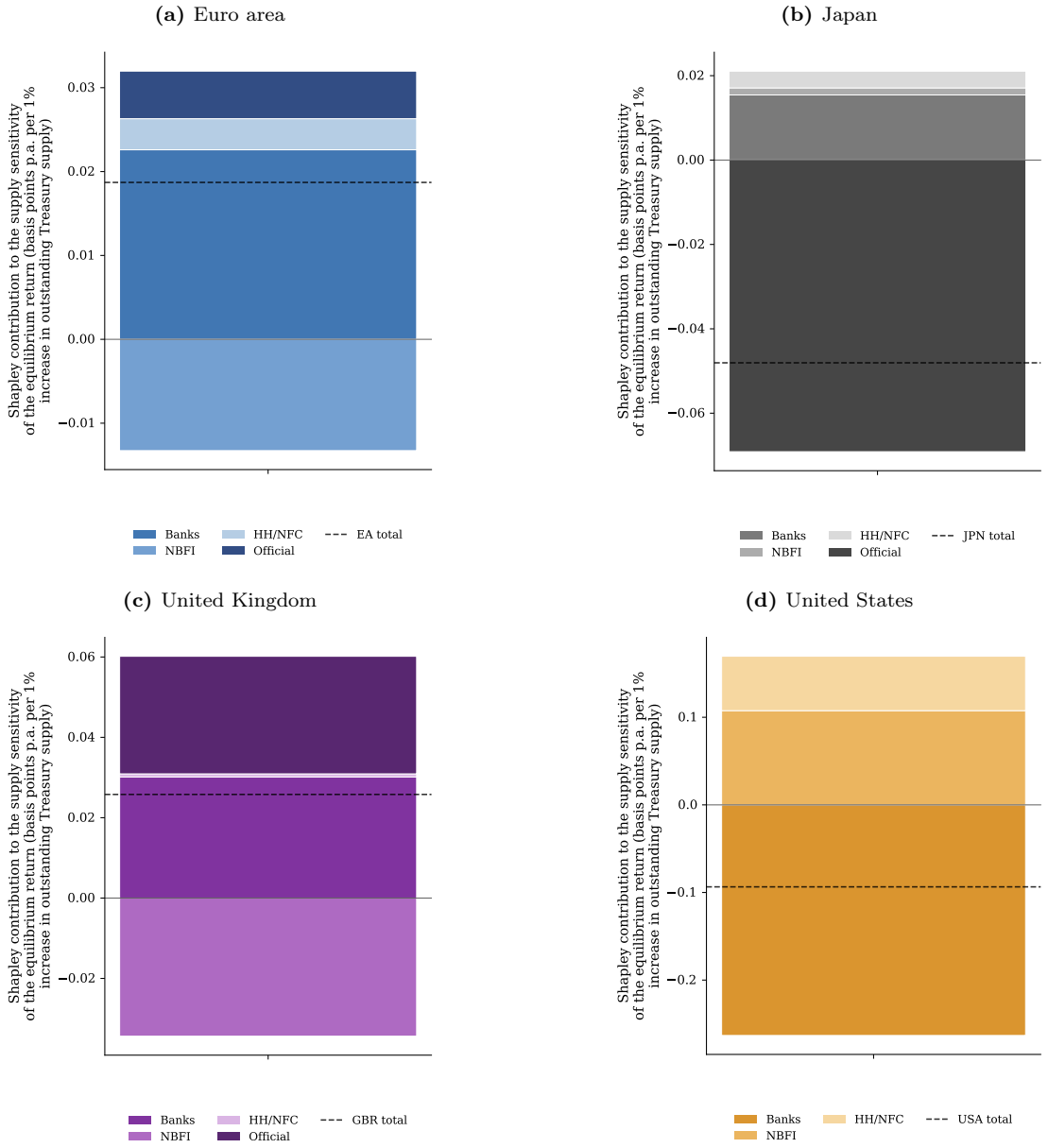


Figure A9. Supply sensitivity decomposition by sector

Notes: Shapley contributions of each sector's cell to the difference between the supply sensitivity of the equilibrium return, $\mathrm{d}R^*_{US,t+1}/\mathrm{d}\log S_t$, with and without Treasury preferences, in basis points per annum per one percent increase in Treasury supply, time-series averages over Q4 2013–Q4 2025, for the four countries with a sector split; positive values mean that the sector's Treasury preferences raise the supply sensitivity. Positive and negative contributions are stacked on opposite sides of zero. The dashed line marks the country total. Vertical scales differ across panels. Shapley values are computed quarter by quarter by exact enumeration of all 2^{17} coalitions of cells, re-solving the market-clearing condition (9) for each coalition. China, Canada and the U.S. official sector are not shown; cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

F Robustness of the experiments

This appendix repeats the experiments of Section 6 and Appendix E under three tests. The first two address the two weaknesses of the estimates discussed in Section 5: parameters that are not statistically significant, and cells whose instruments are weak. The insignificant-parameter test repeats every experiment. The weak-instrument test repeats the equilibrium experiments only, because the portfolio experiments are computed cell by cell and are unchanged for the surviving cells. Neither of the two re-estimates the model: every parameter that is not set to zero or excluded keeps its baseline point estimate, from the same specification as the main results, and every repeated experiment is recomputed exactly as in the baseline. The third test replaces the sectoral partition of the panel with a coarser official-versus-private one, which removes the estimation's dependence on the assumptions that allocate a country's private holdings across sectors. Unlike the other two, it does re-estimate the model, on a smaller set of cells and with a much narrower instrument block.

F.1 Setting insignificant parameters to zero

Implementation. Every level parameter $\psi_{0,c,i}$ and every curvature parameter $\psi_{1,c,i}$ whose two-sided 90% confidence interval, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987), contains zero, is set to zero before the experiments are recomputed. Parameters already at their bound of zero are exempt. Risk aversion is left unchanged in every cell: each risk-aversion estimate away from its bound is significant. The level parameter is set to zero in 7 cells: euro-area banks, euro-area households and non-financial corporations, euro-area official sector, Japanese banks, Japanese non-bank financial institutions, Japanese official sector and U.S. banks. Every cell that loses its level parameter already has its curvature at the bound, so its Treasury preferences are switched off entirely. The curvature parameter is removed in 2 cells, British households and non-financial corporations and China; both are risk neutral, and their demand thereby becomes perfectly elastic at their indifference return.

Results. The aggregate premium is robust. Treasury preferences still raise the equilibrium return in all 49 quarters, and the average premium, −79.2 basis points per annum, is close to the baseline's −77.7, as is the risk component, −7.7 basis points (Figure A12). China remains the largest contributor, and its share of the premium rises to 51.4% (Figure A13). The contributions of U.S. banks and of Japan all but vanish, since their level parameters are among those set to zero (Figure A14). Across sectors, non-bank financial institutions become even more prominent, with 38.8% of the premium, while the shares of banks and of the official cells excluding China shrink to 3.2 and 1.4%. The main finding of Section 6.4 therefore survives: the premium is driven by a few large and elastic holders, led by China. Because China's holdings are predominantly official, the effective role of official demand survives as well, even though the official cells of the euro area, Japan and the United Kingdom contribute little.

The portfolio experiments change only for the cells whose parameters are modified (Figures A10 and A11). Where Treasury preferences are switched off entirely, the estimated and counterfactual markers coincide. For British households and non-financial corporations and for China, removing the curvature parameter leaves a step demand at the estimates, so their sensitivity is zero in every quarter. The euro-area and Japanese cells that keep a significant level parameter, euro-area non-bank financial institutions and Japanese households and non-financial corporations, have unmodified parameters, so their results are those of the baseline. The difference between their estimated and counterfactual sensitivities is the composition effect of Section 6.2: removing the

level parameter raises the desired share, which reaches the upper bound of one in more quarters, and each constrained quarter enters the average with a zero, although $\psi_{1,c,i}$ is unchanged.

Treasury preferences amplify the response of the equilibrium return to supply in this variant as well, and China is still the main contributor. However, the total effect is much larger, 5.6 basis points per one percent of supply against 0.7 in the baseline, because China, whose demand becomes a step function without its curvature, and U.S. banks, whose demand already is one, sit at a corner rather than at their indifference return: China holds essentially no Treasuries in most quarters, and U.S. banks, without their level parameter, hold the maximum share in every quarter, so neither absorbs additional supply as they do in the baseline (Figure A17).

The only result that does not survive is the ranking of convenience shock absorption. The level parameter of U.S. banks is set to zero, so their absorption share falls from 44.9% in the baseline to 0.0%, and the largest average absorption share passes to U.S. non-bank financial institutions, at 23.0% (Figure A16). With its curvature removed, China's demand becomes perfectly elastic at its indifference return, but its absorption share falls only to 20.4%, from 26.8. On average, U.S. non-bank financial institutions and euro-area non-bank financial institutions also become the main counterparties to other cells' changes in position, taking 26.7 and 19.4% (Figure A15).

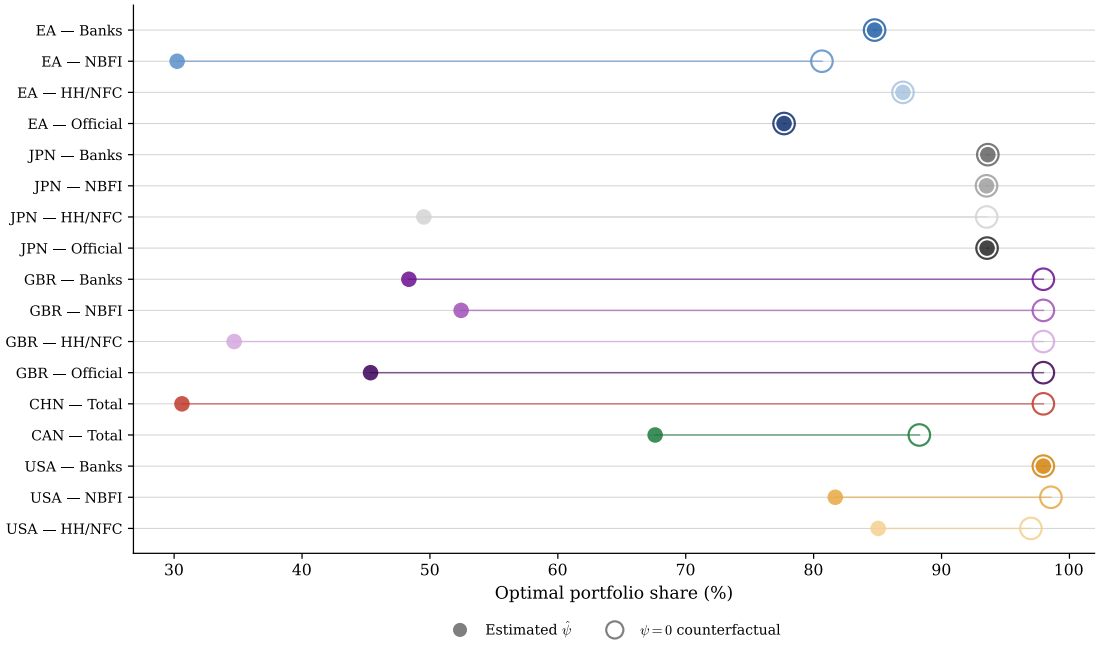


Figure A10. Insignificant-parameter test: portfolio share experiment

Notes: Time-series average of the optimal Treasury portfolio share $s^*_{c,i,t}$, in percent, at the estimated preference parameters $(\hat{\psi}_{0,c,i}, \hat{\psi}_{1,c,i})$ (filled markers) and in the counterfactual $\psi_{0,c,i} = \psi_{1,c,i} = 0$ (open markers), Q4 2013–Q4 2025. Shares are evaluated quarter by quarter, one cell at a time, from equation (5) at the cell's observed excess return $E_{c,t}$, variance $V_{c,t}$ and covariance $C_{c,t}$, with estimated risk aversion $\hat{\gamma}_{c,i}$, and are bounded between zero and one. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution of Appendix A.4. Every $\psi_{0,c,i}$ and $\psi_{1,c,i}$ whose 90% confidence interval, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987), contains zero is set to zero (Appendix F.1); other parameters keep their baseline estimates. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

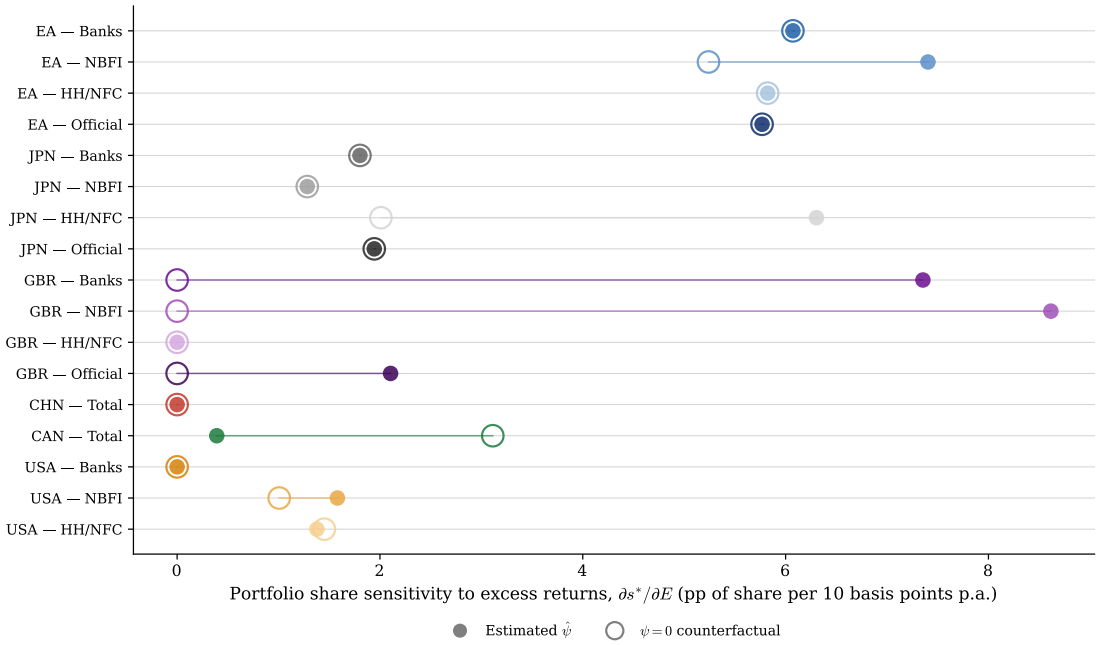


Figure A11. Insignificant-parameter test: portfolio sensitivity experiment

Notes: Time-series average of the sensitivity of the optimal Treasury portfolio share to the excess return, $\mathrm{d}s^*_{c,i,t}/\mathrm{d}E_{c,t}$ (Proposition 2), in percentage points of portfolio share per ten basis points per annum of excess return, at the estimated preference parameters (filled markers) and in the counterfactual $\psi_{0,c,i} = \psi_{1,c,i} = 0$ (open markers), Q4 2013–Q4 2025. The derivative is evaluated quarter by quarter, one cell at a time, at the share that is optimal under each set of parameters and at the cell's observed $(E_{c,t}, V_{c,t}, C_{c,t})$, and is set to zero in quarters in which that share is at zero or one. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution of Appendix A.4; where their curvature is also zero, their sensitivity is zero in every quarter. Every $\psi_{0,c,i}$ and $\psi_{1,c,i}$ whose 90% confidence interval, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987), contains zero is set to zero (Appendix F.1); other parameters keep their baseline estimates. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

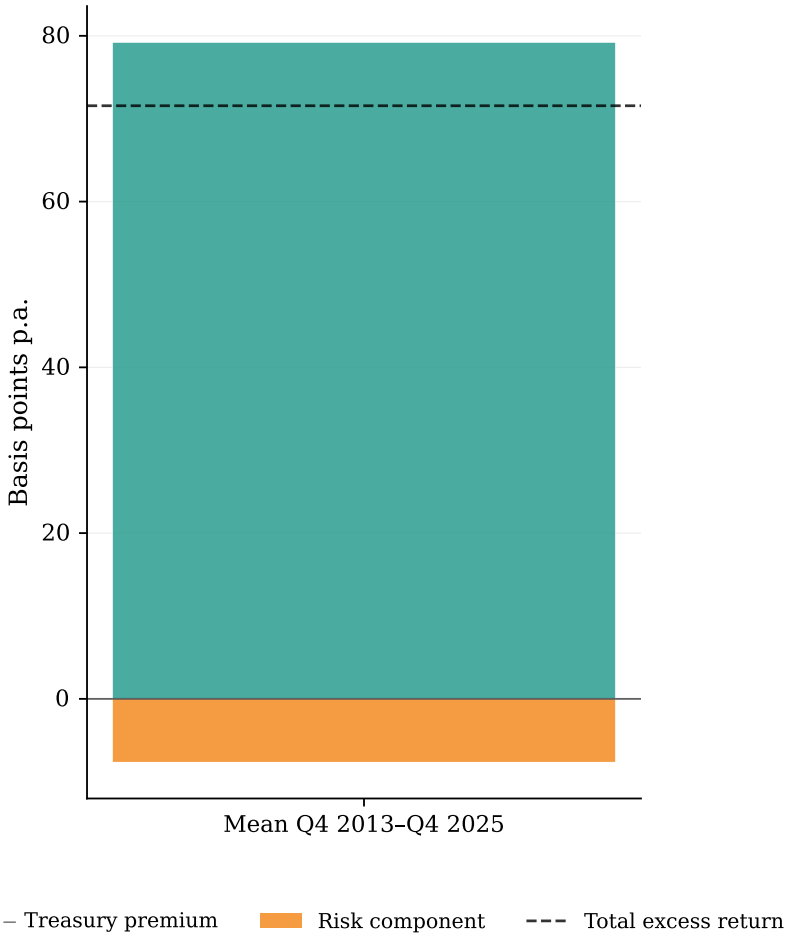


Figure A12. Insignificant-parameter test: excess return decomposition

Notes: Time-series averages, in basis points per annum, Q4 2013–Q4 2025. The dashed line is the model-implied equilibrium excess return of U.S. Treasuries, $E_t^* = R_{US,t+1}^* - \bar{R}_t^F$, where $\bar{R}_t^F$ averages the cells' benchmark returns with weights equal to each country's number of cells. It is split into the Treasury-premium component $-TP_t = R_{US,t+1}^* - R_{US,t+1}^*\big|_{\psi=0}$, positive when Treasury preferences raise the equilibrium return, and the risk component $R_{US,t+1}^*\big|_{\psi=0} - \bar{R}_t^F$. Both equilibrium returns solve the market-clearing condition (9) quarter by quarter, with each cell's own benchmark return; $\bar{R}_t^F$ is used for reporting only. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). Every $\psi_{0,c,i}$ and $\psi_{1,c,i}$ whose 90% confidence interval, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987), contains zero is set to zero (Appendix F.1); other parameters keep their baseline estimates. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

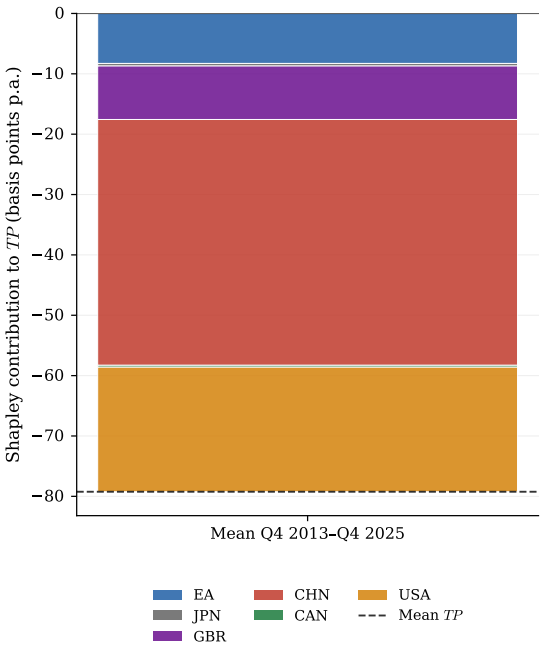


Figure A13. Insignificant-parameter test: Treasury premium decomposition by country

Notes: Shapley contributions of each country's cells to the model-implied Treasury premium $TP_t = R^*_{US,t+1}\big|_{\psi=0} - R^*_{US,t+1}$, time-series averages in basis points per annum, Q4 2013–Q4 2025. The dashed line marks the average TP_t; a negative contribution means that the country's preferences raise the equilibrium Treasury return. Shapley values are computed quarter by quarter by exact enumeration of all 2^{17} coalitions of cells, re-solving the market-clearing condition (9) for each coalition; cell contributions sum to TP_t, and a country's contribution is the sum of its cells' contributions. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). Every $\psi_{0,c,i}$ and $\psi_{1,c,i}$ whose 90% confidence interval, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987), contains zero is set to zero (Appendix F.1); other parameters keep their baseline estimates. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

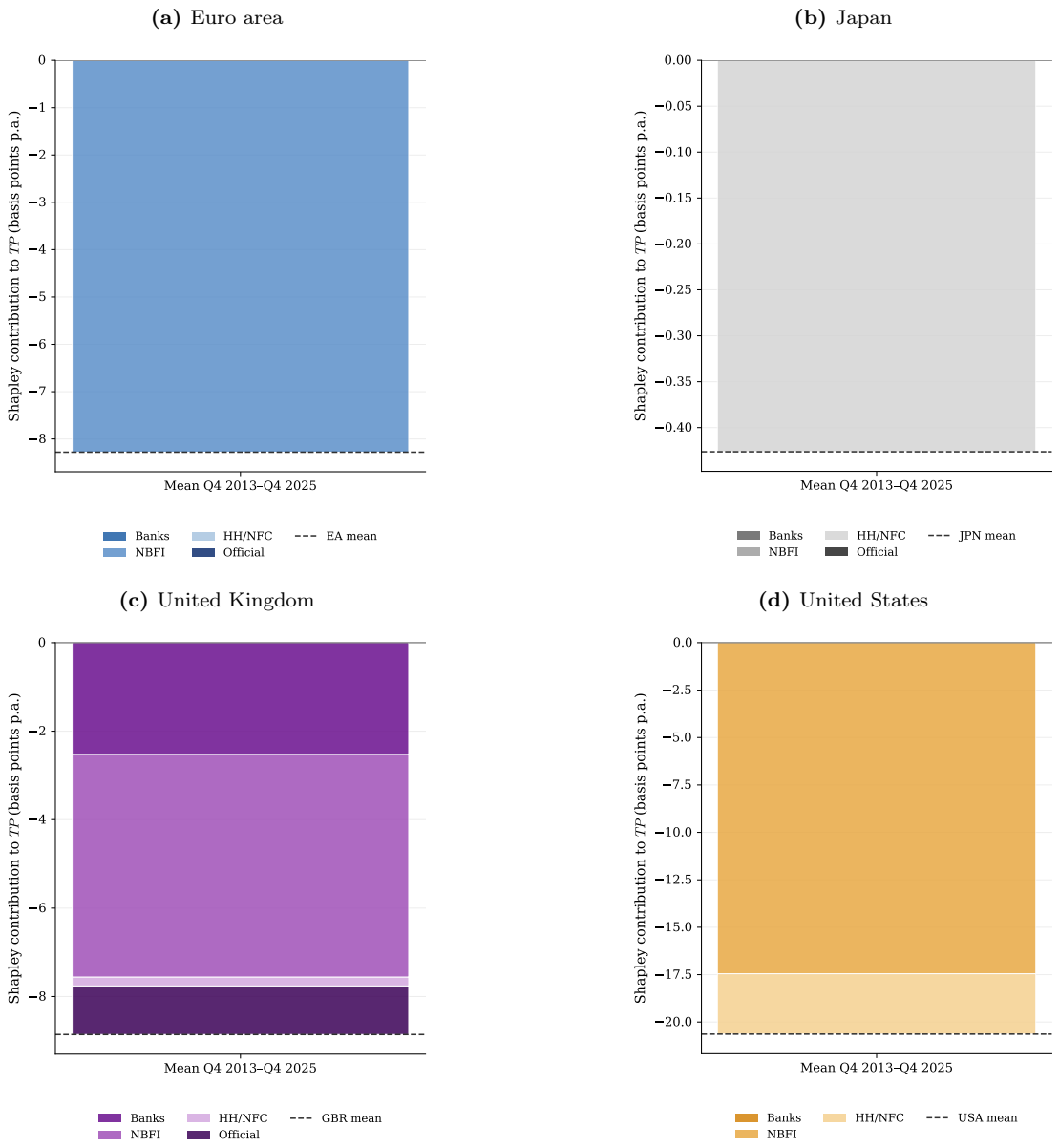


Figure A14. Insignificant-parameter test: Treasury premium decomposition by sector

Notes: Shapley contributions of each sector's cell to the model-implied Treasury premium $TP_t = R^*_{US,t+1}\big|_{\psi=0} - R^*_{US,t+1}$, time-series averages in basis points per annum, Q4 2013–Q4 2025, for the four countries with a sector split; negative values raise the equilibrium return. The dashed line marks the country total. Vertical scales differ across panels. Shapley values are computed quarter by quarter by exact enumeration of all 2^{17} coalitions of cells, re-solving the market-clearing condition (9) for each coalition; cell contributions sum to TP_t. China, Canada and the U.S. official sector are not shown; cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). Every $\psi_{0,c,i}$ and $\psi_{1,c,i}$ whose 90% confidence interval, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987), contains zero is set to zero (Appendix F.1); other parameters keep their baseline estimates. Cells with preferences switched off entirely contribute zero. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

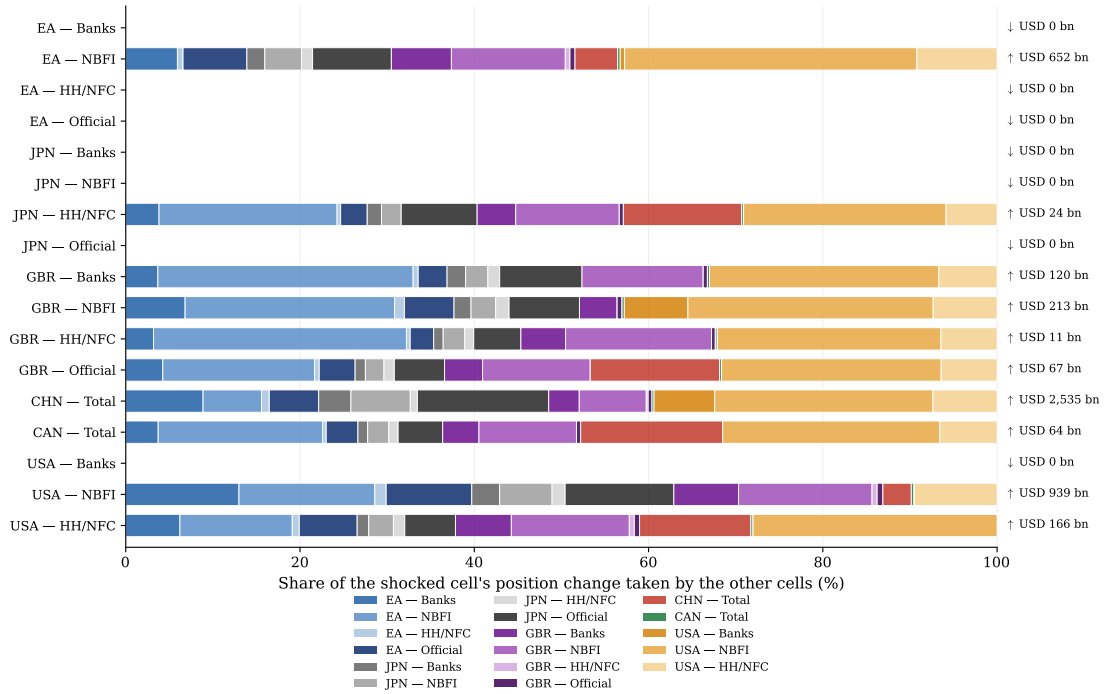


Figure A15. Insignificant-parameter test: counterparties to a cell's change in position

Notes: Each bar refers to the cell on the vertical axis, whose Treasury preferences alone are set to zero, Q4 2013–Q4 2025. Segments show the share of that cell's change in Treasury holdings taken by each other cell in the re-solved equilibrium, in percent, averaged over quarters with weights equal to the absolute size of the shocked cell's own change; segments sum to one hundred. The label on the right gives the direction and the average size of the shocked cell's own change, in USD billions. Quarters in which the shocked cell is at the same bound of $[0, 1]$ with and without its preferences are excluded. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). Every $\psi_{0,c,i}$ and $\psi_{1,c,i}$ whose 90% confidence interval, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987), contains zero is set to zero (Appendix F.1); other parameters keep their baseline estimates. Cells whose preferences are switched off entirely have no change in position and do not appear as shocked cells. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

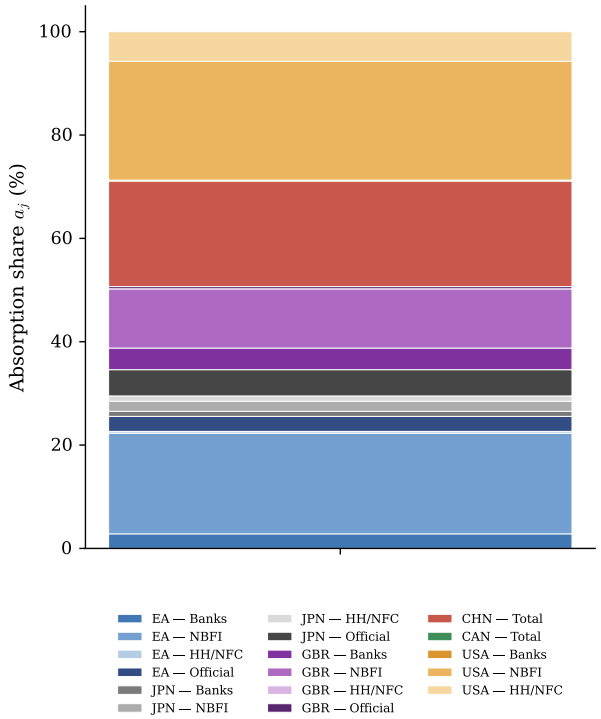


Figure A16. Insignificant-parameter test: absorption shares

Notes: Time-series averages of the absorption shares $a_{j,t} = (W_{j,t}/D_{j,t})/\mathcal{A}_t$ of Section 2.4, in percent, at the equilibrium of this test, Q4 2013–Q4 2025. In any quarter, a cell whose share is at a bound of $[0, 1]$ has an absorption share of zero, and a cell with a step demand that holds an interior share has an absorption share of one hundred percent. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). Every $\psi_{0,c,i}$ and $\psi_{1,c,i}$ whose 90% confidence interval, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987), contains zero is set to zero (Appendix F.1); other parameters keep their baseline estimates. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

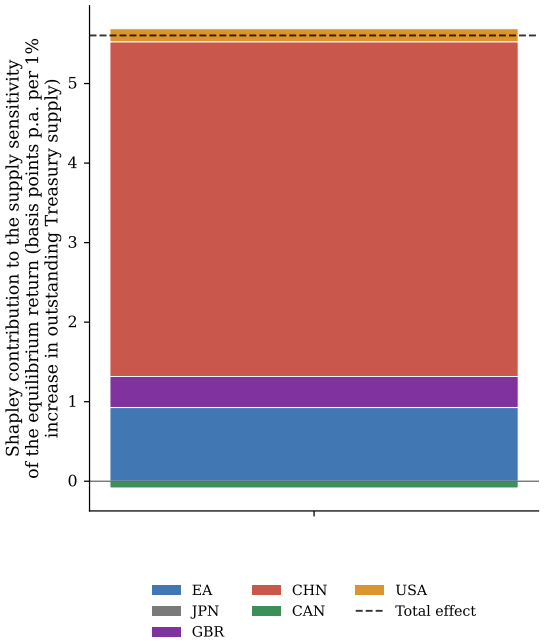


Figure A17. Insignificant-parameter test: supply sensitivity decomposition by country

Notes: Shapley contributions of each country's cells to the difference between the supply sensitivity of the equilibrium return, $\mathrm{d}R^{*}_{US,t+1}/\mathrm{d}\log S_t$, with and without Treasury preferences, in basis points per annum per one percent increase in Treasury supply, time-series averages over Q4 2013–Q4 2025. The dashed line is the total difference; a positive value means that Treasury preferences raise the sensitivity. Positive and negative contributions are stacked on opposite sides of zero, so the dashed line equals their net sum rather than the top of the bar. Shapley values are computed quarter by quarter by exact enumeration of all 2^{17} coalitions of cells, re-solving the market-clearing condition (9) for each coalition. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). Every $\psi_{0,c,i}$ and $\psi_{1,c,i}$ whose 90% confidence interval, based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987), contains zero is set to zero (Appendix F.1); other parameters keep their baseline estimates. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

F.2 Excluding cells with weak instruments

Implementation. A cell is excluded if the Olea and Pflueger (2013) effective F-statistic falls below the rule-of-thumb threshold of 10 for either of its two instrumented variables, $x_{c,i,t}$ or $1/s_{c,i,t}$ (Table A4). This excludes the 7 cells that are weak in at least one test (Appendix D.2): euro-area banks; euro-area households and non-financial corporations; British banks; British non-bank financial institutions; British households and non-financial corporations; British official sector; U.S. non-bank financial institutions. In every quarter, the Treasury holdings of an excluded cell are reassigned to the inelastic residual investor H_t^{Res}, so they still absorb supply but no longer respond to the return, and the Shapley decompositions run over the remaining cells only. The surviving cells keep their baseline estimates, whose instruments continue to use the excluded cells as donors. The common benchmark of the excess-return decomposition averages over the surviving cells. The portfolio experiments of Sections 6.1 and 6.2 are computed cell by cell, so for the surviving cells they are identical to the baseline and are not repeated.

Results. The aggregate premium is again robust, at -76.7 basis points per annum on average, and the risk component remains small, at -11.5 basis points (Figure A18). China's share of the premium rises to 56.4%, from 37.6 in the baseline, and it remains by far the largest contributor (Figure A19). Most of the excluded cells' share is absorbed there. The U.S. share falls to 22.4%, from 34.2, because U.S. non-bank financial institutions are excluded; the United Kingdom loses all four of its cells and thus contributes nothing; and the remaining countries' shares stay close to their baseline values, 7.5% against 7.9 for the euro area, 13.2 against 12.0 for Japan and 0.5 against 0.2 for Canada. The sector ranking changes (Figure A20). Once the excluded non-bank cells are removed, banks, with -16.0 basis points, overtake non-bank financial institutions, with -8.3, as the largest sector outside China and Canada, while the contribution of the official cells, -5.9 basis points, is close to the baseline's -6.0. As in the baseline, Treasury preferences raise the sensitivity of excess returns to debt supply by 1.9 basis points per one percent of supply, again mainly through China, whose Shapley contribution is 1.39 basis points (Figure A23). The largest average absorption share is that of U.S. banks, at 46.9%, and China's rises to 35.4% (Figure A22); U.S. banks and China remain the main counterparties to other cells' changes in position (Figure A21).

Across these two tests, four results of Section 6 are robust: the negative sign and the size of the model-implied Treasury premium; China's position as its largest driver, and hence the large effective role of official demand, mainly driven by China; and the amplification of excess returns' sensitivity to debt supply. In contrast, the relative contributions of banks and non-bank financial institutions and the U.S. share of the premium depend on cells that are imprecisely estimated or weakly identified, and should be read with corresponding caution.

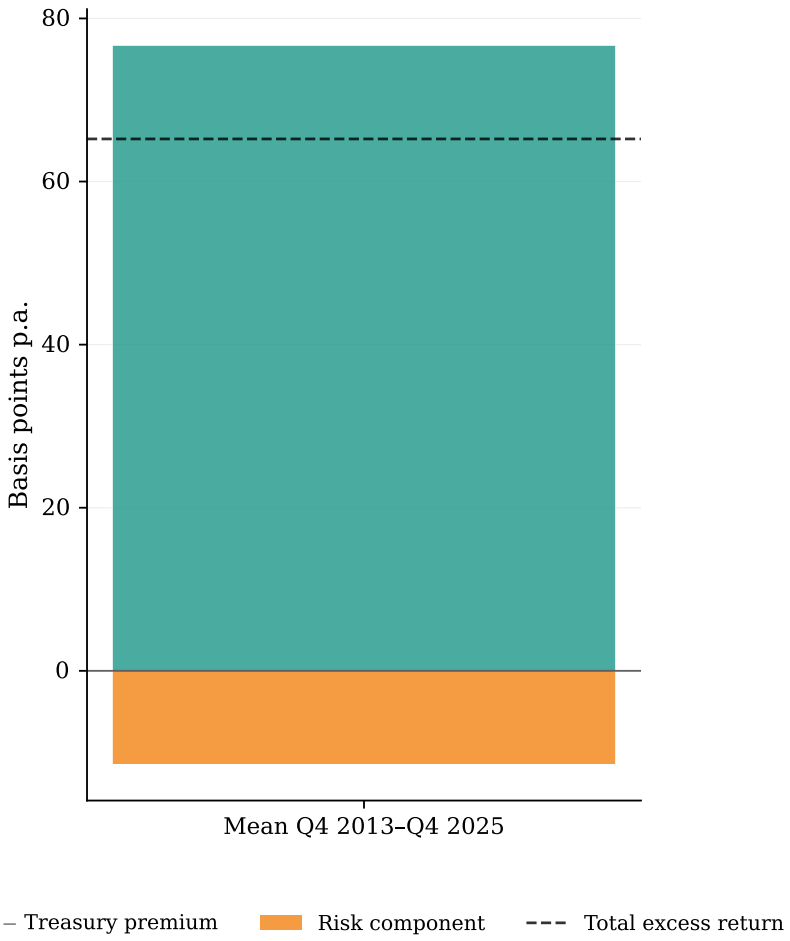


Figure A18. Weak-instrument test: excess return decomposition

Notes: Time-series averages, in basis points per annum, Q4 2013–Q4 2025. The dashed line is the model-implied equilibrium excess return of U.S. Treasuries, $E_t^* = R_{US,t+1}^* - \bar{R}_t^F$, where $\bar{R}_t^F$ averages the surviving cells' benchmark returns with weights equal to each country's number of surviving cells. It is split into the Treasury-premium component $-TP_t = R_{US,t+1}^* - R_{US,t+1}^*\big|_{\psi=0}$, positive when Treasury preferences raise the equilibrium return, and the risk component $R_{US,t+1}^*\big|_{\psi=0} - \bar{R}_t^F$. Both equilibrium returns solve the market-clearing condition (9) quarter by quarter, with each cell's own benchmark return; $\bar{R}_t^F$ is used for reporting only. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The 7 cells with an effective F-statistic below 10 for either $x_{c,i,t}$ or $1/s_{c,i,t}$ are excluded (Appendix F.2); their holdings are reassigned to the inelastic residual investor and the other cells keep their baseline estimates. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

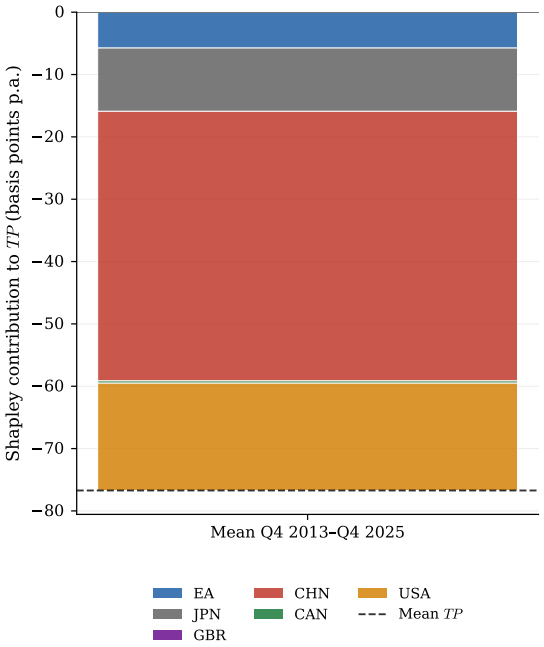


Figure A19. Weak-instrument test: Treasury premium decomposition by country

Notes: Shapley contributions of each country's cells to the model-implied Treasury premium $TP_t = R^*_{US,t+1}\big|_{\psi=0} - R^*_{US,t+1}$, time-series averages in basis points per annum, Q4 2013–Q4 2025. The dashed line marks the average TP_t; a negative contribution means that the country's preferences raise the equilibrium Treasury return. Shapley values are computed quarter by quarter by exact enumeration of all coalitions of the surviving cells, re-solving the market-clearing condition (9) for each coalition; cell contributions sum to TP_t, and a country's contribution is the sum of its cells' contributions. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The 7 cells with an effective F-statistic below 10 for either $x_{c,i,t}$ or $1/s_{c,i,t}$ are excluded (Appendix F.2); their holdings are reassigned to the inelastic residual investor and the other cells keep their baseline estimates. Excluded cells: euro-area banks, euro-area households and non-financial corporations, all British cells and U.S. non-bank financial institutions; a country with no surviving cell contributes zero. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

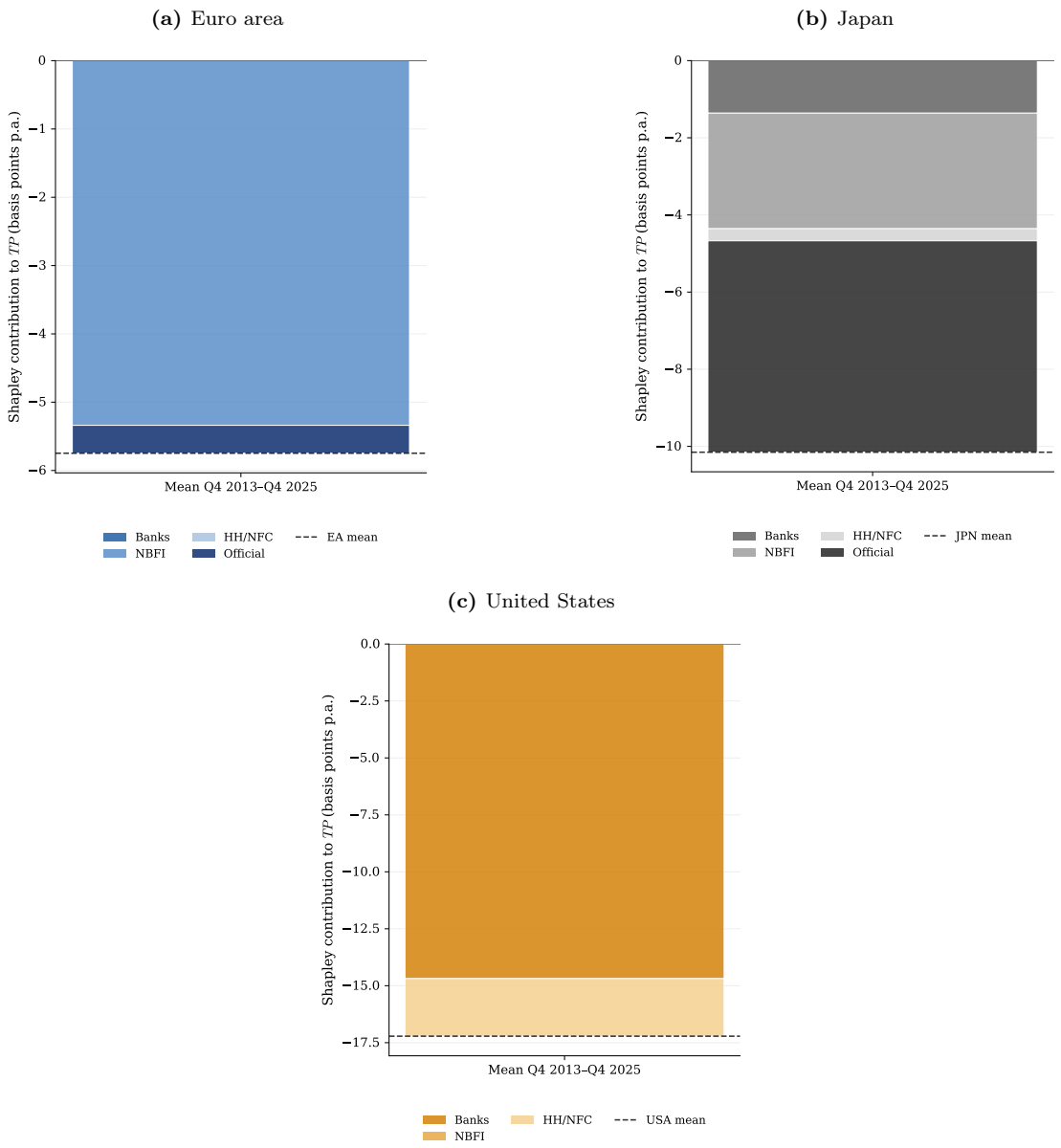


Figure A20. Weak-instrument test: Treasury premium decomposition by sector

Notes: Shapley contributions of each sector's cell to the model-implied Treasury premium $TP_t = R^*_{US,t+1}\big|_{\psi=0} - R^*_{US,t+1}$, time-series averages in basis points per annum, Q4 2013–Q4 2025, for the countries with a sector split; negative values raise the equilibrium return. The dashed line marks the country total. Vertical scales differ across panels. Shapley values are computed quarter by quarter by exact enumeration of all coalitions of the surviving cells, re-solving the market-clearing condition (9) for each coalition; cell contributions sum to TP_t. China, Canada and the U.S. official sector are not shown; cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The 7 cells with an effective F-statistic below 10 for either $x_{c,i,t}$ or $1/s_{c,i,t}$ are excluded (Appendix F.2); their holdings are reassigned to the inelastic residual investor and the other cells keep their baseline estimates. Excluded cells contribute zero; a country with no surviving cell has no panel. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

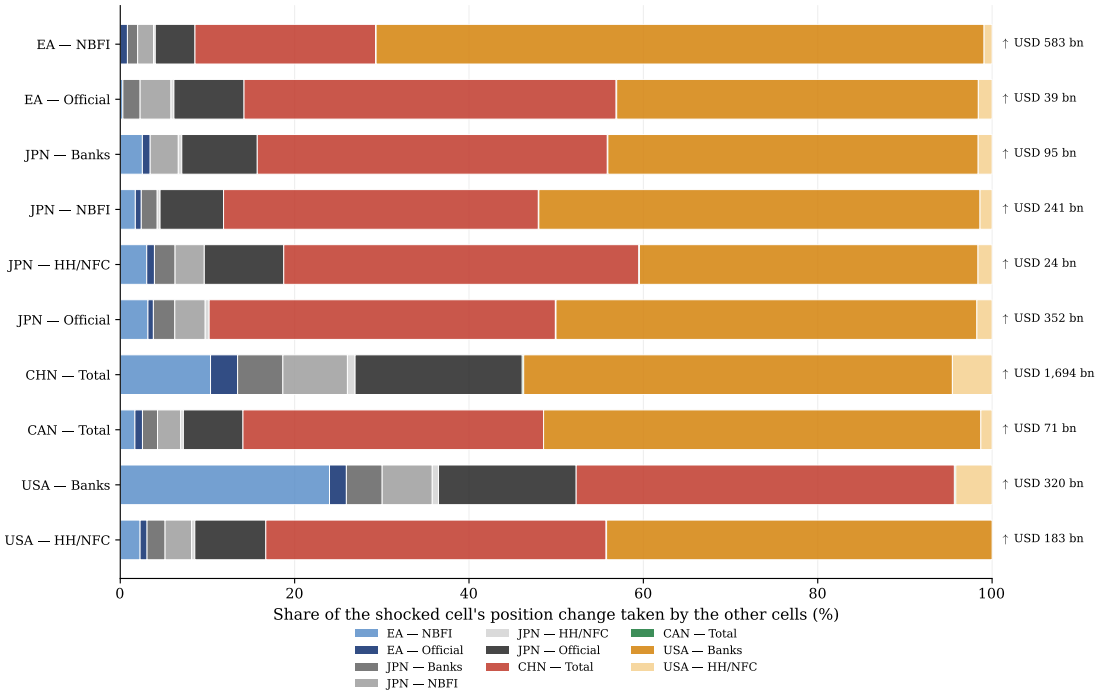


Figure A21. Weak-instrument test: counterparties to a cell's change in position

Notes: Each bar refers to the cell on the vertical axis, whose Treasury preferences alone are set to zero, Q4 2013–Q4 2025. Segments show the share of that cell's change in Treasury holdings taken by each other cell in the re-solved equilibrium, in percent, averaged over quarters with weights equal to the absolute size of the shocked cell's own change; segments sum to one hundred. The label on the right gives the direction and the average size of the shocked cell's own change, in USD billions. Quarters in which the shocked cell is at the same bound of $[0, 1]$ with and without its preferences are excluded. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The 7 cells with an effective F-statistic below 10 for either $x_{c,i,t}$ or $1/s_{c,i,t}$ are excluded (Appendix F.2); their holdings are reassigned to the inelastic residual investor and the other cells keep their baseline estimates. Only surviving cells are shown.
Sources: Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

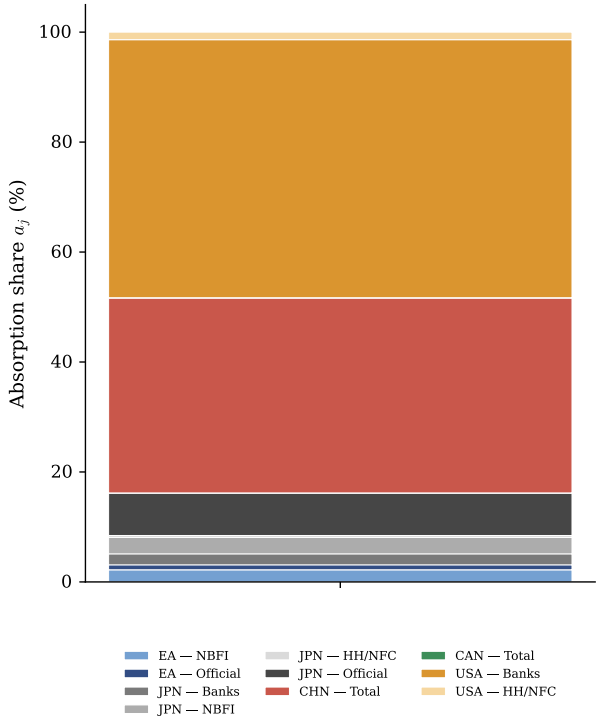


Figure A22. Weak-instrument test: absorption shares

Notes: Time-series averages of the absorption shares $a_{j,t} = (W_{j,t}/D_{j,t})/\mathcal{A}_t$ of Section 2.4, in percent, at the equilibrium of this test, Q4 2013–Q4 2025. In any quarter, a cell whose share is at a bound of $[0, 1]$ has an absorption share of zero, and a cell with a step demand that holds an interior share has an absorption share of one hundred percent. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The 7 cells with an effective F-statistic below 10 for either $x_{c,i,t}$ or $1/s_{c,i,t}$ are excluded (Appendix F.2); their holdings are reassigned to the inelastic residual investor and the other cells keep their baseline estimates. Only surviving cells are shown. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

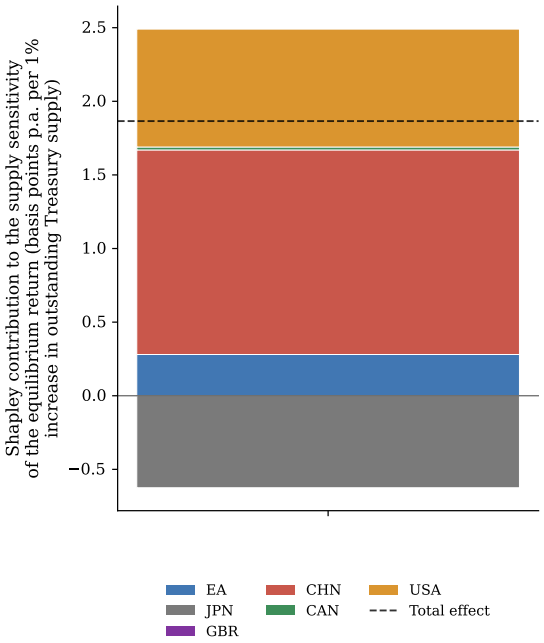


Figure A23. Weak-instrument test: supply sensitivity decomposition by country

Notes: Shapley contributions of each country's cells to the difference between the supply sensitivity of the equilibrium return, $\mathrm{d}R^*_{US,t+1}/\mathrm{d}\log S_t$, with and without Treasury preferences, in basis points per annum per one percent increase in Treasury supply, time-series averages over Q4 2013–Q4 2025. The dashed line is the total difference; a positive value means that Treasury preferences raise the sensitivity. Positive and negative contributions are stacked on opposite sides of zero, so the dashed line equals their net sum rather than the top of the bar. Shapley values are computed quarter by quarter by exact enumeration of all coalitions of the surviving cells, re-solving the market-clearing condition (9) for each coalition. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The 7 cells with an effective F-statistic below 10 for either $x_{c,i,t}$ or $1/s_{c,i,t}$ are excluded (Appendix F.2); their holdings are reassigned to the inelastic residual investor and the other cells keep their baseline estimates. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

F.3 Collapsing the sectoral split to official and private

Implementation. This test replaces the sectoral partition of the panel with an official-versus-private one and re-estimates the model on the resulting 9 cells: the euro area, Japan and the United Kingdom each split into an official and a private cell, the United States collapsed into a single private cell, since it has no official cell in the baseline, and China and Canada left as country aggregates. Its purpose is to remove the estimation's dependence on the assumptions that allocate a country's private Treasury holdings across banks, non-bank financial institutions, and households and non-financial corporations. In the baseline those three series rely on the necessary but unpalatable assumption that the relative sectoral split of U.S. Treasury holdings mirrors that of overall U.S.-issued debt, which I relax in this robustness test.

Figure A24 shows the resulting official/private split, the counterpart of Figure 2. Because the official and private series sum to the restated Treasury total, the same denominator against which the official shares quoted in Section 3.2 are measured, the official share for the euro area and Japan is the one reported there: it falls from 37.4% to 20.8% in the euro area, whose growth in holdings is mostly private, and stays between 45.7% and 68.1% in Japan, where official holdings exceed private ones in most quarters, while in the United Kingdom it never exceeds 24.7%.

Unlike the other two tests, this one re-estimates the model. The instruments are rebuilt on the official/private taxonomy under the same leave-own-country-and-own-sector donor rule, which leaves between 4 and 8 donor cells per target, against 10 to 16 in the baseline, and hence between 3 and 7 instruments, against 9 to 15. Everything else matches the baseline specification.

Results. The sign pattern of the estimates is consistent with the baseline. The level parameter is negative in all 9 cells, and Treasury preferences raise rather than lower the required excess return in all 9 of them, by between 43.7 and 138.2 basis points per annum with a median of 68.2, the largest effect being that of the U.S. private sector and the smallest that of the euro-area official sector. Risk aversion is pinned to zero for the United Kingdom and China, as in the baseline, while in the other cells it ranges from 0.42 to 8.40, the latter for the United States, a wider range than the baseline's 1.41 to 4.56.

The aggregate premium is of the same sign and order of magnitude as in the baseline, at -74.4 basis points per annum on average against -77.7 in the baseline, the average equilibrium excess return is 68.9 basis points and the risk component is small, -5.5 basis points (Figure A27). China remains the largest single contributor, with 36.8% of the premium, followed by the euro area with 20.8%, the United States with 16.5, Japan with 14.8, the United Kingdom with 10.7 and Canada with 0.3 (Figure A28). The official cells together, excluding China, account for 10.8% of the premium and the private cells for 52.0%, the remaining 37.1% coming from the Chinese and Canadian aggregates (Figure A29). Treasury preferences again raise the sensitivity of the equilibrium return to supply, by 0.9 basis points per one percent of supply, and China is again the largest contributor, at 0.77 basis points, while Japan and the United Kingdom contribute negatively (Figure A30).

In the portfolio experiments, removing a cell's Treasury preferences raises its desired share in all 9 cells, by between 13.8 and 50.9 percentage points, least for the U.S. private sector and most for the euro-area private sector (Figure A31). The sensitivity of portfolio shares to excess returns falls for all cells except for the United States, consistent with the baseline except for the case of Canada. The largest average absorption share is that of the Chinese aggregate, at 39.4% (Figure A33), and the Chinese aggregate and the euro-area private sector are the main counterparties to other cells' changes in position, taking 42.4 and 27.9% on average (Figure A34).

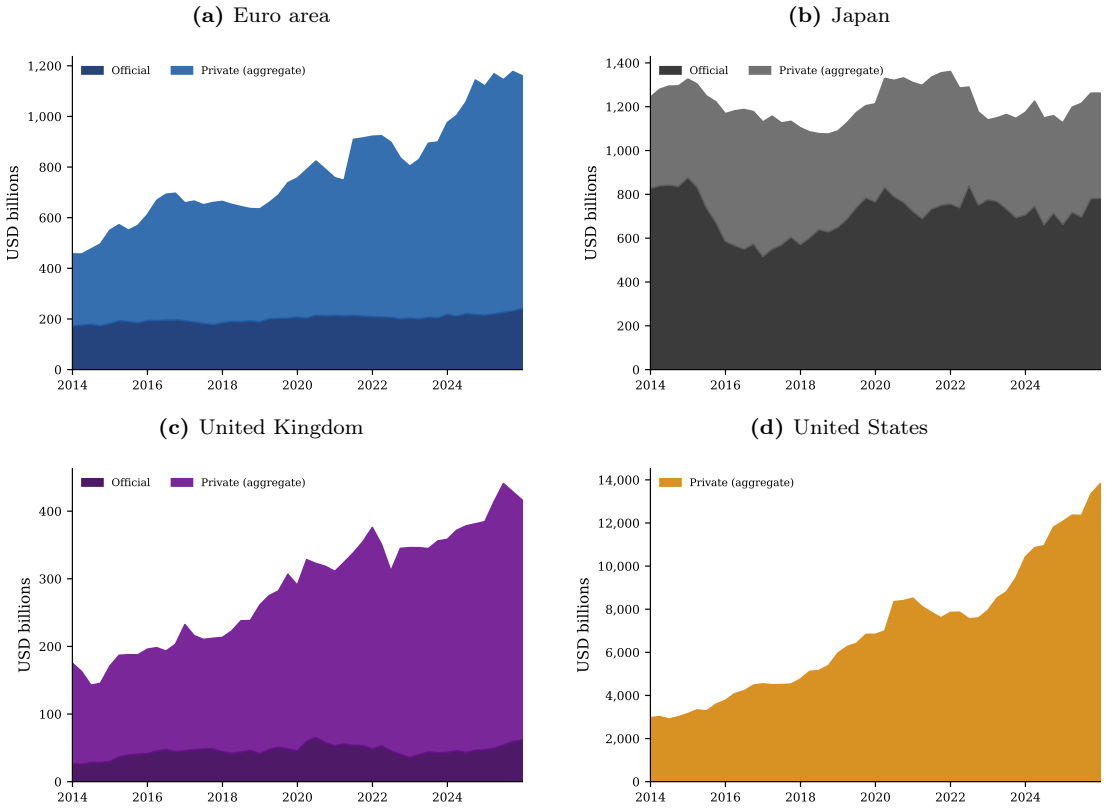


Figure A24. Official/private test: holdings of U.S. Treasuries by official and private sector

Notes: Quarterly holdings of U.S. Treasury securities by official and private sector, stacked with official holdings at the bottom, in USD billions, Q4 2013–Q4 2025. Official holdings are measured as in the baseline sector split. The private aggregate is the sum of the three private sectors for Japan and the United Kingdom, and the restated Treasury total net of official holdings for the euro area. Panel (d) shows the sum of the three modelled private sectors; U.S. official holdings, including the Federal Reserve's System Open Market Account, are excluded. China and Canada, country aggregates in both taxonomies, are not shown. *Sources:* Author's calculations based on U.S. Treasury (TIC); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts); BIS; Bank of England; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026).

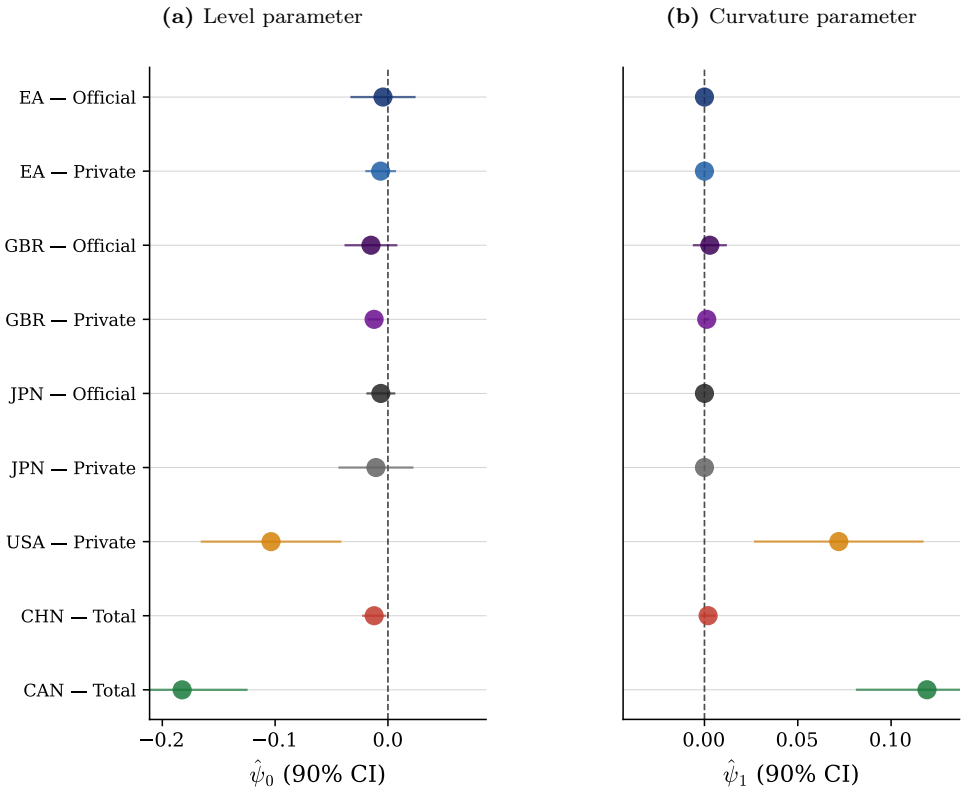


Figure A25. Official/private test: estimated convenience parameters

Notes: Point estimates of the level parameter $\hat{\psi}_{0,c,i}$ and the curvature parameter $\hat{\psi}_{1,c,i}$ with 90% confidence intervals for the 9 cells of the official/private taxonomy, estimated cell by cell by two-step GMM, Q4 2013–Q4 2025. Confidence intervals are based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987). Estimates at the bound $\psi_{1,c,i} = 0$ are plotted at zero without a confidence interval. The model is re-estimated on the 9 cells of the official/private taxonomy (Appendix F.3). *Sources:* Author's estimates based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

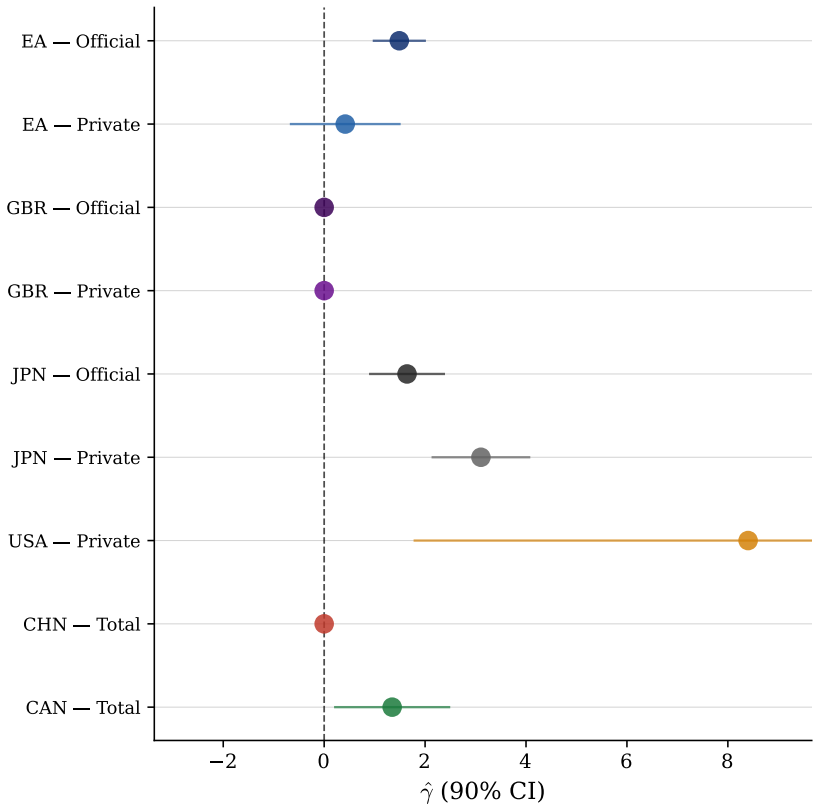


Figure A26. Official/private test: estimated risk aversion

Notes: Point estimates of risk aversion $\hat{\gamma}_{c,i}$ with 90% confidence intervals for the 9 cells of the official/private taxonomy, estimated cell by cell by two-step GMM, Q4 2013–Q4 2025. Confidence intervals are based on Newey–West (HAC) standard errors with four lags (Newey and West, 1987). Estimates at the bound $\gamma_{c,i} = 0$ are plotted at zero without a confidence interval. The model is re-estimated on the 9 cells of the official/private taxonomy (Appendix F.3).
Sources: Author's estimates based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

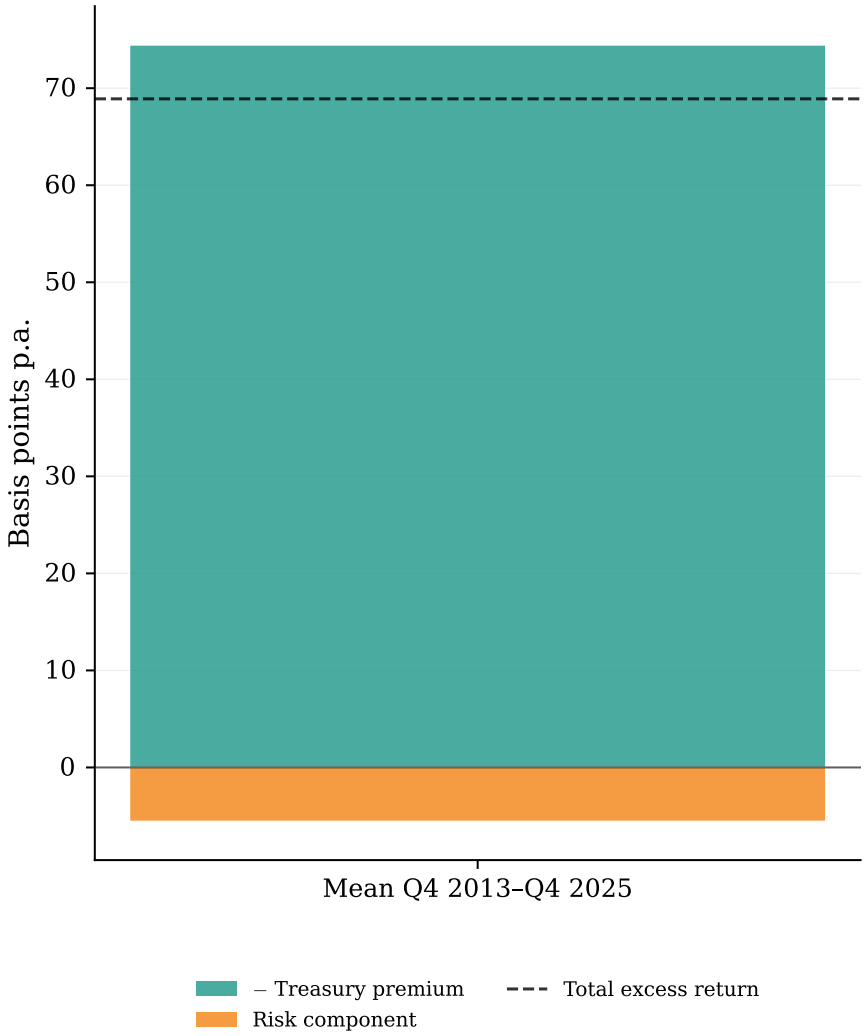


Figure A27. Official/private test: excess return decomposition

Notes: Time-series averages, in basis points per annum, Q4 2013–Q4 2025. The dashed line is the model-implied equilibrium excess return of U.S. Treasuries, $E_t^* = R_{US,t+1}^* - \bar{R}_t^F$, where $\bar{R}_t^F$ averages these cells' benchmark returns with weights equal to each country's number of cells. It is split into the Treasury-premium component $-TP_t = R_{US,t+1}^* - R_{US,t+1}^*\big|_{\psi=0}$, positive when Treasury preferences raise the equilibrium return, and the risk component $R_{US,t+1}^*\big|_{\psi=0} - \bar{R}_t^F$. Both equilibrium returns solve the market-clearing condition (9) quarter by quarter, with each cell's own benchmark return; $\bar{R}_t^F$ is used for reporting only. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The model is re-estimated on the 9 cells of the official/private taxonomy (Appendix F.3). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

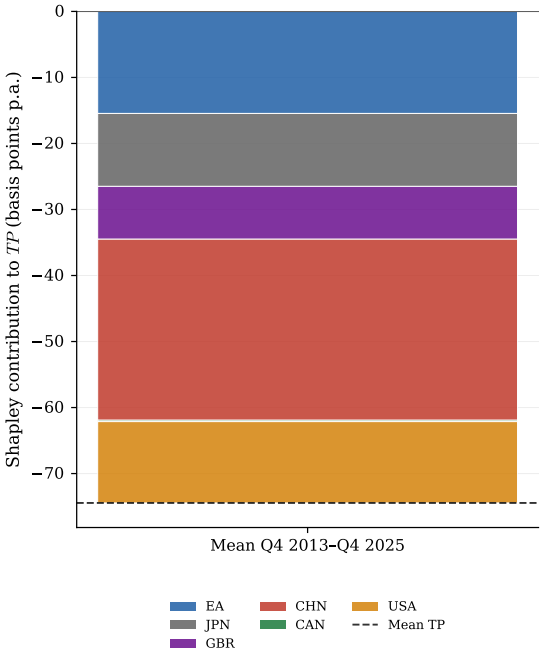


Figure A28. Official/private test: Treasury premium decomposition by country

Notes: Shapley contributions of each country's cells to the model-implied Treasury premium $TP_t = R^*_{US,t+1}\big|_{\psi=0} - R^*_{US,t+1}$, time-series averages in basis points per annum, Q4 2013–Q4 2025. The dashed line marks the average TP_t; a negative contribution means that the country's preferences raise the equilibrium Treasury return. Shapley values are computed quarter by quarter by exact enumeration of all 2^9 coalitions of cells, re-solving the market-clearing condition (9) for each coalition; cell contributions sum to TP_t, and a country's contribution is the sum of its cells' contributions. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The model is re-estimated on the 9 cells of the official/private taxonomy (Appendix F.3). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

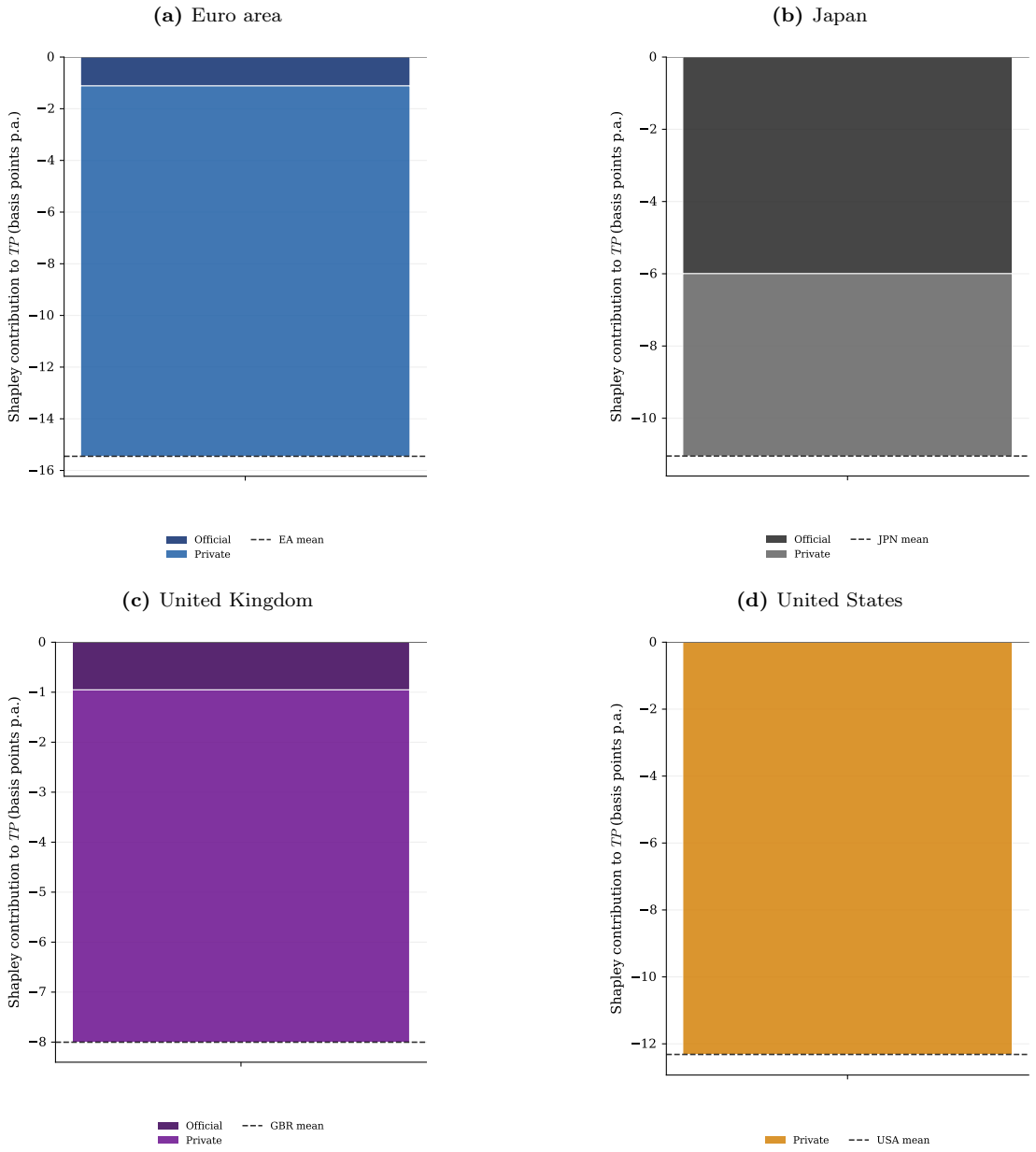


Figure A29. Official/private test: Treasury premium decomposition by sector

Notes: Shapley contributions of each sector's cell to the model-implied Treasury premium $TP_t = R^*_{US,t+1}\big|_{\psi=0} - R^*_{US,t+1}$, time-series averages in basis points per annum, Q4 2013–Q4 2025, for the euro area, Japan, the United Kingdom and the United States; negative values raise the equilibrium return. The dashed line marks the country total. Vertical scales differ across panels. Shapley values are computed quarter by quarter by exact enumeration of all 2^9 coalitions of cells, re-solving the market-clearing condition (9) for each coalition; cell contributions sum to TP_t. China, Canada and the U.S. official sector are not shown; cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The model is re-estimated on the 9 cells of the official/private taxonomy (Appendix F.3). The United States has only a private cell. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

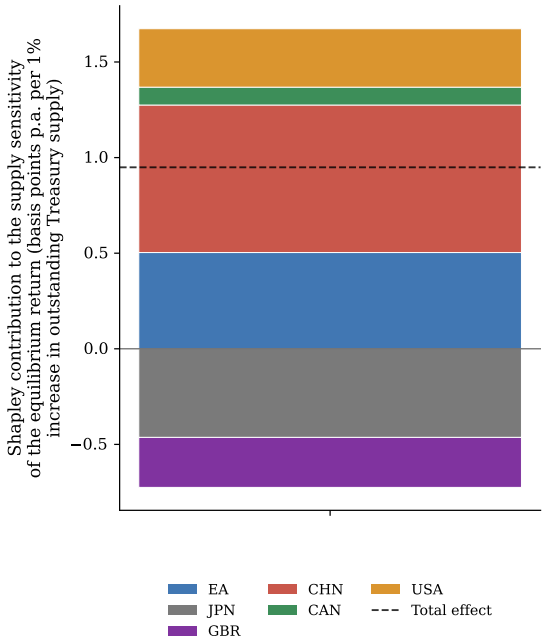


Figure A30. Official/private test: supply sensitivity decomposition by country

Notes: Shapley contributions of each country's cells to the difference between the supply sensitivity of the equilibrium return, $\mathrm{d}R^*_{US,t+1}/\mathrm{d}\log S_t$, with and without Treasury preferences, in basis points per annum per one percent increase in Treasury supply, time-series averages over Q4 2013–Q4 2025. The dashed line is the total difference; a positive value means that Treasury preferences raise the sensitivity. Positive and negative contributions are stacked on opposite sides of zero, so the dashed line equals their net sum rather than the top of the bar. Shapley values are computed quarter by quarter by exact enumeration of all 2^9 coalitions of cells, re-solving the market-clearing condition (9) for each coalition. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The model is re-estimated on the 9 cells of the official/private taxonomy (Appendix F.3). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

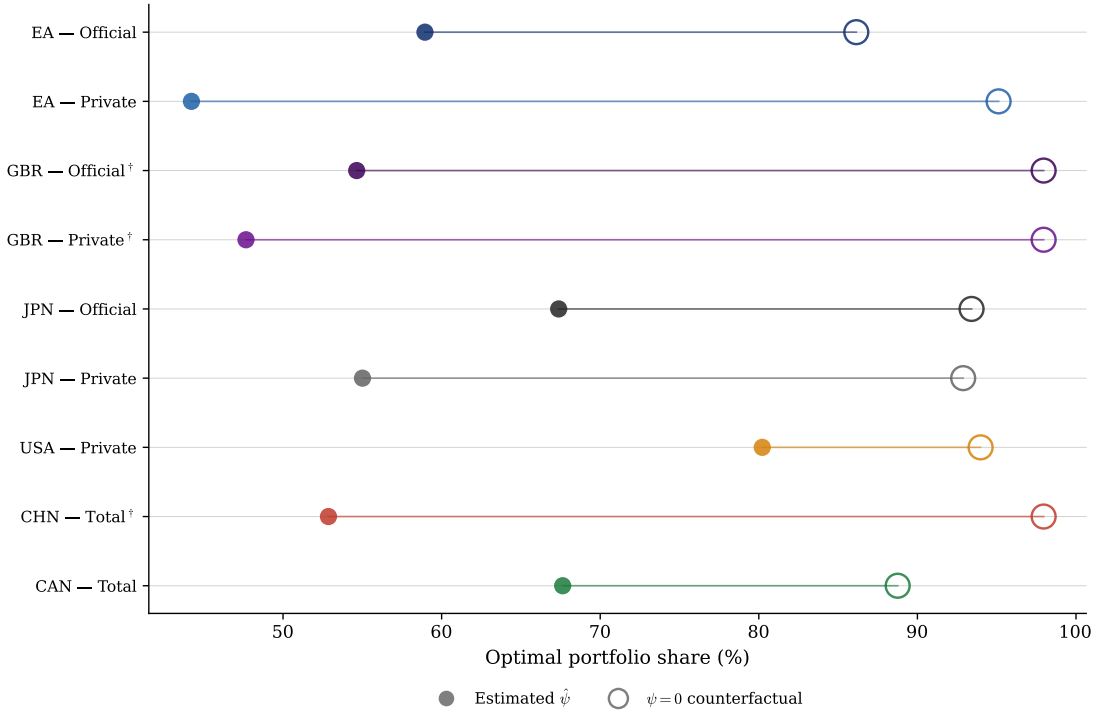


Figure A31. Official/private test: portfolio share experiment

Notes: Time-series average of the optimal Treasury portfolio share $s^*_{c,i,t}$, in percent, at the estimated preference parameters $(\hat{\psi}_{0,c,i}, \hat{\psi}_{1,c,i})$ (filled markers) and in the counterfactual $\psi_{0,c,i} = \psi_{1,c,i} = 0$ (open markers), Q4 2013–Q4 2025. Shares are evaluated quarter by quarter, one cell at a time, from equation (5) at the cell's observed excess return $E_{c,t}$, variance $V_{c,t}$ and covariance $C_{c,t}$, with estimated risk aversion $\hat{\gamma}_{c,i}$, and are bounded between zero and one. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution of Appendix A.4. The model is re-estimated on the 9 cells of the official/private taxonomy (Appendix F.3). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

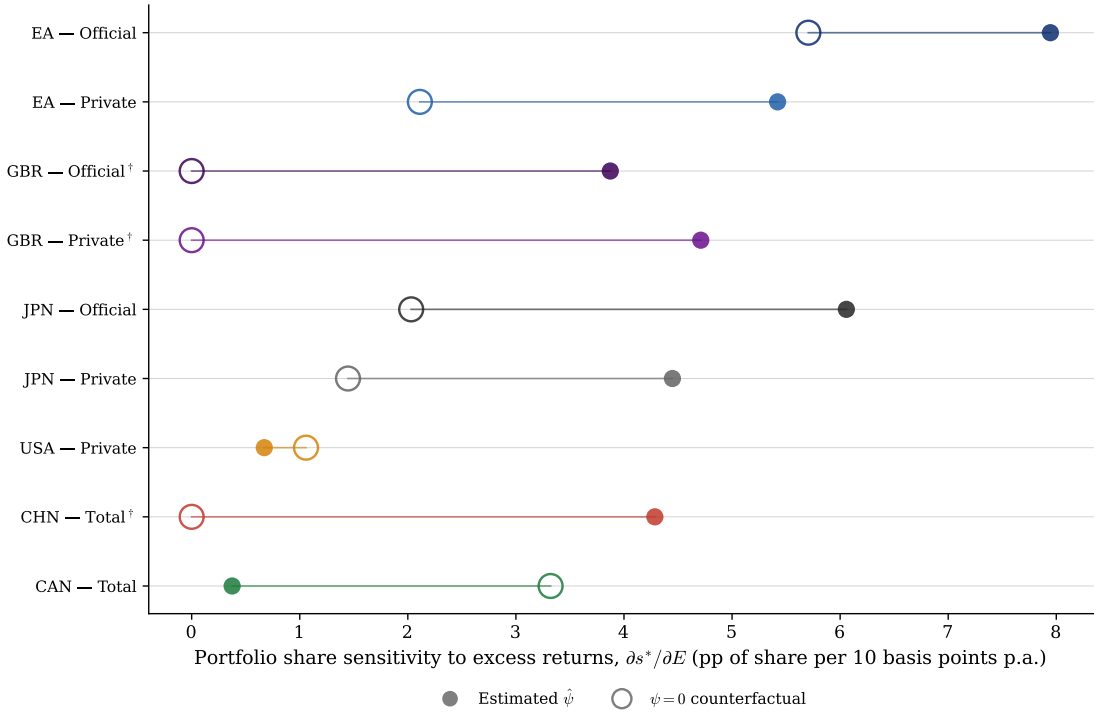


Figure A32. Official/private test: portfolio sensitivity experiment

Notes: Time-series average of the sensitivity of the optimal Treasury portfolio share to the excess return, $\mathrm{d}s^*_{c,i,t}/\mathrm{d}E_{c,t}$ (Proposition 2), in percentage points of portfolio share per ten basis points per annum of excess return, at the estimated preference parameters (filled markers) and in the counterfactual $\psi_{0,c,i} = \psi_{1,c,i} = 0$ (open markers), Q4 2013–Q4 2025. The derivative is evaluated quarter by quarter, one cell at a time, at the share that is optimal under each set of parameters and at the cell's observed $(E_{c,t}, V_{c,t}, C_{c,t})$, and is set to zero in quarters in which that share is at zero or one. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution of Appendix A.4; where their curvature is also zero, their sensitivity is zero in every quarter. The model is re-estimated on the 9 cells of the official/private taxonomy (Appendix F.3). In the 3 cells with risk aversion at the bound of zero, the counterfactual sensitivity is zero in every quarter. *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED.

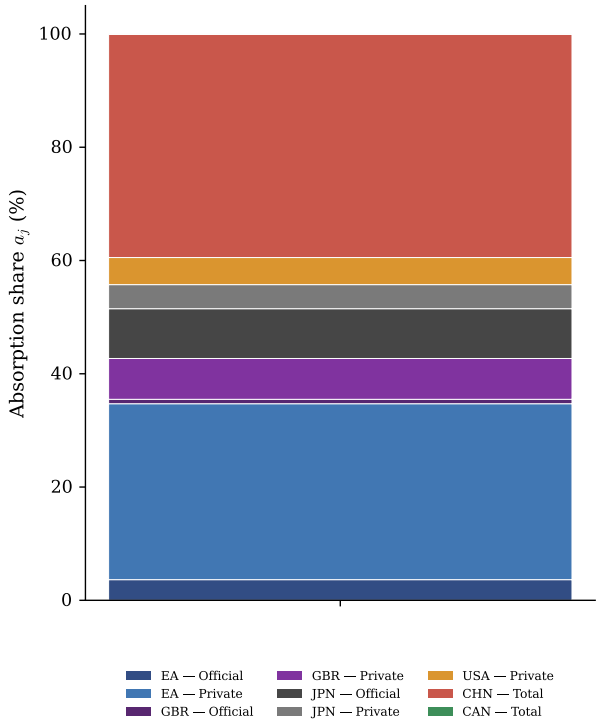


Figure A33. Official/private test: absorption shares

Notes: Time-series averages of the absorption shares $a_{j,t} = (W_{j,t}/D_{j,t})/\mathcal{A}_t$ of Section 2.4, in percent, at the equilibrium of this test, Q4 2013–Q4 2025. In any quarter, a cell whose share is at a bound of $[0, 1]$ has an absorption share of zero, and a cell with a step demand that holds an interior share has an absorption share of one hundred percent. Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The model is re-estimated on the 9 cells of the official/private taxonomy (Appendix F.3). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.

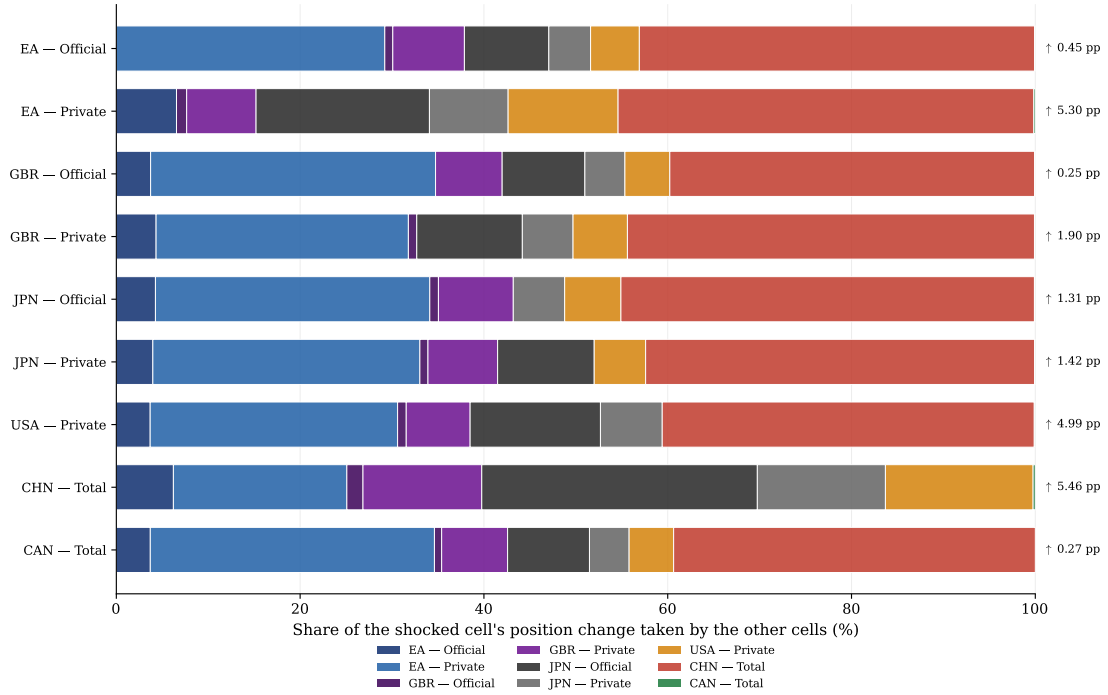


Figure A34. Official/private test: counterparties to a cell's change in position

Notes: Each bar refers to the cell on the vertical axis, whose Treasury preferences alone are set to zero, Q4 2013–Q4 2025. Segments show the share of that cell's change in Treasury holdings taken by each other cell in the re-solved equilibrium, in percent, averaged over quarters with weights equal to the absolute size of the shocked cell's own change; segments sum to one hundred. The label on the right gives the direction and the average size of the shocked cell's own change, in USD billions. Quarters in which the shocked cell is at the same bound of $[0, 1]$ with and without its preferences are excluded (6 of 441 cell-quarters). Cells with $\hat{\gamma}_{c,i} = 0$ use the risk-neutral solution (Appendix A.4). The model is re-estimated on the 9 cells of the official/private taxonomy (Appendix F.3). *Sources:* Author's model calculations based on U.S. Treasury (TIC, Monthly Statement of the Public Debt); ECB (SHS, SHSS, exchange rates); IMF (CPIS, PIP, IIP, IRFCL); Board of Governors of the Federal Reserve System (Financial Accounts; H.10 and H.15, via FRED); BIS; Bank of England; Statistics Canada; Bank of Canada; State Administration of Foreign Exchange; Ito and McCauley (2020) and euro-area central bank reports; Beck et al. (2026); FTSE Russell (WGBI) and Markit (CDS), via LSEG Datastream; CBOE (VIX) and OECD (exchange rates), via FRED; Federal Reserve Bank of Dallas.